

# FINITE FAILURE SCHEMES AND THE RECONSTRUCTION OF QUADRATIC PENCILS

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**ABSTRACT.** We prove that an unmarked, ungraded finite local algebra associated with a complex quadratic pencil determines the pencil up to congruence. The truncation order  $n^2 + 2n - 4$  is sharp, and no regularity assumption is imposed. Multiplication recovers the first relation, its determinant rulings, and the orientation that identifies the pencil coefficient line. The construction extends to families over arbitrary complex bases. On the effectively rigidified failure stack, the recovered source carries a scalar-normalized apolar algebra bundle. The associated multiple-incidence boundary admits explicit multi-Rees equations and relative lifting conditions. The sharp inverse is proved independently of this boundary application; its relative, covering, recognition, and spectral extensions are retained in the appendices. The boundary application now includes formal contact neighbourhoods and non-equidimensional Hilbert fibres, in the explicitly effective deformation category. On the first nonreduced corank-two boundary, the effective-family application determines the whole Hilbert fibre, its two component types, their intersections, and its nilpotent ideal.

## 1. INTRODUCTION

Let  $V$  be a complex vector space of dimension  $n \geq 3$ , and let  $R \subset \operatorname{Sym}^2 V$  have dimension two. Set  $N = \binom{n+1}{2}$ ,  $p = N - 2$ , and  $d = n + 2p = n^2 + 2n - 4$ . The universal graph matrix  $T$  and the quotient map  $\gamma_R$  define the finite local algebra

$$A_R = \mathbb{C}[t_{ij}] / ((\det T)I_p(\gamma_R \operatorname{Sym}^2 T) + \mathfrak{m}^{d+1}).$$

Our principal assertion is

$$A_R \simeq A_{R'} \text{ as unmarked, ungraded algebras} \iff R' = gR \quad (g \in \operatorname{PGL}(V)).$$

Every smaller truncation is independent of the pencil (Theorems 8.3 and 7.1). This includes singular pencils. Neither a chosen grading nor a preferred system of generators is part of the input.

**1.1. The inverse.** The maximal ideal filtration recovers the cotangent space and the first nonzero relation space. The common determinant divisor of that relation recovers the two tensor rulings. The residual coefficient module distinguishes them by unequal support ranks; its remaining coefficient line is the Pluecker line of  $R$ . The proof must show that each of these operations survives ungraded isomorphisms, including nonlinear generator changes. The first seven sections give this argument and its sharpness. The scheme-level coefficient map is a closed immersion over arbitrary complex bases (Theorem 7.3), which also gives the family form of the inverse.

The main proof ends with the coefficient-support orientation and local algebra inverse. The appendices contain all relative, covering, recognition, rigidification, and spectral extensions, together with their proofs. They are not additional hypotheses of the sharp inverse. This separates the central reconstruction argument from consequences that require further geometric inputs.

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**1.2. An effective boundary application.** The sharp first relation already determines the source projective space before the final pencil-coefficient step. Its first-jet bundle and Cartan relations define a source-normalized algebra bundle. Scalar weights cancel, so this construction descends on projective source torsors even when a global  $\mathcal{O}(1)$  does not exist. Proposition E.1 states the resulting morphism of stacks. It is made on the effective pencil-failure stack, after the two specified ineffective kernels are divided out, and not on the stack of all Artin algebras of the same length. The case  $h = 1$  requires no extra exponent. The envelope is not asserted to be a subquotient of  $A_R$ .

The companion computes the full Hilbert modification over the open of regular pencils with any number of distinct reduced corank-two incidences and no higher corank. Its exact centre is  $\text{Fitt}_0(q_*\mathcal{O}_D)$ , where  $D$  is the zero scheme of  $J_{n-1}$  on the intrinsic pencil line. Etale locally the centres have independent parameters  $(u_i, v_i)$ . Theorem E.2 gives the multi-Rees equations  $v_i U_i - u_i V_i$  and descends the computed modification to the effective stack. Proposition E.3 describes the obstruction to lifting a *fixed* small base deformation by the classes  $[v'_i - u'_i \tilde{w}_i] \in J/u'_i J$ . This relative question is different from the unobstructed absolute deformation problem on a smooth quotient stack. It is also different from the local hypersurface deformation module of a singular universal curve.

The recovered pencil line remains an essential intermediate object of this application. A closed algebra determines the space of boundary directions; a family supplies its chosen direction. The multi-Rees equations and relative lifting conditions make the boundary calculation explicit without attributing to the closed algebra a direction that it does not contain.

**1.3. Nonreduced contact neighbourhoods.** The companion's Theorem II.6.2 replaces selected Jordan slices by full formal contact coordinates. Theorem II.8.2 identifies open charts of the ambient Hilbert modification and their mixed-thickness collision law. At a single contact with exponents  $(2, 2)$  the fibre has a dimension-two component containing the thick line-tail jets and a distinct component of dimension at least four containing reduced conic limits (Theorem II.A.1). Corollary F.3 transports these statements to the effective failure image. This uses the inverse on families; it does not assign a chosen tail to a closed algebra before reconstruction.

Proposition F.1 gives an algebraic classifying stack for the fppf forms of the specified envelope bundle. Proposition F.2 identifies the earlier slope defects with the relative cotangent-complex obstruction and writes their transition on the opposite chart. These supplementary results are not premises of the sharp inverse.

**1.4. Dependencies and historical scope.** The sharp inverse uses neither complete quadrics nor the Hilbert boundary theorem. Conversely, the companion's power-ideal and boundary proofs begin with a vector space and a pencil and do not use this inverse. Their interpretation as invariants of unmarked failure algebras uses our source and pencil reconstruction. The applications state these cross-paper inputs rather than hiding them in embedded numerical references.

Classical symmetric-pencil completeness is accessible in [De Teran–Dmytryshyn–Dopico(2018), Theorem 2.1]. A minimal-index block has size  $2\varepsilon + 1$ , explaining the finite range of syzygy maps. Both spectral positions and their local partitions are needed, modulo one common projective reparametrization. Ballico's available quadratic-normality failure-cycle theorem [Ballico(1996), Theorem 0.2] concerns existence of finite subschemes explaining a failure on a projective surface; it is not used as a reconstruction theorem here. This comparison does not replace the still unavailable theorem/proof-level text of [1]. The latter comparison remains documentary-limited; no claim of anticipation or nonanticipation is inferred from it.

**1.5. Complete fibres of effective families.** The companion's Theorem II.9.2 determines the entire first nonreduced contact fibre, rather than only its primitive open and a family in its

complement. Its reduced components are  $\mathbb{P}^4$  and  $\mathbb{F}_2$ , meeting along the doubled-line  $\mathbb{P}^1$ . The local equation  $(eA, eB, e^2C)$  also exhibits a square-zero nilpotent ideal. Theorem II.10.1 then determines all fibres whose smaller corank-two Smith exponents are at most two, with arbitrary larger exponents and any number of distinct contacts. Corollary G.3 transports this scheme-level description through the effective family inverse. This transfer is an application, not a premise of the sharp inverse or an assertion about every raw Artin-algebra deformation. Proposition G.2 specifies the linearization splitting the envelope automorphism extension. No external independent proof audit of the whole sharp inverse is claimed by the present manuscript.

## 2. FIRST RELATIONS, UNGRADED ISOMORPHISMS, AND INVERSE SYSTEMS

The passage from a finite local algebra to its first relation space is independent of pencils. We isolate it before undertaking the geometric inverse. In this section  $E$  denotes a space of degree-one *generators*; thus it is a cotangent space, rather than the dual normal space used later. Let  $e = \dim E$ ,  $d \geq 2$ , and  $K \subset \text{Sym}^d E$ . Set

$$A(K) = \text{Sym } E / ((K) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = (E), \quad \mathfrak{n} = \mathfrak{m}A(K). \quad (2.1)$$

The grading in this presentation is not supplied to an isomorphism.

**Proposition 2.1** (The first-relation orbit principle). *The abstract algebra  $A(K)$  recovers  $E$  and  $K$ , up to linear isomorphism, by*

$$E = \mathfrak{n}/\mathfrak{n}^2, \quad K = \ker[\text{Sym}^d(\mathfrak{n}/\mathfrak{n}^2) \rightarrow \mathfrak{n}^d]. \quad (2.2)$$

Consequently  $A(K) \simeq A(K')$  as ungraded complex algebras if and only if some  $a : E \xrightarrow{\sim} E'$  satisfies  $(\text{Sym}^d a)K = K'$ .

More precisely, after fixing the homogeneous inclusions of generators, all isomorphisms with tangent map  $a$  are exactly the substitutions

$$x \mapsto a(x) + u(x), \quad u \in \text{Hom}_{\mathbb{C}}(E, \mathfrak{n}^2).$$

These assertions about substitutions remain valid after scalar extension to any commutative complex algebra.

*Proof.* The positive ideal is the unique maximal ideal, and  $\mathfrak{n}^{d+1} = 0$ . Multiplication of  $d$  lifts of cotangent vectors is independent of their lifts: replacing any one by an element of  $\mathfrak{n}^2$  changes the product in  $\mathfrak{n}^{d+1}$ . There are no relations of smaller degree, and the degree- $d$  quotient is  $\text{Sym}^d E/K$ . This proves (2.2). In particular every isomorphism induces the stated linear equivalence.

Conversely substitute  $a(x) + u(x)$  for each generator. A relation of degree  $d$  changes, beyond its linear substitution, only in degrees at least  $d + 1$ . Those terms vanish. The relations of degree  $d + 1$  also vanish. The induced map on associated graded algebras is the isomorphism induced by  $a$ ; a finite filtration then makes the algebra map invertible. Every map with this tangent map has precisely such generator images. These polynomial identities hold over any commutative complex coefficient algebra. The intrinsic preservation of the augmentation under that scalar extension is verified in the next proof, so no preservation of a chosen origin has tacitly been added to the isomorphism problem.  $\square$

**Proposition 2.2** (The full automorphism group scheme). *Write  $\ell = \dim_{\mathbb{C}} A(K) = \binom{e+d}{d} - \dim K$  and  $H_K = \text{Stab}_{\text{GL}(E)}(K)$ , with its closed subgroup scheme structure. The functor  $D \mapsto \text{Aut}_D(A(K) \otimes_{\mathbb{C}} D)$  is an affine group scheme of finite type and fits into a split exact sequence of affine group schemes over  $\mathbb{C}$ :*

$$1 \longrightarrow U_K \longrightarrow \text{Aut}(A(K)) \longrightarrow H_K \longrightarrow 1. \quad (2.3)$$

*The kernel is unipotent. Choosing the homogeneous lifting of  $E$  identifies its underlying scheme, but not generally its group law, with the affine space associated to  $\mathrm{Hom}_{\mathbb{C}}(E, \mathfrak{n}^2)$ . Its dimension is  $e(\ell - 1 - e)$ . Intrinsically,  $U_K$  acts simply transitively on the affine space of linear splittings of  $\mathfrak{n} \twoheadrightarrow \mathfrak{n}/\mathfrak{n}^2$ . The splitting of (2.3) uses a choice of such a lifting; it is not an additional intrinsic grading.*

*Proof.* Choose a basis of the finite-dimensional vector space  $A = A(K)$ . Inside  $\mathrm{GL}(A)$ , the equations  $f(1) = 1$  and  $f(xy) = f(x)f(y)$ , imposed on basis elements, are polynomial equations. They represent the automorphism functor, including its values on nonreduced complex algebras.

The augmentation can be recovered without mistaking the absolute nilradical after scalar extension for the augmentation ideal. If  $M_z$  denotes multiplication by  $z$ , the homogeneous filtration gives

$$\mathrm{tr}(M_z) = \ell \epsilon(z). \quad (2.4)$$

Indeed multiplication by a positive-degree element is strictly triangular, whereas multiplication by a scalar has that scalar on every diagonal entry. Identity (2.4) holds over every commutative complex algebra  $D$ . Algebra automorphisms conjugate multiplication matrices, so they preserve  $\epsilon = \ell^{-1} \mathrm{tr}(M_{(-)})$ , its kernel  $\mathfrak{n}_D$ , and all its powers. Thus the tangent map is a morphism of group schemes into  $H_K$ , not merely a map on complex points.

Linear substitutions give a morphism  $H_K \rightarrow \mathrm{Aut}(A)$  sectioning this tangent map. By Proposition 2.1, its kernel has, functorially in  $D$ , the unique parametrization  $x \mapsto x + u(x)$  with  $u : E_D \rightarrow \mathfrak{n}_D^2$ . Reading generator images and making these substitutions are inverse polynomial maps. On a basis ordered by the positive filtration all kernel matrices are unitriangular, proving unipotence. Multiplication is substitution, which need not be addition of corrections. The dimension follows from  $\dim \mathfrak{n}^2 = \ell - 1 - e$ .

Finally two splittings of  $\mathfrak{n} \twoheadrightarrow E$  differ by exactly a map  $E \rightarrow \mathfrak{n}^2$ . Any splitting generates  $A$ , and its degree- $d$  products impose exactly  $K$ ; the preceding substitution argument therefore gives the unique element of  $U_K$  relating any two splittings. This is the asserted coordinate-free torsor description.  $\square$

**2.1. The inverse-system description and its precise role.** Put  $E^\vee = \mathrm{Hom}_{\mathbb{C}}(E, \mathbb{C})$ . Use the apolar pairing in which dual divided monomials satisfy  $\langle x^\alpha, y^\beta / \beta! \rangle = \delta_{\alpha\beta}$ . Here the pairing means evaluation at the origin after applying the constant-coefficient differential operator. For  $I = (K) + \mathfrak{m}^{d+1}$ , its Macaulay inverse system is

$$I^\perp = \bigoplus_{j < d} \mathrm{Sym}^j E^\vee \oplus K^\perp, \quad K^\perp \subset \mathrm{Sym}^d E^\vee. \quad (2.5)$$

To check this equality, the relations of degree  $d + 1$  bound the degree of every inverse-system element by  $d$ . The degree- $d$  relations pair only with its top component. All lower components are unrestricted. Conversely that direct sum is stable under every partial derivative: differentiating its top part lands in the full lower-degree space. It is therefore annihilated by  $I$ , proving the displayed equality. Thus the orbit problem for  $K$  is equivalently the contragredient orbit problem for  $K^\perp$ . This uses the classical inverse-system correspondence; it does not assert a new canonical-grading theorem.

Elias and Rossi prove that Gorenstein local algebras with Hilbert function  $(1, e, e, 1)$  are canonically graded [7, Theorem 3.3 and Corollaries 3.4–3.5]. Their proof removes a lower-degree term of the inverse-system generator by a nonlinear coordinate change. In their later work, compressed Gorenstein algebras of socle degree at most four are canonically graded, with counterexamples beyond the stated range [8, Theorem 3.1 and Examples 3.3–3.5]. In contrast, (2.1) is homogeneous by construction, and its truncation makes all higher substitutions invisible

to its first relation. We do not deduce that an arbitrary filtered deformation with the same Hilbert function is graded.

For the pencil algebras,  $e = n^2$ ,  $d = n^2 + 2n - 4 \geq 11$ , and  $\dim K = \binom{N}{2}$ . They are not Gorenstein: their socles contain  $\mathfrak{n}^d$ , whose dimension is  $\binom{e+d-1}{d} - \binom{N}{2} > 1$ . For example, the first binomial is at least  $\binom{e+1}{2}$ , while  $e = n^2 > N$ . The issue left by Proposition 2.1 is accordingly not the removal of a nonhomogeneous inverse-system term. It is the recovery of the matrix factors and defining coefficient data from the *unrestricted*  $\mathrm{GL}(E)$ -orbit of  $K$ . The next sections establish that recovery and its global descent.

### 3. INTRINSIC TENSOR RULINGS

The first step of the inverse problem is to recognize the two tensor factors of a determinant cone without an ambient marking. The following scheme-theoretic ruling calculation and descent lemma supply precisely this step. They apply to the pencil argument independently of the web and exterior-contraction constructions in the separate research archive.

**Lemma 3.1** (The relative scheme of maximal rank-one spaces). *Let  $B$  be any complex scheme and  $E, F$  vector bundles of rank  $n \geq 2$ . In  $\mathrm{Gr}_B(n, E \otimes F)$ , let  $Z(E, F)$  be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle  $\mathcal{T}$ . Equivalently it is the zero scheme of*

$$\mathrm{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

*The natural morphism*

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (3.1)$$

*is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.*

*Proof.* We first work over  $\mathbb{C}$ . Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- $n$  Veronese embedding on  $\ell^{\otimes n} \otimes \det F$ , respectively  $\det E \otimes m^{\otimes n}$ , followed by its natural linear inclusion into  $\bigwedge^n(E \otimes F)$ . This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through  $Z(E, F)$ .

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors to fix the same factor throughout. A space of dimension  $n$  is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not  $n$ .

We verify that these pointwise statements omit no infinitesimal structure. At  $\ell \otimes F$ , choose a generator of  $\ell$  and a complement in  $E$ . A Grassmannian tangent vector is a map  $\phi : F \rightarrow (E/\ell) \otimes F$ . For each coordinate of  $E/\ell$ , write its component as  $A : F \rightarrow F$ . Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on  $\mathrm{Sym}^2 F$ , so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that  $A$  is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar

$A$  satisfies all the linearized equations. Thus the Zariski tangent space to  $Z(E, F)$  is precisely  $\text{Hom}(\ell, E/\ell)$ , of dimension  $n - 1$ . The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least  $n - 1$ , whereas the tangent calculation gives embedding dimension exactly  $n - 1$ . The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over  $\mathbb{C}$ .

For general  $B$ , trivialize  $E, F$  on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (3.1) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation.  $\square$

**Lemma 3.2** (Oriented relative Segre descent). *Let  $B$  be a connected projective complex variety with  $H^0(B, \mathcal{O}_B) = \mathbb{C}$ , and let  $V, V'$  have the same dimension  $n \geq 2$ . Let  $\mathcal{U}, \mathcal{U}'$  be vector bundles of rank  $n$  on  $B$ . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

*preserves the rank-one cones and the ruling whose parameter bundle is  $\mathbb{P}(V) \times B$ . The induced isomorphism  $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$  is regular and is induced by one constant projective isomorphism  $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ . Locally  $\alpha(T) = gTh^{-1}$ , with the same  $[g]$  on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.*

*Proof.* Lemma 3.1, applied to  $E = V \otimes \mathcal{O}_B$  and  $F = \mathcal{U}^*$ , identifies the closed relative Fano scheme with  $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$ . In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism  $\alpha$  induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial  $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative  $\mathcal{O}(1)$  has the form  $\mathcal{O}(1) \otimes p^*L$  for a line bundle  $L$  on the base. To see this, tensor by  $\mathcal{O}(-1)$ ; the resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward  $\mathcal{O}(1)$  now gives an isomorphism of the corresponding trivial rank- $n$  bundles, with that line-bundle twist. On charts trivializing  $L$ , it is an invertible matrix  $g_i$  of regular functions. On overlaps,  $g_j = u_{ij}g_i$  for an invertible regular function  $u_{ij}$ . The classes  $[g_i]$  thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with  $\text{PGL}_n$ . This accounts explicitly for the possible absence of a global linear lift.

The group  $\text{PGL}_n$  is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from  $B$  to an affine scheme factors through  $\text{Spec } H^0(B, \mathcal{O}_B) = \text{Spec } \mathbb{C}$ , so the projective map is constant. Choose a constant linear lift  $g$ . Each local  $g_i$  then differs from  $g$  by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts  $h$ . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree  $a$  by its scalar to the power  $a$  (or  $-a$  on its dual). It cannot alter a projective coefficient line. The conclusion applies to every homogeneous degree. In particular, no right change of frame can replace the single projective left map by separate transformations on different coefficient spaces.  $\square$

#### 4. ACTUAL FACTOR DESCENT AND MOVING COEFFICIENT SPACES

We now separate the linear-preserver input, the descent of vector bundles, and the recovery of coefficients. The resulting principle allows several varying coefficient spaces, of arbitrary prescribed ranks, rather than only the line associated with a pencil.

**Lemma 4.1** (Linear preservers of the determinant cone). *Let  $U, V, U', V'$  be complex vector spaces of dimension  $n \geq 2$ . An invertible linear map  $\Phi : \text{Hom}(U, V) \rightarrow \text{Hom}(U', V')$  carrying the determinant hypersurface onto the determinant hypersurface has one of the forms*

$$T \mapsto gTh^{-1}, \quad T \mapsto gT^{\mathfrak{t}}h^{-1},$$

where the second formula is written after choosing identifications for transposition. The two cases respectively preserve and exchange the two rank-one rulings. In the first case the pair  $(g, h)$  is unique up to a common nonzero scalar.

*Proof.* At a matrix of rank  $r$ , an invertible  $r$ -minor reduces the local rank condition to the rank condition on a generic  $(n - r)$ -square matrix. The singular locus of the rank-at-most  $s$  variety is therefore the rank-at-most  $s - 1$  variety for  $1 \leq s < n$ : on the rank- $s$  stratum the Schur-complement entries give independent equations, whereas on a lower stratum the linear terms of its defining  $(s + 1)$ -minors vanish. Starting with the determinant cone and iterating reduced singular loci hence recovers the rank-one cone. For  $n = 2$  that cone is already the determinant cone.

If  $v \otimes u$  and  $w \otimes z$  have rank one, their sum has rank at most one only if  $v, w$  or  $u, z$  are dependent. It follows that each maximal linear space of rank-one tensors has one factor fixed. The two families are  $\mathbb{P}(V)$  and  $\mathbb{P}(U^*)$ . The induced Segre automorphism must preserve these families or exchange them. In the preserving case its restriction to a ruling gives projective linear maps on each factor. Remove linear lifts of both maps. The remaining linear transformation fixes each decomposable line. In tensor bases its eigenvalues at  $v_i \otimes u_j$  agree along each row and each column by applying it to  $(v_i + v_k) \otimes u_j$  and  $v_i \otimes (u_j + u_l)$ . They are all equal, so the remaining map is scalar. This proves the first form and its uniqueness. Composing with transposition reduces the exchanging case to the first.

This is the square invertible case of the classical rank-one preserver theorem of Marcus and Moyls [9, Theorem 1, pp. 1218–1219]; the argument above specifies the reduction from the determinant hypersurface that is needed here.  $\square$

**Lemma 4.2** (An actual tensor map removes the line twist). *Let  $B$  be a reduced complex variety,  $V$  a complex vector space with  $\dim V \geq 2$ , and  $\mathcal{A}_1, \mathcal{A}_2$  vector bundles. Suppose an actual bundle isomorphism*

$$\Psi : V \otimes \mathcal{A}_1 \xrightarrow{\sim} V \otimes \mathcal{A}_2$$

*preserves the rank-one cones and induces the identity on the  $\mathbb{P}(V)$ -parameter space of rulings. There is a unique bundle isomorphism  $a : \mathcal{A}_1 \rightarrow \mathcal{A}_2$  with  $\Psi = 1_V \otimes a$ .*

*Equivalently, the intrinsic operation of taking the commutant of the constant matrix algebra gives*

$$\text{Hom}_{\text{End}(V) \otimes \mathcal{O}_B}(V \otimes \mathcal{A}_1, V \otimes \mathcal{A}_2) \simeq \text{Hom}_{\mathcal{O}_B}(\mathcal{A}_1, \mathcal{A}_2). \quad (4.1)$$

*Thus after a constant left projective map has been lifted linearly, no residual line-bundle twist is left in the recovery of the right factor from  $\Psi$ .*

*Proof.* For every constant line  $\mathbb{C}v \subset V$ , the corresponding ruling is fixed, so on every geometric fibre  $\Psi$  maps  $v \otimes \mathcal{A}_1$  onto  $v \otimes \mathcal{A}_2$ . Choose a basis  $v_1, \dots, v_n$  for the purpose of checking this assertion as an equality of bundle maps. The rulings for  $v_i$  force all off-diagonal blocks of  $\Psi$  to vanish. The rulings for  $v_i + v_j$  force its diagonal blocks to agree. Since  $B$  is reduced, fibrewise zero morphisms of vector bundles are zero. Therefore  $\Psi$  commutes with every constant matrix unit, hence with  $\text{End}(V) \otimes \mathcal{O}_B$ .

For clarity, commutant recovery itself is canonical and does not use these chosen matrix units as extra data. Tensoring a bundle map  $a$  with  $1_V$  gives the forward map in (4.1). Commutation with matrix units shows locally that it is bijective, with inverse reading any diagonal block. This proves that it is an isomorphism of sheaves, independent of the checking basis. Its inverse applied to the global section  $\Psi$  produces a global section  $a$ , not projective local sections requiring arbitrary scalar lifts. Applying the same operation to  $\Psi^{-1}$  proves that  $a$  is invertible. Uniqueness follows from injectivity of  $a \mapsto 1_V \otimes a$ .

Projective-factor data alone would permit replacing a bundle by a line twist. The additional datum used here is the actual linear map of the tensor bundles. It is this datum, recovered from the dual of the first nilradical quotient, that makes the commutant argument applicable.  $\square$

**4.1. A general finite-order coefficient inverse.** Fix  $n \geq 2$ ,  $q \geq 1$ , and integers  $0 \leq r_\lambda \leq m_\lambda = \dim \mathbb{S}_\lambda \mathbb{C}^n$  for partitions  $\lambda$  of  $q$  of length at most  $n$ . Assume that for at least one partition

$$0 < r_\lambda < m_\lambda. \quad (4.2)$$

Let  $B$  be a smooth connected projective complex variety,  $V$  a constant  $n$ -dimensional space, and  $\mathcal{U}$  a rank- $n$  bundle. For each  $\lambda$ , choose a subbundle with locally free quotient

$$\mathcal{L}_\lambda \subset \mathbb{S}_\lambda V^* \otimes \mathcal{O}_B, \quad \text{rank } \mathcal{L}_\lambda = r_\lambda.$$

Zero and full subbundles are permitted. In the matrix bundle  $\mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_B)$ , the multiplicity-one Cauchy decomposition supplies the subbundle

$$\mathcal{J}_q = \bigoplus_{\lambda} \mathcal{L}_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^q \mathcal{E}^*. \quad (4.3)$$

Let  $\mathcal{D}_n \subset \text{Sym}^n \mathcal{E}^*$  be the determinant line, put  $D = n + q$ , and define

$$\mathcal{K} = \mathcal{D}_n \mathcal{J}_q, \quad \mathcal{A} = \text{Sym } \mathcal{E}^* / ((\mathcal{K}) + \mathfrak{a}^{D+1}), \quad Y = \text{Spec}_B \mathcal{A}, \quad (4.4)$$

where  $\mathfrak{a}$  is the positive homogeneous ideal. The notation  $D$  in this subsection is an integer, not a failure locus. The scheme  $Y$  is finite over  $B$ , and need not be zero-dimensional over  $\mathbb{C}$ .

**Theorem 4.3** (Unmarked descent for moving coefficient spaces). *For fixed  $n, q, (r_\lambda)$  satisfying (4.2), the abstract complex scheme  $Y$  recovers its coefficient data as follows. Two such schemes  $Y_1, Y_2$  are isomorphic if and only if there are an isomorphism  $\sigma : B_1 \rightarrow B_2$ , a constant linear isomorphism  $g : V_1 \rightarrow V_2$ , and an actual bundle isomorphism  $h : \mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$  such that, for every  $\lambda$ ,*

$$\mathcal{L}_{1,\lambda} = (\mathbb{S}_\lambda g)^* \sigma^* \mathcal{L}_{2,\lambda}. \quad (4.5)$$

*Every unmarked isomorphism induces a normal-bundle map  $T \mapsto gTh^{-1}$ ; for that map the pair  $(g, h)$  is unique up to a common nonzero constant scalar. Conversely every such triple lifts homogeneously to an isomorphism. The scheme is finite locally free over its reduced base of rank*

$$\binom{n^2 + D}{D} - \sum_{\lambda} r_{\lambda} m_{\lambda}.$$

*No nontriviality hypothesis on  $\mathbb{P}_B(\mathcal{U}^*)$  is required.*

*Proof. Recovering the intrinsic linear data.* Multiplication by a determinant polynomial is an injective linear map of polynomial spaces. In local bundle frames its matrix is constant over  $\mathbb{C}$  and its cokernel is free. The Cauchy summands and the  $\mathcal{L}_\lambda$  are subbundles, so  $\mathcal{K}$  is a subbundle of rank  $\sum r_\lambda m_\lambda$ . The graded pieces of  $\mathcal{A}$  below  $D$  are the full symmetric powers; its top piece is  $\text{Sym}^D \mathcal{E}^*/\mathcal{K}$ . This proves local freeness and the rank formula.

The nilradical  $\mathfrak{n}$  of  $\mathcal{O}_Y$  is its positive ideal, since  $B$  is reduced. Thus  $Y_{\text{red}} = B$  and  $\mathfrak{n}/\mathfrak{n}^2 = \mathcal{E}^*$  intrinsically. The multiplication kernel

$$\ker[\text{Sym}^D(\mathfrak{n}/\mathfrak{n}^2) \rightarrow \mathfrak{n}^D]$$

is precisely  $\mathcal{K}$ , by the proof of Proposition 2.1 in local frames. An abstract isomorphism consequently gives the base isomorphism  $\sigma$  and an actual linear normal-bundle isomorphism carrying  $\mathcal{K}_1$  to  $\mathcal{K}_2$  contravariantly. The normal map is linear over the reduced base even when the original scheme map does not preserve the chosen graph projection. Indeed the action on  $\mathfrak{n}/\mathfrak{n}^2$  is linear over  $\mathcal{O}_Y/\mathfrak{n}$ .

*Recovering and orienting the factors.* Generate the homogeneous ideal ( $\mathcal{K}$ ) in the recovered symmetric algebra; do not confuse its cone with the finite truncation. Its radical is the determinant ideal. To verify this, at an invertible matrix  $T$ , the map  $\mathbb{S}_\lambda T$  is invertible. For a partition with  $r_\lambda > 0$ , a nonzero covector in  $\mathcal{L}_\lambda$  has a nonzero matrix coefficient there. Thus the residual coefficients cannot all vanish off the determinant. All generators vanish on the determinant. The relative determinant hypersurface is reduced over the smooth base; the equality of radicals follows on affine charts from this equality of closed points. Successive reduced singular loci and Lemma 3.1 recover its two unordered tensor rulings with their relative scheme structures.

Cancellation of the determinant line is intrinsic and recovers the ideal generated by  $\mathcal{J}_q$ , and hence its degree- $q$  piece. In a Cauchy summand the two contraction supports of  $\mathcal{L}_\lambda \otimes \mathbb{S}_\lambda \mathcal{U}$  have ranks  $(r_\lambda, m_\lambda)$ . Separate factor changes preserve these ranks. By Lemma 4.1, an exchange of the two matrix factors interchanges the ranks in each summand. At a partition satisfying (4.2), the result  $(m_\lambda, r_\lambda)$  cannot equal  $(r_\lambda, m_\lambda)$  for the other datum. Thus no fibre permits an exchange. Twisting local factor lifts by lines changes no support rank. The  $V$ -ruling is thereby oriented intrinsically, including when both projective ruling bundles are trivial.

*Descending the actual map.* By Lemma 3.2, the induced morphism  $B_1 \rightarrow \text{Isom}(\mathbb{P}(V_1), \mathbb{P}(V_2))$  is constant: the target is affine and  $H^0(B_1, \mathcal{O}_{B_1}) = \mathbb{C}$ . Choose a constant linear lift  $g$ . Remove it from the actual normal map. Lemma 4.2, applied to  $\mathcal{A}_1 = \mathcal{U}_1^*$  and  $\mathcal{A}_2 = (\sigma^* \mathcal{U}_2)^*$ , supplies the unique global right factor. Dualizing its inverse gives the actual isomorphism  $h$ . This step uses more than an isomorphism of projective bundles; it is exactly what excludes an undetected line twist. Replacing  $g$  by  $cg$  replaces  $h$  by  $ch$ . These are all ambiguities because the left lift is constant.

*Recovering every coefficient subbundle.* The oriented Cauchy decomposition is now functorial for the recovered left-right map. Project the residual degree- $q$  module onto each summand and contract against the full right dual factor. The image is precisely  $\mathcal{L}_\lambda$ , as a subbundle of the constant left module, not just a collection of fibrewise projective subspaces. The right map acts invertibly on the full Schur factor and hence does not change this image. Pullback by the normal map therefore gives (4.5) for the same constant  $g$  in all summands. Scalar choices in the determinant line do not change these subbundles. This proves necessity.

Conversely  $T \mapsto gTh^{-1}$  carries the determinant line, all coefficient subbundles, and every power of the positive ideal to the corresponding objects. It induces the required homogeneous scheme isomorphism. The stated uniqueness refers to the induced normal map; it does not eliminate the higher-order automorphisms of that scheme.  $\square$

**4.2. Identical fibres and graded bundles do not determine a family.** The global conclusion has content even when neither fibre isomorphism classes nor the underlying graded vector bundles

vary. The following example uses the general coefficient theorem, not a further spectral readout of a reconstructed pencil.

**Example 4.4** (Two isotrivial thickenings separated by coefficient descent). Take  $B = \mathbb{P}^1$ ,  $V = \mathbb{C}^3$ ,  $\mathcal{U} = \mathcal{O}_B^3$ ,  $q = 1$ , and  $r_{(1)} = 1$ . The two everywhere nonzero columns

$$(s^3, t^3, 0)^t, \quad (s^3 + st^2, t^3, 0)^t$$

define subbundles  $\mathcal{L}_0, \mathcal{L}_1 \subset V^* \otimes \mathcal{O}_B$ , both isomorphic to  $\mathcal{O}_B(-3)$ . Let  $Y_0, Y_1$  be their order-4 thickenings (4.4). They are finite flat of rank 712 over  $B$  and Zariski locally trivial with the same Artin algebra as fibre. Their nilradical graded vector bundles are isomorphic in every degree, although  $Y_0$  and  $Y_1$  are not isomorphic as abstract complex schemes.

*Proof.* Neither column vanishes, since its second entry is nonzero when  $t \neq 0$ , and its first is nonzero at  $t = 0$ . The left coefficient map to its image line  $\mathbb{P}^1 \subset \mathbb{P}(V^*)$  is, in the coordinate  $z = s/t$ , respectively

$$f_0(z) = z^3, \quad f_1(z) = z^3 + z.$$

A local frame completing a generator of  $\mathcal{L}_i$  to a basis of  $V^* \otimes \mathcal{O}$  changes the coefficient line to a fixed line. This gives Zariski local product presentations of both thickenings. Alternatively, equality of the fibre isomorphism classes follows from transitivity of  $\mathrm{GL}(V)$  on nonzero coefficient lines. The rank formula is  $\binom{13}{4} - 3 = 712$ .

Let  $E = V^* \otimes \mathbb{C}^3$ , of dimension nine. For  $j < 4$ , the degree- $j$  nilradical quotient is  $\mathrm{Sym}^j E \otimes \mathcal{O}_B$ . Multiplication by the fixed determinant embeds  $E$  as a fixed nine-dimensional subspace of  $\mathrm{Sym}^4 E$ . Choosing a constant linear complement gives, for the top quotient,

$$\mathfrak{n}_i^4 \simeq \mathcal{O}_B^{486} \oplus ((V^* \otimes \mathcal{O}_B)/\mathcal{L}_i)^{\oplus 3}.$$

Each column has only its first two entries nonzero, and these two entries have no common zero. The rank-one quotient of  $\mathcal{O}_B^2$  by  $\mathcal{O}_B(-3)$  is consequently  $\mathcal{O}_B(3)$ , by determinants. Thus the displayed top bundle is  $\mathcal{O}_B^{489} \oplus \mathcal{O}_B(3)^{\oplus 3}$  for both  $i$ . The assertions about every graded vector bundle follow. They are not assertions that the graded *algebra* structures agree.

If an abstract isomorphism  $Y_0 \simeq Y_1$  existed, Theorem 4.3 would identify their coefficient maps by one automorphism of  $B$  and one constant projective transformation of  $V^*$ . Both maps have as image the same projective line, so the latter transformation would restrict to a projective automorphism of that line. Hence  $f_0$  and  $f_1$  would be equivalent by independent projective changes in source and target. This is impossible. The map  $f_0$  has two ramification points, each of index three. The map  $f_1$  has the two simple zeros of  $3z^2 + 1$ , each of index two, and infinity, of index three. Ramification indices are invariant under those projective changes. Thus the unmarked algebra scheme distinguishes these two globally different coefficient maps even though its fibre algebras, conormal bundle, abstract first-relation bundle, and all graded vector bundles agree.  $\square$

**4.3. Pencils and the category of their automorphisms.** For pencils, take  $q = 2p$ ,  $p = N - 2$ , and  $F(H) = \bigwedge^{N-2} \mathrm{Sym}^2 H$ . By Lemma 6.2 this is a single Schur module of dimension  $M = \binom{N}{2}$ , and Lemma K.1 identifies its coefficient subbundle with  $\mathcal{L}_{\mathcal{R}} = \bigwedge^p (\mathrm{Sym}^2 V/\mathcal{R})^*$ . Its rank is one, so the strict-support hypothesis holds for every pencil. Exterior duality recovers the subbundle  $\mathcal{R}$  from that line. Theorem 4.3 therefore proves the unrestricted moving-pencil inverse, including its actual source-bundle conclusion. This is a simultaneous reconstruction theorem for a map into a coefficient Grassmannian, not just a classification of its individual fibres.

**Corollary 4.5** (Algebraic automorphism sequence for a pencil). *Let  $\mathfrak{A}_R^{[d]}$  be the unmarked local algebra of Theorem 8.3, and let  $G_R = \{g \in \mathrm{GL}(V) : (\mathrm{Sym}^2 g)R = R\}$ . There is a split exact*

sequence of complex affine algebraic groups

$$1 \longrightarrow U_R \longrightarrow \operatorname{Aut}(\mathfrak{A}_R^{[d]}) \longrightarrow (G_R \times \operatorname{GL}(U))/\mathbb{G}_m \longrightarrow 1, \quad (4.6)$$

where  $\mathbb{G}_m$  is the diagonal scalar subgroup. Its intrinsic quotient is the closed stabilizer of  $K_R$  in the general linear group of the cotangent space. In that form the sequence represents automorphisms after extension to every commutative complex algebra. Its splitting is homogeneous after choosing generators, and  $U_R$  has the intrinsic description and dimension in Proposition 2.2.

*Proof.* Proposition 2.2 gives the represented sequence with quotient  $H_{K_R}$ . The coefficient inverse and Lemma 4.1 identify its complex points with the image of the left-right action; transposition is excluded by the support ranks  $(1, M)$ . Use the contragredient of the normal action to obtain a homomorphism on the cotangent space. Its kernel is exactly the diagonal scalar subgroup: a tensor product acting identically has both factors scalar, with reciprocal actions on the dual factors. The image is a closed algebraic subgroup and all the groups in question are smooth in characteristic zero [10, Lemma 39.8.2]. Thus the stabilizer and this image agree as closed subgroup schemes, not only on complex points. The induced quotient identification is an isomorphism of algebraic groups: after division by the kernel it is a bijective separable homomorphism between smooth groups. The splitting and kernel are those already constructed by substitutions.  $\square$

For a positive-dimensional projective reduction, Theorem K.4 is used as an exact sequence of groups of complex scheme automorphisms, with its stated filtered kernel. We do not identify that sequence with a finite-type affine group scheme or a moduli-stack equivalence. Likewise, the marked construction over nonreduced parameter bases later in the paper retains its specified relative socle: it does not replace that ideal by the absolute nilradical. These distinctions preserve the full unmarked geometric assertion while specifying its category.

## 5. AN INTRINSIC COEFFICIENT PRINCIPLE

Extracting coefficient data from an abstract scheme and reconstructing a defining relation space from those data are distinct operations. The criterion below isolates the first. Its hypotheses will be verified directly for the multiplication schemes and then on curves.

**5.1. The structural criterion.** Let  $X$  be a finite-type complex scheme with a closed subvariety  $B$  specified by a construction preserved by every abstract scheme isomorphism. Write  $\mathfrak{b} \subset \mathcal{O}_X$  for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \operatorname{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers  $n \geq 2, m \geq 1$ .

- (i)  $B$  is connected and projective,  $H^0(B, \mathcal{O}_B) = \mathbb{C}$ , and  $\mathcal{G}_1$  is locally free. Its dual admits a presentation  $V \otimes \mathcal{U}^*$ , where  $\dim V = \operatorname{rank} \mathcal{U} = n$ .
- (ii) The reduction of  $\operatorname{Spec}_B \mathcal{G}$  in its degree-one ambient bundle is the generic determinant cone. The bundle  $\mathbb{P}_B(\mathcal{U}^*)$  is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If  $\mathcal{D} \subset \mathcal{A}$  is the ideal of that reduced cone and  $\mathcal{L} = \mathcal{D}_n$ , then  $\mathcal{I} = \mathcal{D}\mathcal{J}$ , where  $\mathcal{J}$  is generated in degree  $m$ . Under the Cauchy decomposition in any local frame of  $\mathcal{U}$ ,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \operatorname{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (5.1)$$

where  $\Lambda$  is a specified collection of partitions of  $m$  of length at most  $n$ , and the left subspaces are constant over  $B$ . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations  $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$ .

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

**Theorem 5.1** (Intrinsic coefficient reconstruction). *Suppose  $X, B$  satisfy (i)–(iii). The abstract scheme  $X$  determines the collection  $(L_\lambda)_{\lambda \in \Lambda}$  up to one common projective coordinate transformation on  $V$ . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism  $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$  such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

*The equality means equality of subspaces; scalar changes of a linear lift of  $[g]$  do not affect it.*

*Proof.* An isomorphism transports  $B$ , every power of  $\mathfrak{b}$ ,  $\mathcal{G}_1$ , and the surjection  $\mathcal{A} \twoheadrightarrow \mathcal{G}$ . In particular it transports  $\mathcal{I}$  in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 3.1 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 3.2 then gives one constant projective map  $[g]$ . After choosing a constant linear lift and local frames on both bases, the ambient bundle map has the form  $T \mapsto gTh^{-1}$ . These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally  $\mathcal{D} = (d)$  with  $d = \det T$ . Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence  $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$ , and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under  $T \mapsto gTh^{-1}$ , the Cauchy coefficient map transports a tensor by  $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$ . Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly  $L'_\lambda$  to  $L_\lambda$ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of  $[g]$  act by scalar units. Since every partition has size  $m$ , this ambiguity is common in degree and disappears on subspaces. This proves the assertion.  $\square$

## 6. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

**6.1. Automatic coefficient extraction for quadratic multiplication.** Let  $\dim V = n \geq 3$ ,  $W = \text{Sym}^2 V$ ,  $N = n(n+1)/2$ , and  $R \in \text{Gr}(r, W)$ , where  $1 \leq r < N - n$ . Put  $p = N - r > n$ ,  $S = W/R$ , and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map  $\gamma : \text{Sym}^2 V \rightarrow S$ . Write  $\mathcal{K}$  for the tautological bundle on  $X_R$ , and use the same definition (H.1) of  $\widehat{D}_R$ , with  $A_R$  in place of  $B_R$ . Thus  $\widehat{D}_R$  is the failure scheme of generating  $A_R$  by a varying  $n$ -plane and the unit. The notation in this section agrees with the web construction when  $(n, r) = (4, 4)$ .

Let  $\Lambda_{n,p}$  denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (6.1)$$

Every partition has size  $2p$ ; terms of length greater than  $n$  are omitted. Choose a nonzero volume form on  $S$  and let  $\beta_R \in F(V)^*$  be its pullback by  $\bigwedge^p \gamma$ . Its component in  $\mathbb{S}_\lambda V^*$  is denoted by  $\beta_{R,\lambda}$ . Only the lines of the nonzero components will be used, so the volume choice is immaterial.

**Theorem 6.1** (Automatic intrinsic coefficient extraction). *For every  $R \in \text{Gr}(r, \text{Sym}^2 V)$  in the stated range, the abstract scheme  $\widehat{D}_R$  determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

*up to one common projective transformation of  $V$ , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base  $B = \text{Gr}(n, S)$ , its degree-one normal-cone bundle, and the residual ideal generated in degree  $2p$  with coefficient module*

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (6.2)$$

*where  $\mathcal{U}$  is the tautological rank- $n$  bundle on  $B$ . Consequently all geometric hypotheses of Theorem 5.1 hold for this entire Grassmannian family, not just for smooth webs.*

*Proof.* On a frame chart of  $\mathcal{K}$ , write its inclusion as  $\binom{M}{A_0}$ , with  $M$  an  $n \times n$  matrix. Since  $A_R$  is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \text{Sym}^2 M \end{pmatrix}. \quad (6.3)$$

A nonzero maximal minor must use the scalar column and all  $n$  linear columns: these are the only columns meeting the first  $1+n$  rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M) I_p(\gamma \text{Sym}^2 M). \quad (6.4)$$

The same expansion works for every  $m \geq 2$ . Where  $M$  is invertible, the quadratic map is surjective. Hence the reduction of  $\widehat{D}_R$  is exactly the Schubert divisor where  $M$  is singular, independent of  $R$  after identifying the vector spaces  $S$ .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is  $B = \text{Gr}(n, S)$ . For clarity, the usual determinantal assertion here is local: at a projection-rank- $k$  point, an invertible  $k$ -minor splits away, leaving a generic square  $(n-k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in  $V$  modulo its image. Thus its reduced singular locus is the next rank locus. Since  $p > n$ , the final base is nonempty, connected and projective, with  $H^0(B, \mathcal{O}_B) = \mathbb{C}$ .

Near  $B$ , a Grassmannian chart consists of a varying plane in  $S$  and its graph  $T : \mathcal{U} \rightarrow V$ . Formula (6.4) is independent of the other graph coordinates and homogeneous in  $T$ . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \text{Tot Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T) \mathcal{J}, \quad \mathcal{J} = I_p(\gamma \text{Sym}^2 T). \quad (6.5)$$

There are no degree-one equations. The ambient bundle in (6.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal  $\mathcal{D}$ . The determinant polynomial is monic for a suitable monomial order, so cancellation gives  $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$ . This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 3.1. They are  $\mathbb{P}(V) \times B$  and  $\mathbb{P}_B(\mathcal{U}^*)$ . Exactly the first is projectively trivial. Indeed a Schubert line in  $\text{Gr}(n, S)$  restricts  $\mathcal{U}^*$  to  $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$ . A trivial projective bundle on that line would make this bundle a line-bundle twist of  $\mathcal{O}^n$ ; its determinant would then have degree divisible

by  $n$ , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 3.2 applies: an abstract isomorphism gives a morphism  $B \rightarrow \mathrm{PGL}(V)$ , constant because  $B$  is connected projective and  $\mathrm{PGL}(V)$  is affine. Local linear lifts and right changes of frame give  $T \mapsto gTh^{-1}$  for this one fixed  $[g]$ .

It remains to prove (6.2), rather than assume it as a condition on the example. The minors are the image, for fixed  $\beta_R$ , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \mathrm{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (6.6)$$

In characteristic zero, the polynomial functor  $F$  has the multiplicity-free decomposition (6.1); see [3, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal  $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$  once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify  $U$  with  $V$ , evaluate at  $T = 1$ , and choose  $\beta, \xi$  in that summand with  $\beta(\xi) \neq 0$ . Schur's lemma then identifies its image with the specified Cauchy summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping  $\beta_R$  fixed proves (6.2). All generators have degree  $2p$ , and at least one minor is nonzero since  $\gamma$  is surjective.

Under  $T \mapsto gTh^{-1}$ , the right map acts within the full factor  $\mathbb{S}_\lambda U$ . Contracting against its dual therefore leaves exactly the left coefficient line transformed by  $\mathbb{S}_\lambda(g)^*$ . Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem.  $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved in the separate research archive. For pencils that issue disappears altogether.

## 6.2. Every pencil, without a regularity hypothesis.

**Lemma 6.2** (The exterior square of the quadratic representation). *For  $n \geq 2$ ,  $\bigwedge^2 \mathrm{Sym}^2 V = \mathbb{S}_{(3,1)} V$  is irreducible. Consequently  $\bigwedge^{N-2} \mathrm{Sym}^2 V$  is irreducible as well.*

*Proof.* The vector  $x_1^2 \wedge x_1 x_2$  is a highest-weight vector of weight  $(3, 1)$ . By complete reducibility it generates a copy of  $\mathbb{S}_{(3,1)} V$ . The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities.  $\square$

**Theorem 6.3** (Intrinsic reconstruction of all quadratic pencils). *Let  $n \geq 3$  and let  $R, R'$  be two-dimensional subspaces of  $\mathrm{Sym}^2 V$ . For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function  $(1, n, N-2)$ ,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V). \quad (6.7)$$

*No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.*

*Proof.* Here  $p = N - 2 > n$ . By Theorem 6.1 and Lemma 6.2, the normal cone has only one left coefficient line to recover: the entire line  $\mathbb{C}\beta_R$ . The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (6.8)$$

sends that line to the Pluecker line of  $R$ , with the same projective transformation  $g$ . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel  $R$ . The common

determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as  $\{w \in W : w \wedge w_R = 0\}$ . Thus an abstract scheme isomorphism implies  $R' = gR$ .

Conversely such a linear coordinate change induces the quotient isomorphism  $S_R \rightarrow S_{R'}$ , the algebra isomorphism  $A_R \rightarrow A_{R'}$ , and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (6.4).  $\square$

**6.3. Discriminants, elementary divisors, and hyperelliptic moduli.** A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [4, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

**Corollary 6.4** (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension  $n \geq 3$ , the abstract scheme  $\widehat{D}_R$  determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

*Proof.* The first implication follows by reconstructing  $R$ , then choosing an invertible member  $A$  and a complementary member  $B$ . For completeness, the relevant classical sufficiency can be seen directly. Put  $C = A^{-1}B$ ; it is self-adjoint for  $A$ . If  $hC = C'h$  identifies the Jordan forms for another pair  $(A', B')$ , set

$$H = A^{-1}h^t A' h.$$

Then  $H$  is invertible,  $A$ -self-adjoint, and commutes with  $C$ . There is a polynomial  $k$  in  $H$  with  $k^2 = H$ : on each nonzero eigenvalue choose a square root and truncate its Taylor series to the size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus  $k$  is invertible, self-adjoint, and commutes with  $C$ . The map  $hk^{-1}$  is an isometry from  $A$  to  $A'$  and intertwines  $C, C'$ , so it simultaneously identifies  $A, B$  with  $A', B'$ .

Changing the basis of the pencil replaces  $C$  by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (6.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme.  $\square$

**Example 6.5** (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero  $\mu_3, \dots, \mu_n$ , and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (6.9)$$

At the unique double root the ranks are respectively  $n - 2$  and  $n - 1$ , so no simultaneous projective equivalence exists. The two root partitions are  $(1, 1)$  and  $(2)$ . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

**Corollary 6.6** (Hyperelliptic isomorphism classes). *Fix  $g \geq 2$  and  $n = 2g + 2$ . On the locus of pencils with simple discriminant,  $\widehat{D}_R$  gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus  $g$ . All these failure schemes have the same reduction; their nonreduced structures distinguish a  $(2g - 1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 6.5, where their singular discriminant double covers alone do not.*

*Proof.* The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at  $\det(sA + tB) = 0$ ; see [6, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of  $A^{-1}B$  have mutually  $A$ -orthogonal eigenlines, each nonisotropic, so simultaneous diagonalization identifies the pencil with  $\langle \sum x_i^2, \sum \lambda_i x_i^2 \rangle$ . The unordered branch divisor modulo  $\mathrm{PGL}_2$  therefore determines the pencil, and every reduced divisor of degree  $2g + 2$  arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree  $(2, 2)$  in  $\mathbb{P}^1 \times \mathbb{P}^1$ , whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 6.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension  $2g + 2$ . Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension  $2g - 1$ . Equality of all reductions follows from (6.4). Finally the two binary forms in (6.9) differ only by a scalar, a square over  $\mathbb{C}$ , and define isomorphic double covers, whereas their failure schemes have already been separated.  $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

## 7. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let  $n \geq 3$ ,  $W = \mathrm{Sym}^2 V$ , and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (7.1)$$

For a pencil  $R \subset W$ , put  $S_R = W/R$  and  $B = \mathrm{Gr}(n, S_R) \subset \widehat{D}_R$ . This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by  $A_R$ , as in Section 6. Write  $\mathfrak{b} \subset \mathcal{O}_{\widehat{D}_R}$  for the ideal of  $B$ , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\widehat{D}_R} / \mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (7.2)$$

Thus  $Z_R^{[0]} = B$ . No embedding or identification of its reduction is supplied with the abstract scheme  $Z_R^{[k]}$ .

**Theorem 7.1** (Sharp finite-neighbourhood reconstruction). *For any two pencils  $R, R' \subset \text{Sym}^2 V$ , the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \widehat{D}_R \simeq \widehat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \text{PGL}(V). \end{aligned} \quad (7.3)$$

*In contrast, for every  $0 \leq k < d$ , all the schemes  $Z_R^{[k]}$  are isomorphic. Consequently  $d$  is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.*

**7.1. The first relation in the conormal algebra.** Let  $\mathcal{U}$  be tautological on  $B$ . The open set of  $X_R$  where projection to  $S_R$  is injective is canonically  $\text{Tot } E$ , where  $E = \text{Hom}(\mathcal{U}, V)$ : a plane is the graph of a unique map from its image in  $S_R$  to  $V$ . In particular this is one open neighbourhood of all of  $B$ , not merely a collection of unrelated formal charts. Let  $\mathfrak{a}$  be its zero-section ideal. The exact multiplication calculation (6.4) gives

$$\mathcal{I}_R = (\det T) I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (7.4)$$

It is generated in degree  $d$ , with a nonzero degree- $d$  piece  $K_R$ . Indeed each quadratic minor has degree  $2p$ , and an invertible  $T$  makes  $\gamma_R \text{Sym}^2 T$  surjective. There are no terms of lower degree, even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

**Lemma 7.2** (Intrinsic recovery from the first relation). *The abstract scheme  $Z_R^{[d]}$  determines, as a graded  $\mathcal{O}_B$ -algebra, the complete normal-cone algebra  $\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$ .*

*Proof.* The reduction of  $Z_R^{[d]}$  is  $B$ , since its zero-section ideal is nilpotent and  $B$  is reduced. Hence its nilradical  $\mathfrak{n}$  is intrinsic. Put  $L = \mathfrak{n}/\mathfrak{n}^2$ . Formula (7.4) and  $d > 1$  identify  $L = E^*$ . Multiplication in the given finite scheme recovers

$$K_R = \ker[\text{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (7.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because  $\mathcal{I}_R$  is generated in this single degree,

$$\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \text{Sym}_{\mathcal{O}_B} L/(K_R). \quad (7.6)$$

Here  $(K_R)$  denotes the homogeneous ideal generated by  $K_R$ , not its radical. The right side is intrinsic to the finite scheme, proving the assertion.  $\square$

*Proof of Theorem 7.1.* For  $k < d$ , equations (7.4) and (7.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \text{Sym } E^* / \mathfrak{a}^{k+1}.$$

A linear identification  $S_R \simeq S_{R'}$  identifies  $B$ , its tautological bundle, and  $E$ . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order  $d$  induces an isomorphism of reductions, of conormal bundles, and, by Lemma 7.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 6.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings  $\mathbb{P}(V) \times B$  and  $\mathbb{P}_B(\mathcal{U}^*)$ . Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective

base. Local lifts have the form  $T \mapsto gTh^{-1}$  for this one  $[g]$ . The determinant ideal line is intrinsic, so its ideal quotient in (7.4) recovers the residual ideal. Finally Lemma 6.2 identifies the entire left coefficient line  $\mathbb{C}\beta_R$ , and exterior duality (6.8) recovers the Pluecker line of  $R$ . These are operations on the recovered normal cone; we do not assume that its isomorphism extends to  $X_R$  or  $\widehat{D}_R$ . They imply  $R' = gR$ .

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (7.3), also using Theorem 6.3. For sharpness, there are inequivalent pencils in every dimension under consideration:  $\langle x_1^2, x_1x_2 \rangle$  has a common linear factor, whereas  $\langle x_1^2, x_2^2 \rangle$  does not. Their neighbourhoods are nevertheless isomorphic at every order below  $d$ . Thus no smaller order classifies all pencils.  $\square$

**7.2. A closed immersion of the full parameter scheme.** The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary  $n$ -dimensional vector space  $U$ , and put

$$F(V) = \bigwedge^p \text{Sym}^2 V, \quad F(U) = \bigwedge^p \text{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \text{Hom}(U, V).$$

Choose a nonzero scalar representative  $\delta$  of the determinant polynomial on  $H$ ; its scalar has no effect on the subspaces below. Let  $K_R \subset \text{Sym}^d H^*$  be the degree- $d$  space of  $(\det T)I_p(\gamma_R \text{Sym}^2 T)$  in these fixed coordinates.

**Theorem 7.3** (Faithful finite-order parameterization). *The assignment*

$$\Phi : \text{Gr}(2, \text{Sym}^2 V) \longrightarrow \text{Gr}(M, \text{Sym}^d H^*), \quad R \longmapsto K_R, \quad (7.7)$$

*is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over  $\mathbb{C}$ . In particular it separates families over nonreduced bases and has injective differential at every pencil.*

**Lemma 7.4** (Tensoring a line with a fixed space). *For finite-dimensional vector spaces  $A, B_0$ , with  $b = \dim B_0 > 0$ , the map  $\mathbb{P}(A) \rightarrow \text{Gr}(b, A \otimes B_0)$ ,  $\ell \mapsto \ell \otimes B_0$ , is a closed immersion.*

*Proof.* Choose bases  $a_0, \dots, a_t$  and a basis of  $B_0$ . Above the projective chart  $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$ , the subspace is the graph of the block matrices  $x_i I_b$ . The corresponding Grassmannian chart has arbitrary  $b \times b$  blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any  $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant  $a_0^b$ . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion.  $\square$

*Proof of Theorem 7.3.* The coefficient map (6.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \text{Sym}^{2p} H^*.$$

It is injective: by Lemma 6.2 its source is irreducible for  $\text{GL}(V) \times \text{GL}(U)$ , and evaluation at an isomorphism  $U \simeq V$  shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial  $\delta$  is injective. Thus  $\nu = \delta\mu$  is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (7.8)$$

In particular  $\dim K_R = M$ , also on every singular pencil. By exterior duality the quotient-volume map  $R \mapsto \mathbb{C}\beta_R$  is the Pluecker closed immersion  $\text{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$ . Factor  $\Phi$  through this

map, the closed immersion of Lemma 7.4, and the Grassmannian closed immersion induced by  $\nu$ . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points.  $\square$

**Corollary 7.5** (A flat family of finite normal algebras). *Over  $\mathrm{Gr}(2, W)$ , the algebras*

$$Q_R = \mathrm{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \mathrm{Sym}^j H^*, \quad (7.9)$$

*form a finite locally free family of rank*

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

*In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree  $d$  are constant families.*

*Proof.* Theorem 7.3 makes the spaces  $K_R$  a rank- $M$  subbundle of the constant degree- $d$  bundle, with locally free quotient. In degree  $j < d$  the algebra is  $\mathrm{Sym}^j H^*$ , and its degree- $d$  piece is  $\mathrm{Sym}^d H^* / K_R$ ; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient in degree  $d$  recovers  $K_R$ , and  $\Phi$  recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem.  $\square$

### 7.3. Classical moduli at a sharp order.

**Corollary 7.6** (Hyperelliptic classes in finite neighbourhoods). *Let  $g \geq 2$ ,  $n = 2g + 2$ , and  $d = 4g^2 + 12g + 4$ . For pencils with simple discriminant,  $Z_R^{[d]}$  and  $Z_{R'}^{[d]}$  are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same  $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order  $d$ .*

*Proof.* Combine Theorem 7.1 with Corollary 6.6, and substitute  $n = 2g + 2$  in (7.1).  $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant  $V$ -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 7.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

## 8. COEFFICIENT-SUPPORT ORIENTATION AND A LOCAL ALGEBRAIC INVERSE

For pencils there is a second, entirely local way to distinguish the two tensor rulings. It uses the asymmetry of the residual coefficient module rather than the topology of a projective bundle. Consequently the nontrivial-projectivization hypothesis in the preceding family theorem can be removed, and a single unmarked Artin local algebra already recovers the pencil.

**8.1. The coefficient module orients its own tensor factors.** If  $K \subset A \otimes B$  is a linear subspace, define its two factor supports by

$$s_A(K) = \text{im}(K \otimes B^* \longrightarrow A), \quad s_B(K) = \text{im}(K \otimes A^* \longrightarrow B).$$

The maps here are contractions. Their ranks are invariant under separate changes of basis and are exchanged by interchanging the factors. In particular, if  $K = L \otimes B$ , then  $s_A(K) = L$  and  $s_B(K) = B$ . The support ranks are therefore  $(\dim L, \dim B)$ , not the common dimension  $\dim K$ .

**Lemma 8.1** (Asymmetric coefficient support). *Suppose a homogeneous ideal in the matrix space  $\text{Hom}(U, V)$ , with  $\dim U = \dim V = n \geq 3$ , has reduced support the determinant cone. Suppose its intrinsic residual ideal after determinant cancellation has, in one multiplicity-one Cauchy summand of degree  $q$ , the coefficient subspace*

$$L \otimes \mathbb{S}_\lambda U \subset \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U, \quad 0 < \dim L < \dim \mathbb{S}_\lambda U.$$

*A linear isomorphism carrying this ideal to another ideal with a coefficient subspace of the same displayed type cannot interchange the two matrix rulings. The conclusion is unchanged by line twists of the locally chosen tensor factors. It also applies fibrewise to subbundles of this type over a reduced base.*

*Proof.* A linear isomorphism of determinant cones preserves their successive rank loci and therefore the rank-one cone. Its action on the two ruling components either preserves them or interchanges them. To make the familiar two alternatives explicit, identify the rank-one projectivization with  $\mathbb{P}(V) \times \mathbb{P}(U^*)$ . An automorphism preserving its two families of maximal linear spaces acts separately on the two projective factors. Each factor map is projective linear, by restricting the ambient linear map to a ruling. Removing their linear lifts leaves a map fixing every decomposable line; sums of tensors with one factor in common show that this remaining map is scalar. The preserving alternative is thus a left-right matrix map. In the interchanging alternative, compose with transposition to obtain the preserving alternative.

The determinant ideal is carried to the determinant ideal, so its ideal quotient is carried to the residual ideal. The Cauchy decomposition is functorial for separate changes of matrix factors; transposition exchanges the factors in its  $\lambda$ -summand. Put  $m = \dim \mathbb{S}_\lambda U$ , and  $r = \dim L < m$ . The displayed subspace has support ranks  $(r, m)$ . Separate factor maps preserve these ranks, while transposition changes them to  $(m, r)$ . An ideal of the prescribed type on the other side has support ranks  $(r', m)$  with  $r' < m$ . It cannot be the transposed image. This excludes interchanging the rulings.

Tensoring a factor by a line changes its Schur module by a line power, not its contraction rank. The argument therefore does not require a specified linear lift of either projective factor. For bundle maps it applies to each geometric fibre; an interchange on any fibre would contradict the same strict rank inequality there.  $\square$

For pencils, take  $F(H) = \bigwedge^{N-2} \text{Sym}^2 H$ . Lemma 6.2 makes this a single irreducible Schur module, of dimension  $M = \binom{N}{2} > 1$ . The residual module is precisely  $\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U})$ , by Lemma K.1. Its support ranks are  $(1, M)$ . The relevant strict inequality holds for every pencil, including singular pencils; it is not a generic-rank condition.

**Theorem 8.2** (Removal of the projective-bundle restriction). *In Theorem K.2, the hypothesis that  $\mathbb{P}_B(\mathcal{U}^*)$  is nontrivial can be omitted. The coefficient-subbundle reconstruction, the smallest uniform order, and the automorphism exact sequence remain valid for every rank- $n$  source bundle on a smooth connected projective complex base. In particular the base is allowed to be a point.*

*Proof.* The first-relation construction, the recovered homogeneous cone, and cancellation of the determinant use neither projective nontriviality nor positive dimension of the base. They recover

both the unordered rulings and the residual coefficient module before a ruling is chosen. At this stage apply Lemma 8.1 with support ranks  $(1, M)$ . It excludes the transposed alternative and hence orients the left ruling even when both ruling bundles are trivial. This replaces exactly the nontrivial-projectivization step in the proof of Theorem K.2.

The left ruling is the trivial bundle  $\mathbb{P}(V) \times B$ . Projectivity and connectedness still make its projective-matrix map constant. Choose its constant linear lift; uniqueness of the right bundle map on overlaps gives the same actual isomorphism  $h$ , and coefficient contraction recovers the entire pencil subbundle. The converse uses the same homogeneous map. The lower-order algebras and the two inequivalent constant pencils in Corollary K.3 are unchanged. Finally, the proof of Theorem K.4 uses only the resulting classification of graded data, not the earlier way of orienting the rulings, so it too applies without the restriction.  $\square$

**8.2. Every pencil from one Artin local algebra.** Choose  $n$ -dimensional vector spaces  $U, V$ . Let  $T = (t_{ij})$  be the universal matrix, and put  $\mathfrak{m} = (t_{ij})$ . For a pencil  $R \subset \text{Sym}^2 V$ , let  $\gamma_R$  be its quotient map and set

$$\mathfrak{A}_R^{[k]} = \mathbb{C}[t_{ij}] / ((\det T)I_p(\gamma_R \text{Sym}^2 T) + \mathfrak{m}^{k+1}). \quad (8.1)$$

An isomorphism below means an isomorphism of complex algebras, without a grading, tangent coordinates, matrix decomposition or chosen generators.

**Theorem 8.3** (Unmarked local reconstruction). *For every  $n \geq 3$  and every pair of quadratic pencils,*

$$\mathfrak{A}_R^{[d]} \simeq \mathfrak{A}_{R'}^{[d]} \iff R' = (\text{Sym}^2 g)R \text{ for some } g \in \text{GL}(V). \quad (8.2)$$

*The algebra has embedding dimension  $n^2$  and length  $\binom{n^2+d}{d} - \binom{N}{2}$ . Every lower-order algebra  $\mathfrak{A}_R^{[k]}$ ,  $k < d$ , is independent of  $R$ . Thus  $d = n^2 + 2n - 4$  is the smallest uniform reconstruction order already among these single-point objects.*

*Proof.* Theorem 8.2 applies with  $B = \text{Spec } \mathbb{C}$  and the bundle  $U$ . To spell out what its intrinsic construction uses in this case, the unique maximal ideal  $\mathfrak{n}$  of the finite algebra determines its tangent space and the multiplication kernel

$$K_R = \ker \left[ \text{Sym}^d(\mathfrak{n}/\mathfrak{n}^2) \longrightarrow \mathfrak{n}^d \right].$$

This is precisely the degree- $d$  ideal in the chosen presentation; nonlinear changes of generators contribute only terms above degree  $d$  to a degree- $d$  relation. Hence an ungraded algebra isomorphism identifies the homogeneous cones generated by these kernels. Their determinant reductions recover the unordered matrix factors. The residual support ranks  $(1, M)$  rule out transposition, and the remaining left-right map carries the left coefficient line to the left coefficient line. Exterior duality then gives  $R' = (\text{Sym}^2 g)R$ . Conversely such a map, together with any chosen right identification, induces the algebra isomorphism. There are no relations below degree  $d$ , so the embedding dimension and the lower-order assertion are immediate. The relation subspace at degree  $d$  has dimension  $M$ , giving the asserted length. The inequivalent common-factor and no-common-factor pencils prove uniform minimality as before.  $\square$

In particular the earlier Schubert-line construction is a sufficient geometric realization, not a lower bound on the dimension of a supporting reduced scheme. The local inverse just proved retains no curve at all. Its additional ingredient is the coefficient-support asymmetry, which was not used in the earlier geometric orientation argument. This is a stronger inverse statement, not a conclusion from knowing the discriminant or the reduced determinant cone.

**8.3. The local automorphism kernel is explicit.** Put  $a = n^2$ ,  $\ell = \text{length } \mathfrak{A}_R^{[d]}$ , and  $G_R = \{g \in \text{GL}(V) : (\text{Sym}^2 g)R = R\}$ . The symbol  $\mathfrak{n}^2$  in the next statement is the square of the maximal ideal of the finite algebra.

**Proposition 8.4** (Tangent-identity automorphisms). *There is a split exact sequence of complex automorphism groups*

$$1 \longrightarrow U_R \longrightarrow \text{Aut}_{\mathbb{C}}(\mathfrak{A}_R^{[d]}) \longrightarrow (G_R \times \text{GL}(U))/\mathbb{C}^* \longrightarrow 1, \quad (8.3)$$

where the scalar subgroup is diagonal. The kernel  $U_R$  is a unipotent algebraic group whose underlying variety is an affine space of dimension

$$a(\ell - 1 - a).$$

More explicitly, after choosing the  $a$  degree-one generators  $x_1, \dots, x_a$ , its elements are exactly the substitutions  $x_i \mapsto x_i + u_i$  with arbitrary  $u_i \in \mathfrak{n}^2$ . The affine-space parametrization is not an assertion that this group law is additive.

*Proof.* The local inverse and the support-rank argument identify the linear stabilizer of  $K_R$  with the indicated left-right quotient; transposition is absent. These linear maps lift homogeneously, giving the quotient and splitting. An automorphism in the kernel is the identity on  $\mathfrak{n}/\mathfrak{n}^2$ , and consequently has exactly the displayed generator images.

Conversely choose arbitrary  $u_i \in \mathfrak{n}^2$ . Under this substitution a homogeneous defining relation of degree  $d$  changes only by terms in  $\mathfrak{n}^{d+1} = 0$ , so all defining relations are respected. The induced map is the identity on the associated graded algebra, hence is invertible as a finite filtered linear map and therefore as an algebra map. There are no further conditions on the  $u_i$ . Reading the generator images and performing substitutions are polynomial mutually inverse maps between the kernel and the vector space  $(\mathfrak{n}^2)^{\oplus a}$ . Since  $\dim \mathfrak{n}^2 = \ell - 1 - a$ , this proves the variety and dimension assertions. On a basis ordered by degree every such automorphism is upper unitriangular; the algebraic group is therefore unipotent. Its law includes higher substitution terms, which accounts for its generally noncommutative structure.  $\square$

The positive-dimensional family statement remains useful after the local inverse: it recovers an entire varying subbundle and the actual source bundle up to one global left transformation, whereas separate fibrewise local isomorphisms need not glue. The geometric automorphism statements here still do not identify unmarked moduli stacks or all deformations over nonreduced parameter bases.

#### APPENDIX A. THE INTRINSIC POLARIZATION AND A PROJECTIVE MODULI COMPLETION

Put  $P = \text{Gr}(2, \text{Sym}^2 V)$ ,  $N = \binom{n+1}{2}$ ,  $M = \binom{N}{2}$ , and  $D = n^2 + 2n - 4$ . Fix temporarily an auxiliary  $n$ -space  $U$  and put  $E = \text{Hom}(U, V)^*$ . Write  $L = \mathcal{O}_P(1)$  for the Pluecker line and  $\Gamma = \Phi(P) \subset \text{Gr}(M, \text{Sym}^D E)$  for the closed image of Theorem 7.3. The group in this section is  $\text{SL}(V)$ , acting by congruence on pencils and by its induced action on  $E$ , with  $U$  fixed. It is not the full group  $\text{GL}(E)$ .

**Theorem A.1** (Polarized completion of the effective failure moduli). *There is an  $\text{SL}(V)$ -equivariant isomorphism*

$$\Phi^* \mathcal{O}_{\text{Gr}(M, \text{Sym}^D E)}(1) \simeq L^{\otimes M} \otimes C_U, \quad (\text{A.1})$$

where  $C_U$  is a constant one-dimensional space on which  $\text{SL}(V)$  acts trivially. Consequently the stable and semistable loci of  $\Gamma$ , for the restricted Pluecker linearization, are respectively the images

of  $P^s$  and  $P^{ss}$ . The quotient defined by the complete restricted section ring is canonically

$$\overline{\mathcal{M}}_n^{\text{fail}} = \text{Proj} \left( \bigoplus_{q \geq 0} H^0(P, L^{\otimes q})^{\text{SL}(V)} \right) \simeq \Gamma // \text{SL}(V). \quad (\text{A.2})$$

It is a normal projective variety. It contains the geometric quotient of the simple-discriminant pencils as a dense open subvariety, and has dimension  $n - 3$ . The complement of that open is the image of the semistable discriminant-zero locus.

The intrinsic inverse identifies the effective semistable failure stack with  $[P^{ss} / \text{PGL}(V)]$  and gives a moduli morphism to (A.2). On framed charts the failure-algebra family extends over all  $P$ , remains finite locally free, and commutes with arbitrary base change. These assertions do not require, and do not assert, the existence of a universal ordinary algebra on the coarse quotient or properness of the entire quotient stack.

*Proof.* Let  $\mathcal{K}$  be the tautological relation bundle pulled back by  $\Phi$ . The coefficient factorization (7.8) identifies it with  $\mathcal{B} \otimes F(U)$ , where  $\mathcal{B}$  is the quotient-volume coefficient line. Exterior duality gives  $\mathcal{B} \simeq L^{-1}$  up to a fixed determinant character of  $\text{GL}(V)$ . That character is trivial on  $\text{SL}(V)$ . Taking the dual determinant gives (A.1), with the harmless constant factor  $(\det F(U))^*$  and the corresponding fixed determinant line of  $V$ . A trivialization of the constant factor identifies the section rings; a different trivialization rescales each graded piece and has no effect on its projective spectrum. The section ring for  $L^M$  is the  $M$ th Veronese of the section ring for  $L$ . Taking invariants commutes with this operation. The Hilbert–Mumford weights are multiplied by  $M > 0$ , so stability and semistability are unchanged. This also explains why one uses the complete section ring on  $\Gamma$ , rather than assuming surjectivity of unspecified ambient invariant restrictions.

Finite generation and the projective good quotient are the reductive GIT construction of [25, Chapter 1]. The complete section ring of the normal projective variety  $P$  is integrally closed: it is the intersection of the valuation conditions for rational sections of powers of  $L$  at the prime divisors of  $P$ . Its invariant subring is integrally closed as well. Indeed an element of its fraction field integral over the invariants is integral over the original ring, belongs to that ring by normality, and is invariant. This proves normality of the projective quotient.

For completeness, the simple-discriminant open is nonempty and stable. In a frame  $(Q_0, Q_1)$  of a pencil, take the discriminant of  $\det(sQ_0 + tQ_1)$ . Under a frame change  $h \in \text{GL}_2$  it has weight  $(\det h)^{n(n-1)}$ , and it is invariant under  $\text{SL}(V)$ . Thus it is a nonzero invariant section of  $L^{n(n-1)}$ . It is nonzero at a diagonal pencil with  $n$  distinct roots, so its nonvanishing locus is semistable and saturated. A pencil there has a nonsingular member. Relative to that member, the other member is self-adjoint with distinct eigenvalues; orthogonal diagonalization identifies its congruence class with an unordered configuration of  $n$  distinct points of  $\mathbb{P}^1$ . The latter configurations are stable for the usual binary-form linearization: the root-multiplicity criterion is strict since  $1 < n/2$  for  $n \geq 3$ . Their quotient is geometric and separated. It follows that the congruence orbits are closed in the saturated nonvanishing locus. Their stabilizers are finite: the induced projective transformation permutes a set of at least three points, and the kernel consists of diagonal sign changes modulo scalars. They are therefore stable. This gives a geometric dense open of dimension  $n - 3$ . The complement is the image of the invariant section's zero locus by the good-quotient property. This reasoning uses classical pencil classification and GIT, not a new classification of binary forms; compare [4, 25].

Finally apply Theorem L.2 and the rigidification results of Section M. The Pluecker polarization is taken on the recovered pencil, not on an arbitrary first-relation orbit. A sufficiently divisible power of  $L$ , for example  $L^n$ , has trivial central character and descends from  $\text{SL}(V)$  to  $\text{PGL}(V)$ . A change of auxiliary frame changes the constant factor in (A.1), not the resulting projective

quotient. On the framed Grassmannian the extended algebras are exactly those of Corollary 7.5. Their equivariance yields the stated classifying morphisms. Removing ineffective inertia does not turn these into a universal ordinary algebra on the coarse space, and no such descent is used.  $\square$

*Remark A.2* (What is compactified). The completion is the classical polarized pencil quotient recovered from the intrinsic failure invariant, with its polarization verified by (A.1). It is not a new construction of classical GIT. Its role is to supply a proper coarse target for the effective invariant and a specified class of semistable limiting families. Strictly semistable points represent closed-orbit representatives, not every orbit in their fibres. In particular Segre data on the stack need not descend to an invariant separating all points of a coarse-quotient fibre. For  $n = 3$  the quotient is a point and its discriminant boundary can be empty. The full  $\mathrm{GL}(E)$  common limit of Theorem II.AA.1 is a different construction.

## APPENDIX B. THE INTRINSIC SOURCE AND ITS APOLAR ENVELOPE

The source of a pencil and the coefficient complement in a divisor normalization fibre are different spaces. We construct the native power geometry from the first relation of the failure algebra, without taking a coefficient section of that fibre. The construction produces a canonical projective power diagram and an honest algebra bundle on the intrinsic source projective space. The location of the algebra bundle is essential: forgetting a scalar lift does not give a linear representation of the projective general linear group.

**B.1. A construction from the first relation.** Let  $A$  be an unmarked, ungraded complex algebra in the sharp pencil image, and write  $\mathfrak{m}$  for its maximal ideal. Set

$$E_A = \mathfrak{m}/\mathfrak{m}^2, \quad H_A = \ker\{\mathrm{Sym}^d E_A \longrightarrow \mathfrak{m}^d\}, \quad d = n^2 + 2n - 4. \quad (\text{B.1})$$

Here  $n^2 = \dim E_A$ , so neither  $n$  nor  $d$  is extra marking. The reduced support of the homogeneous ideal generated by  $H_A$  is a determinant hypersurface in  $\mathbb{P}(E_A^*)$ . Its successive reduced singular loci give the rank-one Segre variety. The scheme of maximal rank-one linear spaces has two components. Cancel the determinant factor from  $H_A$  and select the component singled out by coefficient support ranks  $(1, M)$  in Lemma 8.1. Denote the resulting source projective space by  $S_A$ ; with a pencil presentation it is  $\mathbb{P}(V)$ , not  $\mathbb{P}(\mathbb{C}^{n^2-1})$ .

There is a direct jet-theoretic presentation which does not begin with a linear source lift. Let  $L$  be an ample generator of  $\mathrm{Pic}(S_A)$ , and let  $J^1 L$  denote its first principal-parts bundle. Put

$$\mathcal{E}_A = \mathrm{Hom}(J^1 L, L), \quad \mathcal{B}_{A,h} = \mathrm{Sym}(\mathrm{Sym}^2 \mathcal{E}_A) / \langle (u^2)^{h+1} : u \text{ a local section of } \mathcal{E}_A \rangle. \quad (\text{B.2})$$

Scalar automorphisms of  $L$  act trivially on  $\mathcal{E}_A$ , so this bundle is independent of the representative of the ample generator. For  $p > 0$  its projective coefficient space can be constructed as  $\mathbb{P}H^0(S_A, \mathcal{B}_{A,h,p} \otimes L^{-2p})$ ; changing  $L$  changes the linear identification only by a scalar. Multiplication supplies the projective power maps. Thus jets, a Cartan quotient, and projectivization give the construction from the intrinsic source.

**Theorem B.1** (Intrinsic projective power geometry). *On the groupoid of complex sharp pencil-failure algebras and their ungraded isomorphisms, the following construction is functorial. There is a source projective space  $S_A$ , a finite locally free graded algebra  $\mathcal{B}_{A,h}$  on  $S_A$  for each  $h \geq 1$ , and projective spaces  $P_{A,p}$  with a multiplication power diagram. In any source lift  $S_A = \mathbb{P}(V)$  they are*

$$\mathcal{B}_{A,h} = \bigoplus_{p=0}^{nh} B_h(V)_p \otimes \mathcal{O}_{\mathbb{P}(V)}(2p), \quad P_{A,p} = \mathbb{P}(B_h(V)_p). \quad (\text{B.3})$$

*The nondegenerate power graph on  $P_{A,1}$  is the native  $\mathrm{CQ}(V)$ , canonically as a projective diagram, and every rank- $r$  graph is the relative complete-quadric space of Theorem II.D.1. Over a general*

base the projective spaces mean the associated projective schemes, which need not admit a global linear lift. These constructions use only (B.1), determinant cancellation, the oriented ruling, and (II.2.1); no pencil coefficient line or reciprocal-fibre splitting is needed to construct the ambient graph. Extracting that line afterwards supplies the intrinsic pencil line  $\Lambda_A \subset P_{A,1}$ .

All source-lift choices cancel. Once an oriented projective-source bundle is given, the same associated constructions descend over any complex base, including a nonreduced base, and commute with arbitrary base change. This last assertion concerns the associated objects; it does not assert that reduction or graph closure commutes with arbitrary base change on unrelated degenerating families.

*Proof.* The first-relation construction is preserved by every ungraded isomorphism: terms of degree at least two in a change of generators cannot change the degree- $d$  relation modulo  $\mathfrak{m}^{d+1}$ . The determinant and its rank-one locus are therefore intrinsic. Lemma 3.1 identifies the two ruling components as schemes; coefficient-support asymmetry orients them. These are the steps of the local inverse before the final recovery of the pencil coefficient line. Consequently an isomorphism of algebras induces a unique projective isomorphism of the selected source spaces, and compositions give compositions. This establishes the source functor without first choosing a congruence representative of the pencil.

To verify the jet presentation, take  $S_A = \mathbb{P}(V)$  and  $L = \mathcal{O}(1)$ . The evaluation map  $H^0(L) \otimes \mathcal{O} \rightarrow J^1 L$  is an isomorphism: a linear form whose value and first derivatives vanish at a point is zero, and both bundles have rank  $n$ . Hence  $\mathcal{E}_A = V \otimes \mathcal{O}(1)$ , and the Cartan quotient in (B.2) is exactly (B.3). Its degree- $p$  twist by  $L^{-2p}$  is constant, with global section space  $B_h(V)_p$ ; this proves the stated projective coefficient construction. Relative principal parts commute with base change on the locally trivial smooth projective-space bundle. Replacing  $L$  by its tensor product with a line from the base cancels in  $\mathcal{H}om(J^1 L, L)$ .

Fix a source lift temporarily. A scalar  $c$  acts on  $B_h(V)_p$  by  $c^{2p}$  and on  $\mathcal{O}_{\mathbb{P}(V)}(2p)$  by  $c^{-2p}$ . Hence every summand in (B.3), together with its multiplication, is naturally  $\mathrm{PGL}(V)$ -equivariant. If two linear lifts of a projective isomorphism differ by a scalar, their induced algebra-bundle maps are equal. This also proves the cocycle identity on three overlaps; in a family, replacing  $V$  by  $V \otimes L$  cancels  $L^{2p}$  against  $L^{-2p}$ . Thus the algebra bundle descends independently of all lifts. For a projective-source bundle, take fppf local frames of its  $\mathrm{PGL}_n$ -torsor; the scalar-trivial action just proved supplies the cocycle descent even when no global vector-bundle lift exists. The same scalar acts trivially on each  $P_{A,p}$ , and power maps commute with scalar changes and congruence. They therefore define a projective diagram on the source torsor, with the same cocycle compatibility. Its first projective space can alternatively be described as the parameter space of quadric hypersurfaces in the dual projective space  $S_A^\vee$ .

Apply Theorem II.5.1 to identify the nondegenerate graph. The rank- $r$  identification is proved independently in Theorem II.D.1. Equivariance makes both identifications independent of the source lift. Finally the residual left coefficient line, by the multiplicity-one exterior-duality map in the local inverse, gives the Pluecker point of a two-dimensional subspace in  $\mathrm{Sym}^2 V$ . Scalar changes do not change its projective line. This is  $\Lambda_A$ . Associated bundles, their specified equations, and their multiplication commute with any base change; no claim about arbitrary restricted graph closures is required.  $\square$

*Remark B.2* (Exact intrinsic status). The envelope is a functor constructed from multiplication, not a claimed subalgebra or preferred quotient of  $A$ . It is an algebra bundle on  $S_A$ , not an untwisted algebra canonically based at  $\mathrm{Spec} \mathbb{C}$ . This distinction records, rather than suppresses, the scalar ambiguity: for  $n \geq 3$  the central element  $\zeta I \in \mathrm{SL}(V)$  acts on  $B_h(V)_1 = \mathrm{Sym}^2 V$  by  $\zeta^2$ , which need not be one although it acts trivially projectively. Thus the usual linear degree-one action does not descend to  $\mathrm{PGL}(V)$ . The twist in (B.3) removes exactly this obstruction. There

is no selection of an  $n$ -plane inside the auxiliary complement of dimension  $n^2 - 1$ . The latter still defines its own, different coefficient-section algebra in Corollary II.3.2.

### APPENDIX C. UNIVERSAL PROPERTY OF THE INTRINSIC ENVELOPE

The intrinsic envelope is a construction from the multiplication table, not an identification of an auxiliary coefficient section with the original algebra. Its universal property can be stated before choosing a linear source or recovering a representative of the pencil.

**Proposition C.1** (The source-normalized nilpotence envelope). *Let  $S_A$  and  $\mathcal{E}_A = \mathcal{H}om(J^1 L, L)$  be as in Theorem B.1. The algebra  $\mathcal{B}_{A,h}$  represents, on  $S_A$ , graded commutative algebras  $C$  equipped with a linear map  $\mathrm{Sym}^2 \mathcal{E}_A \rightarrow C_1$  for which*

$$(\text{image of } u^2)^{h+1} = 0$$

*for every local section  $u$ , also after base change. More precisely, giving such a map is equivalent to giving a unique graded algebra map  $\mathcal{B}_{A,h} \rightarrow C$ . The degree-zero part maps by the structure map. This property, its representing algebra, and its projectivized power maps descend along the oriented source torsor. The case  $h = 1$  requires no additional integer choice and gives the same native graph as every  $h \geq 1$ .*

*Proof.* The symmetric algebra is free on its degree-one module. In characteristic zero the span of  $(u^2)^{h+1}$  is exactly the image of

$$\mathrm{Sym}^{2h+2} \mathcal{E}_A \longrightarrow \mathrm{Sym}^{h+1}(\mathrm{Sym}^2 \mathcal{E}_A), \quad u^{2h+2} \longmapsto (u^2)^{h+1}.$$

Polarization verifies this assertion on every trivializing open; the required denominators are nonzero integers, hence units on every complex base. Factoring a free-algebra map through the ideal generated by this image is equivalent to the displayed identities. This proves existence and uniqueness and their compatibility with base change.

On  $\mathbb{P}(V)$ , evaluation of linear sections identifies  $J^1 \mathcal{O}(1)$  with  $V^* \otimes \mathcal{O}$ . Thus  $\mathcal{E}_A = V \otimes \mathcal{O}(1)$ . A scalar on  $V$  acts oppositely on  $\mathcal{O}(1)$ , so it acts trivially on  $\mathcal{E}_A$ , the Cartan map, and the quotient. This proves descent without choosing a global  $\mathcal{O}(1)$  on a nonsplit source bundle. The scalar action on each untwisted coefficient space is scalar, and hence also disappears on its projectivization. The graph assertions follow from Theorems II.5.1 and II.D.1.  $\square$

This universal property distinguishes an algebra bundle constructed from the intrinsic source from a preferred linear subquotient of  $A$ . The former is proved; the latter is neither needed nor asserted. The source is extracted by the determinant and oriented-ruling steps of the first-relation inverse, before its final pencil-coefficient step.

### APPENDIX D. THE INCIDENCE CENTRE FROM FAILURE MULTIPLICATION

We specify how the preceding modification is recovered from a family of sharp failure algebras. This is a construction of its centre and Rees algebra, not merely a pullback of the existing finite algebra. For a framed source let  $\Phi : G \rightarrow \mathcal{H}$  be the first-relation closed immersion of Theorem 7.3. Here  $\mathcal{H}$  denotes its relation-space Grassmannian. We use  $\Phi(\mathcal{U})$  with its induced scheme structure, not an arbitrary neighbourhood in  $\mathcal{H}$ .

**Theorem D.1** (Intrinsic incidence ideal and normal direction). *On the effective family of sharp pencil-failure algebras corresponding to  $\mathcal{U}$ , its first relation, oriented source, and multiplication diagram determine  $I_{\Sigma_2}$  and the graded Rees algebra*

$$\bigoplus_{m \geq 0} I_{\Sigma_2}^m. \tag{D.1}$$

Their Proj is the modification in Theorem II.H.2. More explicitly, restrict the power ideal  $J_{h(n-2)+1}$  to the intrinsic pencil line and let  $q : \mathcal{D} \rightarrow \mathcal{U}$  be its zero scheme. Then

$$I_{\Sigma_2} = \ker\{\mathcal{O}_{\mathcal{U}} \longrightarrow q_*\mathcal{O}_{\mathcal{D}}\}. \quad (\text{D.2})$$

The construction is independent of  $h$ . It is equivariant for ungraded failure-algebra isomorphisms and descends along an oriented projective source torsor.

An arc through  $R \in \Sigma_2$  whose first normal derivative is nonzero has exceptional limit given by its line in  $N_{\Sigma_2/\mathcal{U},R}$ . Thus its first-order failure-algebra family determines the embedded Hilbert limit. A fixed closed algebra  $A_R$  alone determines the pencil and its full exceptional parameter space, but not a preferred point of that space. For higher-contact arcs the leading nonzero normal ideal quotient replaces the first normal derivative.

*Proof.* The first-relation inverse recovers the oriented projective source and the intrinsic pencil, with its arbitrary-base closed immersion  $\Phi$ . The source-normalized envelope in Theorem B.1 supplies its power diagram. The universal power identity gives

$$J_{h(n-2)+1} = I_{n-2}^{h-1} I_{n-1} = I_{n-1}$$

on  $P \setminus Z_{n-3}$ , since  $I_{n-2}$  is the unit ideal there. For  $h = 1$  the same formula is  $J_{n-1} = I_{n-1}$ . Lemma II.H.1 identifies its zero scheme on the universal line with  $\Sigma_2$  by a closed immersion. Taking the kernel of the unit map gives its exact scheme ideal, establishing (D.2). Multiplication of this ideal produces (D.1), whose Proj was identified with the normalized Hilbert graph in the preceding theorem. Because  $\Phi$  is a closed immersion onto its scheme image, this also gives

$$\text{Bl}_{\Phi(\Sigma_2)} \Phi(\mathcal{U}) \simeq \text{Bl}_{\Sigma_2} \mathcal{U}.$$

This is an identity on the effective image, not an unjustified assertion about blow-ups of the whole ambient Grassmannian.

In the local coordinates of (II.H.2), an arc with  $u(\tau) = \tau b_1 + O(\tau^2)$  and  $v(\tau) = \tau b_2 + O(\tau^2)$ ,  $(b_1, b_2) \neq (0, 0)$ , lifts to the exceptional point  $[b_1 : b_2]$ . Under (II.H.1) this gives exactly the plane  $a_R + \mathbb{C}b$  and hence the embedded tail. The arbitrary-base inverse identifies first-order pencil deformations with their effective first-relation deformations. Terms of degree at least two in an ungraded generator change do not change the sharp relation, so they do not change this normal direction. Higher-contact arcs are handled by dividing  $u, v$  by their common smallest power of  $\tau$  before specializing. This calculation explains why no preferred tail can be assigned to the closed algebra without its normal direction.

Every operation is invariant under the congruence and source-lift changes already treated in the intrinsic-envelope theorem. Its scalar normalization and projectivization remove the linear-lift ambiguity; fppf descent therefore applies to the incidence ideal, its Rees algebra, and the resulting projective family. The universal Cartier factorization property, rather than an assertion of arbitrary blow-up base change, specifies their functorial meaning on test bases.  $\square$

**Example D.2** (A complete fibre with a tangential residual limit). For  $n = 3$ , take the affine two-parameter family of genuine pencils

$$A_{u,v}(s, t) = \begin{pmatrix} s & 0 & 0 \\ 0 & t + us & vs \\ 0 & vs & t - us \end{pmatrix}.$$

Near the pencil  $R_0 = \langle \text{diag}(1, 0, 0), \text{diag}(0, 1, 1) \rangle$ , the incidence centre is  $(u, v) = (0, 0)$ . Indeed in the chart  $s = 1$  its minor ideal is  $(t, u, v)$ , while near  $s = 0$  the rank is two. The exceptional fibre is the full projective line  $[u : v]$ . The arc  $(u, v) = \tau(\alpha, \beta)$  gives the residual tail

$$[x : y] \longmapsto \begin{pmatrix} x + \alpha y & \beta y \\ \beta y & x - \alpha y \end{pmatrix}, \quad \det = x^2 - (\alpha^2 + \beta^2)y^2.$$

All directions  $(\alpha, \beta) \neq (0, 0)$  occur. For  $\alpha^2 + \beta^2 \neq 0$  the two residual determinant zeros are distinct. For  $[\alpha : \beta] = [1 : i]$  or  $[1 : -i]$  they coalesce into a double zero. The tail remains an embedded reduced line and the total curve remains nodal. These two points of the full incidence fibre are not ordinary transverse residual pencils, and so are not covered by the squarefree hypothesis of Theorem II.J.3. Every power system at these points is still the complete system of Proposition II.H.3. The example separates nodality of the Hilbert fibre from transversality of its boundary contact.

## APPENDIX E. EFFECTIVE BOUNDARY GROUPOIDS AND RELATIVE LIFTING OBSTRUCTIONS

This section places the intrinsic envelope and the boundary modification in a single category. It also distinguishes three deformation questions: deforming the effective failure object, lifting a prescribed base deformation to its boundary modification, and deforming a singularity of the universal curve. These questions have different answers.

**E.1. The category and the envelope functor.** Let  $\mathcal{F}_{\text{pen}}^{\text{eff}}$  denote the fppf stack of Theorem L.2. Starting with the sharp pencil-failure families, its arrows are first divided by the tangent-identity nonlinear generator kernel and then by the ineffective right tensor-factor action, and the resulting groupoid is stackified. Its oriented source is specified by coefficient support, rather than by an arbitrary choice between the determinant rulings. The theorem gives, on every complex base,

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \simeq [G/\text{PGL}(V)]. \quad (\text{E.1})$$

The notation does not denote the stack of all Artin algebras of the same length. In particular deformations leaving the effective pencil image are not included in this equivalence.

**Proposition E.1** (The envelope as a morphism of stacks). *Let  $\mathcal{A}_h$  be the stack whose objects over  $T$  are a projective-space bundle  $S \rightarrow T$  of relative dimension  $n - 1$  and a finite locally free graded algebra bundle on  $S$  locally isomorphic to  $\bigoplus_{p=0}^{nh} B_h(V)_p \otimes \mathcal{O}(2p)$ , with its projectivized power diagram. Isomorphisms preserve the grading, algebra structure, and diagram. The construction of Theorem B.1 defines a morphism*

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \longrightarrow \mathcal{A}_h.$$

*It is compatible with arbitrary complex base change and does not require a global line bundle  $\mathcal{O}_S(1)$ . The case  $h = 1$  is distinguished without an additional exponent choice. The envelope is a functorial algebra bundle on the recovered source, not a claimed subalgebra or quotient of the original finite algebra.*

*Proof.* An object on the right side of (E.1) is a  $\text{PGL}(V)$ -torsor with an equivariant map to  $G$ . On a lift of its source to  $\mathbb{P}(V)$ , the first-jet construction is  $\mathcal{H}om(J^1\mathcal{O}(1), \mathcal{O}(1)) = V \otimes \mathcal{O}(1)$ . A scalar  $\lambda$  acts on  $B_h(V)_p$  with weight  $2p$  and on  $\mathcal{O}(2p)$  with weight  $-2p$ . It therefore acts trivially on their product, on its multiplication, and on the polarized Cartan relation maps. The algebra bundle is genuinely  $\text{PGL}(V)$ -equivariant, not only equivariant up to a scalar. Associate it to the torsor and apply fppf descent. This defines both objects and arrows of the asserted morphism and gives its compatibility with pullback. This argument uses local  $\mathcal{O}(1)$  only to exhibit a scalar-trivial representation; it never posits such a global line bundle on a Brauer-nonsplit projective-space bundle. The universal property of Proposition C.1 verifies the same construction independently of the chosen trivializations.  $\square$

**E.2. A modular modification and its defining equations.** Restrict (E.1) to  $U_{\text{red}}$  of (II.B.1). With  $B_{\text{red}}$  and its invariant Fitting ideal as in (II.B.2), put

$$\mathcal{B}_{\text{red}} = [B_{\text{red}}/\text{PGL}(V)], \quad \mathcal{F}_{\text{red}}^{\text{eff}} = [U_{\text{red}}/\text{PGL}(V)]. \quad (\text{E.2})$$

**Theorem E.2** (The effective multiple-incidence boundary). *The map  $\mathcal{B}_{\text{red}} \rightarrow \mathcal{F}_{\text{red}}^{\text{eff}}$  is representable, proper, birational, and carries the flat embedded curve family of Theorem II.B.3. Both stacks are smooth. This is a modular modification over  $U_{\text{red}}$ , not an assertion that either stack is proper over  $\mathbb{C}$  or has finite stabilizers. On an etale neighbourhood with labelled incidence centres  $I_i = (u_i, v_i)$ , the multi-Rees algebra has presentation*

$$\mathcal{R}(I_1, \dots, I_k) = R[U_1, V_1, \dots, U_k, V_k] / (v_i U_i - u_i V_i)_{i=1}^k, \quad (\text{E.3})$$

where  $\deg U_i = \deg V_i$  is the  $i$ th coordinate vector and  $U_i = u_i T_i$ ,  $V_i = v_i T_i$ . There is no torsion kernel left in this presentation. Its diagonal subalgebra is

$$\bigoplus_{d \geq 0} (I_1 \cdots I_k)^d (T_1 \cdots T_k)^d, \quad (\text{E.4})$$

so its diagonal  $\text{Proj}$  is the computed Hilbert modification. All these objects descend without a global labelling of the points.

*Proof.* For a test scheme  $T \rightarrow [U_{\text{red}}/\text{PGL}(V)]$ , take its defining torsor  $P_T$ . The pullback is the associated space  $(P_T \times_{U_{\text{red}}} B_{\text{red}})/\text{PGL}(V)$ . The projective equivariant morphism descends to a projective morphism over  $T$ . This proves representability and properness; birationality holds over the dense nonincident open. Smoothness of the two atlases and of the acting group proves smoothness of the stacks. The curve family descends by the same construction. No quotient by the stabilizer of a pencil has been mistaken for its fixed-frame scheme fibre.

For the algebra calculation begin in a polynomial ring in the independent parameters  $u_i, v_i$  and complementary smooth coordinates. For one pair, the kernel of  $R[U_i, V_i] \rightarrow R[T_i]$  is generated by  $v_i U_i - u_i V_i$ : comparison of successive powers and cancellation using the regular sequence  $(u_i, v_i)$  proves the assertion. For several pairs the variables are disjoint. Tensoring the one-pair presentations over the ground field preserves their injections into the polynomial ring in the  $T_i$ , giving (E.3). The result persists after the etale flat extension supplied by Lemma II.B.1; flat base change preserves both the coefficient ideals and the displayed kernel. This proves the absence of an additional saturation or torsion relation. The multidegree  $(d, \dots, d)$  part is exactly  $I_1^d \cdots I_k^d (T_1 \cdots T_k)^d$ , which proves (E.4). It gives the actual defining equations before taking the ordinary-power diagonal, rather than treating the formation of ordinary powers as a second recovery theorem.

The ideal itself is  $\text{Fitt}_0(q_* \mathcal{O}_D)$ , where  $D$  is the zero scheme of the distinguished power  $J_{n-1}$  on the intrinsic pencil line. Finite base change identifies the pushforward module, and Fitting ideals commute with arbitrary base change. Its formation is invariant under the group action. The local multi-Rees presentations are labelled charts of this global ideal blow-up; permutation of the labels does not affect their diagonal. Descent therefore follows from the intrinsic Fitting construction, without claiming that graph closure or blow-up commutes with every base change. A pulled-back curve family is defined by its flat pullback; its existence does not require such a graph-closure assertion.  $\square$

**E.3. Relative lifting is not absolute obstruction.** For precision a deformation of an effective object at  $R$  over a local Artin complex algebra  $A$  is an object of  $\mathcal{F}_{\text{pen}}^{\text{eff}}(A)$  with an identification of its closed fibre with  $R$ . A source framing identifies it with a point of the completed pencil parameter scheme; changing the framing acts by  $\text{PGL}(V)(A)$ . A boundary deformation also specifies a lift to  $B_{\text{red}}$  of the closed boundary point.

**Proposition E.3** (Exact relative lifting equations). *Fix a boundary point with  $k$  incidences. In a labelled chart of the base blow-ups write  $v_i = u_i w_i$ . Let  $A' \twoheadrightarrow A$  be a square-zero extension with kernel  $J$ . Fix a lift of the base object to  $A'$ , with coordinates  $u'_i, v'_i$ , and an existing lift to the*

boundary over  $A$ , with slopes  $w_i$ . For any lifts  $\tilde{w}_i \in A'$ , define

$$o_i = [v'_i - u'_i \tilde{w}_i] \in J/u'_i J. \quad (\text{E.5})$$

These classes do not depend on the lifts  $\tilde{w}_i$ . The prescribed base deformation lifts in this chart if and only if all  $o_i$  vanish. When nonempty, its set of framed lifts is a torsor under  $\bigoplus_i \ker(u'_i : J \rightarrow J)$ . At a closed boundary point with slopes  $w_i^0$ , a tangent vector  $u_i = \epsilon a_i$ ,  $v_i = \epsilon b_i$  lifts if and only if  $b_i = w_i^0 a_i$ . The kernel and cokernel of the differential of the fixed-frame blow-up map both have dimension  $k$  there.

The absolute effective and boundary stacks are smooth, with tangent complexes  $[\mathfrak{p}\mathfrak{gl}(V) \rightarrow T_{RU_{\text{red}}}]$  and  $[\mathfrak{p}\mathfrak{gl}(V) \rightarrow T_b B_{\text{red}}]$  in degrees  $-1, 0$ . Thus the relative conditions above do not assert an absolute obstruction to deforming an effective failure object. Nor do they identify all deformations of its raw unmarked Artin algebra.

*Proof.* The defining equation over  $A'$  is  $v'_i = u'_i w'_i$ . Its defect for a chosen lift lies in  $J$  because it vanishes over  $A$ . Replacing that lift by  $\tilde{w}_i + \eta_i$ , with  $\eta_i \in J$ , changes the defect by  $-u'_i \eta_i$ . This proves the well-defined obstruction class, its necessity and sufficiency, and the stated torsor of solutions. The independent coordinate charts introduce no further equations. For dual numbers at the closed point, multiplication of  $\epsilon a_i$  by a change  $\epsilon \eta_i$  is zero, giving precisely the displayed tangent condition. In the chart  $(u_i, w_i) \mapsto (u_i, u_i w_i)$  the differential has rank one for each normal parameter pair and kills its slope direction. Complementary coordinates are unchanged, so both kernel and cokernel have dimension  $k$ .

The tangent complexes are those of the smooth quotient atlases in (E.2), and Theorem L.2 identifies the effective failure deformation groupoid with the first of them over every Artin base. To check explicitly the nonlinear kernel used here, let the coefficient ring be  $\mathbb{C}[\epsilon]/(\epsilon^2)$  and replace a generator  $x$  by  $gx + q(x)$ , with  $q$  of degree at least two. In a homogeneous relation of sharp degree  $d$ , every term containing a factor from  $q$  has degree at least  $d + 1$  and vanishes in the sharp truncation. The surviving relation is exactly the linear action  $\text{Sym}^d(g)$ . This argument uses degrees in the generators, not powers of  $\epsilon$ , and is valid for coefficients in any base algebra. It therefore separates the tangent-identity generator kernel from an actual infinitesimal change of the pencil over the base. Passing to the two prescribed rigidifications gives the claimed categorical interpretation of the coordinate calculation.  $\square$

The ideal cutting out the incidence is scheme-theoretically intrinsic to the effective failure family *with its recovered pencil line*. That line is an essential output of the inverse; no bypass of it, and no canonical choice of a slope from a closed algebra, is asserted. The additional content is the complete multi-Rees presentation and relative lifting equations, and, on the nonreduced-contact slices, the singular graph and its computed cotangent module in Proposition II.C.3. These calculations specify the interaction with boundary singularities rather than merely transporting a pre-existing finite algebra by pullback.

**E.4. An accessible completeness input for singular pencils.** The congruence classification used by Theorem II.E.2 is recalled with explicit blocks in [De Teran–Dmytryshyn–Dopico(2018), Theorem 2.1]. In that statement a symmetric minimal-index block with index  $\varepsilon$  has dimension  $2\varepsilon + 1$ ; hence no index in an  $n \times n$  symmetric pencil exceeds  $\lfloor (n - 1)/2 \rfloor$ . The Toeplitz nullities in that finite range recover the multiplicities by the second-difference formula already proved in our theorem. The regular blocks are indexed by their projective spectral points and local exponent partitions. Accordingly “complete data” includes both the points and the partitions, modulo one common projective reparametrization of the pencil line. This is an accessible theorem-level classical completeness input, not a new symmetric Kronecker classification. It is distinct from the local congruence diagonalization proved in the paper.

## APPENDIX F. ALGEBRAIC ENVELOPE FORMS AND INVARIANT RELATIVE DEFORMATIONS

**Proposition F.1** (An algebraic target for the envelope). *The envelope functor of Proposition E.1 factors through an algebraic stack of finite type over  $\mathbb{C}$ : the stack of fppf forms, over the base, of the fixed projective-space bundle with its graded algebra and power diagram. It is the classifying stack  $BH_h$  of a finite-type affine group  $H_h$ . This statement is about forms of the specified bundle; it is not a boundedness theorem for all graded algebra bundles locally trivial only on their total source spaces.*

*Proof.* On the fixed  $\mathbb{P}(V)$  the degree- $p$  summand is  $B_h(V)_p \otimes \mathcal{O}(2p)$ . Its degree-preserving bundle automorphisms are constant matrices in  $\mathrm{GL}(B_h(V)_p)$ , since  $H^0(\mathcal{O}) = \mathbb{C}$ . Inside their finite product, preservation of the unit, multiplication and indicated polarized maps is a finite set of polynomial equations. It defines a closed affine group  $K_h$  of finite type. The proved  $\mathrm{PGL}(V)$  linearization of the entire bundle lifts every projective source automorphism. Consequently its full automorphism group is  $H_h = K_h \rtimes \mathrm{PGL}(V)$ , with the action specified by that linearization. Its classifying stack is algebraic of finite type. Descent identifies its objects with the stated fppf forms. The envelope associated to a projective-source torsor is such a form, so its functor lands here. No global  $\mathcal{O}(1)$  on a Brauer-nonsplit source is used. The exponent-free choice remains  $h = 1$ .  $\square$

**Proposition F.2** (The invariant lifting class). *For the effective boundary map of Theorem E.2, the chart classes in (E.5) are the local representatives of*

$$\mathrm{Ext}^1(b^*L_{B/U}, J).$$

*Here a deformation of the base and a square-zero extension with kernel  $J$  have been fixed. On a chart  $v = uw$ , the relative cotangent complex is  $[\mathcal{O}e \xrightarrow{u} \mathcal{O}dw]$  in degrees  $-1, 0$ . On the opposite chart  $u = vz$ , its frames change by*

$$z = w^{-1}, \quad e' = -ze, \quad dz = -z^2dw. \quad (\text{F.1})$$

*These formulas identify the local obstruction cokernels and solution torsors. For several independent incidences the complexes form a direct sum, permuted by étale monodromy, and descend to the effective quotient stack.*

*Proof.* The one-equation relative presentation has relation  $v - uw$ ; its relative differential is  $-u dw$ . Its dual with coefficients in  $J$  has cokernel  $J/uJ$ , where the equation defect is  $v' - u'\tilde{w}$ . On the overlap the other relation is  $u - vz = -z(v - uw)$  and  $dz = -z^2dw$ . These give (F.1) and commute with the two differentials because  $v = uw$ . Choosing inverse lifts of  $w, z$  makes the new defect  $-z$  times the old one; any other choice changes it by the corresponding differential image. Tangent corrections change by  $\eta_z = -z^2\eta_w$ . Thus the obstruction and torsor are coordinate-independent. The multiple-centre equations have independent variables, so their relative complexes are a direct sum. Changes of labels and source frames act on this complex and its classes, giving descent. The common Lie-algebra directions cancel in the relative cotangent complex of the two quotient atlases. This is a relative lifting obstruction for a prescribed base deformation, not an absolute obstruction on the smooth effective stack.  $\square$

**Corollary F.3** (The ambient contact boundary for effective failure families). *The versal normal-coordinate, jet-tail and non-equidimensional-fibre statements transfer to the effectively rigidified sharp pencil-failure image via Theorem L.2. In particular, a family with contact exponents  $(2, 2)$  has both the primitive thick line-tail locus and reduced conic-tail limits described in Theorem II.A.1. The closed algebra does not choose one of these limits.*

*Proof.* The equivalence on arbitrary complex bases identifies the framed completed pencil atlas with the effective first-relation image. Pull back the explicit normal-coordinate maps and the

embedded Hilbert families along that equivalence; their source torsors and changes of frame agree. This identifies deformations, not merely closed points, in the stipulated effective category. The multiplication power  $J_{n-1}$  on the intrinsic pencil gives the same exact contact ideal  $(f, g, r)$  and its finite presentation in (II.6.5). The distinguished exponent is one; higher powers give the same graph with additional coefficient systems. The two types of limits have already been realized by genuine pencil arcs, so their effective failure families realize them as well. This argument explicitly uses the sharp inverse. It does not claim a new operation internal to the closed finite algebra before source reconstruction, or an equivalence with the deformation stack of all raw Artin algebras of its length.  $\square$

#### APPENDIX G. TANGENT MAPS, LINEARIZATIONS, AND EFFECTIVE TRANSFER

We separate the source of the contact tangent map from the category in which the boundary is embedded. The elementary linearizations below also specify the splitting used in the envelope automorphism group.

**Lemma G.1** (Framed tangent map and stabilizer). *Let  $A(x)$  be a regular symmetric pencil, expressed as a symmetric map from  $E$  to  $E^*$  with its parameter-line twist trivialized locally. After splitting a unit block, let  $j(x) : W[[x]] \rightarrow E[[x]]$  be the inclusion of its orthogonal residual summand, and let  $D(x)$  be the residual diagonal form. On the atlas of ordered pencil coefficients the contact differential is*

$$(\dot{A}_0, \dot{A}_1) \mapsto [j(x)^*(\dot{A}_0 + x\dot{A}_1)j(x)] \quad \text{in} \quad \frac{\text{Sym}^2 W^*[[x]]}{\{H^t D + DH : H \in \text{End}(W)[[x]]\}}. \quad (\text{G.1})$$

*The restriction to distinct contacts gives the direct sum of these maps. Local frame changes act by congruence and by the corresponding line-bundle unit; they identify these modules and differentials. This formula does not discard the pencil-basis directions.*

*If a nonsingular member is fixed and  $T$  is its self-adjoint pencil operator, the Lie algebra of its orthogonal stabilizer is*

$$\{K : KT = TK, K^* = -K\}.$$

*For Jordan block sizes  $d_{\lambda,i}$  its dimension is  $\sum_{\lambda,i < j} \min(d_{\lambda,i}, d_{\lambda,j})$ . Hence on this fixed-member framed slice the contact quotient has dimension*

$$n + \sum_{\lambda, i < j} \min(d_{\lambda,i}, d_{\lambda,j}).$$

*This is not asserted to be the dimension of the unframed pencil quotient by source congruence and parameter reparametrization.*

*Proof.* For a block matrix with invertible upper block  $U$  and off-diagonal block  $B$ , the inclusion is  $j = (-U^{-1}B, 1)^t$ . Differentiating the Schur complement gives  $j^* \dot{A} j$ : the terms differentiating  $j$  cancel because its image is orthogonal to the unit block. Infinitesimal residual frame changes contribute exactly  $H^t D + DH$ , proving (G.1). The polarizing line of the quadratic form has not disappeared. If  $\ell$  is the tautological line at the pencil point, the untrivialized value is in  $\text{Sym}^2 W^* \otimes \ell^*$ , with the same congruence quotient; choosing a frame of  $\ell$  gives the displayed formula. Changing it multiplies both numerator and congruence image by the same unit.

The ordered-basis atlas has tangent space  $\text{Hom}(\mathbb{C}^2, \text{Sym}^2 E^*)$ . At its image pencil  $R$ , the sequence forgetting the basis is

$$0 \longrightarrow \text{End}(R) \longrightarrow \text{Hom}(R, \text{Sym}^2 E^*) \longrightarrow \text{Hom}(R, \text{Sym}^2 E^*/R) \longrightarrow 0.$$

The rightmost term is the Grassmannian tangent space. A splitting specifies a local basis section; a different splitting may change the parameter-fixed contact coordinates. Thus the differential is first formulated on the framed atlas. The comparison theorem retains these choices and descends

by its transition maps, instead of silently treating the four basis directions as zero contact vectors. Source-frame directions contribute the infinitesimal congruence image. The block computation in Theorem II.6.2 proves surjectivity on this atlas, so it proves formal smoothness of its completed contact map.

For the stabilizer, decompose the self-adjoint operator into orthogonal primary cyclic blocks. Intertwiners between distinct eigenvalues vanish. Between blocks of the same eigenvalue their space has dimension  $\min(d_i, d_j)$ , as homomorphisms of  $\mathbb{C}[t]$ -modules  $\mathbb{C}[t]/(t^{d_i})$  and  $\mathbb{C}[t]/(t^{d_j})$ . An off-diagonal intertwiner  $K_{ij}$  forces  $K_{ji} = -K_{ij}^*$  and supplies one independent copy. On a single cyclic block a commuting endomorphism is a polynomial in  $T$ , hence is self-adjoint. It is skew-adjoint only when it is zero. This proves the dimension formula without assuming semisimplicity. Subtracting the orbit dimension  $n(n-1)/2 - \dim \text{Stab}$  from the dimension  $n(n+1)/2$  of self-adjoint operators gives the last formula.  $\square$

**Proposition G.2** (The specified envelope has a split automorphism extension). *Fix  $V$ ,  $h \geq 1$ , and the graded algebra bundle on  $\mathbb{P}(V)$  with pieces  $B_h(V)_p \otimes \mathcal{O}(2p)$ , including its multiplication and power diagram. Its automorphism functor, allowing automorphisms of  $\mathbb{P}(V)$ , is represented by a finite-type affine complex group  $H_h$ . There is a split exact sequence*

$$1 \longrightarrow K_h \longrightarrow H_h \longrightarrow \text{PGL}(V) \longrightarrow 1.$$

*The splitting is the one induced by the explicit source-normalized linearization, not a general assertion about automorphism extensions. The fppf stack of forms of this fixed object is the algebraic classifying stack  $BH_h$ . This statement permits twisted projective source bundles but does not classify all graded algebra bundles of those ranks, or all raw failure algebras.*

*Proof.* Start with the natural  $\text{GL}(V)$  action on the apolar quotient and on the tautological bundle. A scalar  $t$  acts on  $B_h(V)_p$  by  $t^{2p}$  and on  $\mathcal{O}(2p)$  by  $t^{-2p}$ . It therefore acts trivially on their tensor product and on its multiplication. The action factors through  $\text{PGL}(V)$ , giving a group homomorphism into automorphisms of the specified object whose projection to source automorphisms is the identity. This is the required splitting and defines the conjugation action on  $K_h$ . Without this chosen linearization there would only be an extension, and a semidirect-product notation would require justification.

An automorphism over the identity of  $\mathbb{P}(V)$  is, in each degree, a constant invertible matrix, since  $H^0(\mathbb{P}(V), \mathcal{E}nd(B_h(V)_p \otimes \mathcal{O}(2p))) = \text{End}(B_h(V)_p)$ . There are finitely many nonzero degrees. Preservation of the unit and multiplication is a finite collection of polynomial equations in these matrices and their inverses. The resulting closed subgroup  $K_h$  is affine of finite type, and  $H_h = K_h \rtimes \text{PGL}(V)$  has the same properties. An isomorphism torsor for fppf-local forms of this one object is an  $H_h$ -torsor, and effective descent gives the inverse construction. Thus the category of forms is precisely  $BH_h$ . Affine finite-type groups over  $\mathbb{C}$  are smooth, and the usual classifying-stack atlas establishes the asserted algebraicity.  $\square$

**Corollary G.3** (Complete boundary fibres on the effective failure image). *On the effectively rigidified sharp pencil-failure image, families satisfying (II.10.1) have the complete boundary fibres of Theorem II.10.1, including their nilpotent structures, component dimensions and intersections. The exponent  $h = 1$  defines the envelope without an additional integer choice; other  $h \geq 1$  have the same graph. The assertion is a functorial consequence of the sharp inverse on effective families. It is not an internal boundary construction on a closed algebra before reconstruction and is not a statement about the full raw Artin-algebra deformation stack.*

*Proof.* Use the family inverse to recover the oriented source and pencil, then the source-normalized envelope to obtain its multiplication ideals. On the corank-two rank open the relevant graph is determined by  $I_{n-1}$ ; the universal power identity gives the same graph for every  $h$ . The completed comparison and the full-fibre theorem identify its embedded family. Their transition

maps commute with the effective equivalence and the source linearization, so all the scheme structures descend, including the local ideal  $(eA, eB, e^2C)$  and its products. This proves the claim in the stated category and no larger one.  $\square$

## APPENDIX H. DETAILED RECONSTRUCTION STATEMENTS AND COMPARISONS

A degeneracy locus usually forgets the map that defines it. We study an inverse problem in which neither the ambient embedding nor the normal directions are retained: can the abstract nonreduced failure scheme recover a quadratic pencil? The answer is affirmative for every complex pencil, including singular pencils. The first relation, rather than its reduced determinant support, carries this information. The inverse is also global: a thickening over a projective reduction recovers an entire varying pencil subbundle and its actual source bundle, up to one constant left transformation. No fibrewise choice of coordinates or projective line-bundle ambiguity is left. A single unmarked local algebra suffices for the constant-pencil problem; a Schubert line gives another geometric realization.

Let  $V$  be a complex vector space of dimension  $n \geq 3$ , and put

$$W = \operatorname{Sym}^2 V, \quad N = \binom{n+1}{2}, \quad R \in \operatorname{Gr}(2, W), \quad S_R = W/R, \quad p = N - 2.$$

The cube-zero algebra  $A_R = \mathbb{C} \oplus V \oplus S_R$  has multiplication  $\operatorname{Sym}^2 V \rightarrow S_R$ , with  $(V \oplus S_R)S_R = 0$ . On  $X_R = \operatorname{Gr}(n, V \oplus S_R)$ , let  $\mathcal{K}$  denote the tautological bundle. For any  $m \geq 2$ , set

$$\widehat{D}_R = V(\operatorname{Fitt}_0 \operatorname{coker}[\operatorname{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow A_R \otimes \mathcal{O}]). \quad (\text{H.1})$$

This is the maximal-minor scheme of multiplication, not merely its support. It is independent of  $m \geq 2$ . Its reduction is the Schubert divisor where  $\mathcal{K} \rightarrow V$  drops rank. Write  $B_R = \operatorname{Gr}(n, S_R)$ , let  $\mathfrak{b}_R$  be its ideal in  $\widehat{D}_R$ , and define

$$Z_R^{[k]} = V(\mathfrak{b}_R^{k+1}) \subset \widehat{D}_R, \quad d = n + 2p = n^2 + 2n - 4.$$

An isomorphism of these neighbourhoods means an abstract complex scheme isomorphism; it is supplied with no ambient or tensor marking.

**Theorem H.1** (Sharp intrinsic reconstruction). *For all quadratic pencils  $R, R' \subset \operatorname{Sym}^2 V$ ,*

$$Z_R^{[d]} \simeq Z_{R'}^{[d]} \iff R' = gR \text{ for some } g \in \operatorname{PGL}(V).$$

*For every  $0 \leq k < d$ , the isomorphism class of  $Z_R^{[k]}$  is independent of  $R$ . Thus  $d$  is the smallest uniform reconstruction order. The same statements hold after restriction, using the graph projection, to any Schubert line in  $B_R$ . The resulting abstract curve-supported neighbourhood is independent of the choice of that line.*

The Grassmannian statement is Theorem 7.1. The curve-supported statement is Corollary J.2. Its more general form, Theorem J.1, applies over any smooth connected projective curve with a rank- $n$  bundle having nontrivial projectivization. For example,  $\mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$  works on every such pointed curve. The curve is intrinsic as the reduction; a chosen embedding of the curve is not part of the isomorphism data.

**H.1. A moving family, not only its individual fibres.** The same finite object also reconstructs a pencil that varies over its projective reduction. Let  $B$  be a smooth connected projective complex variety, let  $\mathcal{U}$  have rank  $n$ , and let  $\mathcal{R}$  be a rank-two subbundle of  $\operatorname{Sym}^2 V \otimes \mathcal{O}_B$  with locally free quotient. Use the local ideal  $(\det T)I_p(\gamma \operatorname{Sym}^2 T)$  in  $\operatorname{Hom}(\mathcal{U}, V)$  and truncate at order  $d$ . Theorem 8.2 states that two resulting abstract finite schemes are isomorphic exactly when their data are related by an isomorphism of reduced bases, one constant linear left map, and an actual

isomorphism of the source bundles, carrying one pencil subbundle to the other. The projection and both tensor factors are forgotten in the input. The varying coefficient line is recovered as a subbundle, not by pointwise choices of projective coordinates.

This strengthens the constant-pencil curve theorem in two directions: the reduced base and source bundle may vary in the comparison, and the pencil need not be constant on that base. Example K.6 has a nonconstant unordered cross-ratio and six spectral collision points, all retained by the same finite scheme. Theorem K.4 then identifies the remaining freedom in an isomorphism. Its quotient is the group of the recovered linear data; its kernel acts trivially on the associated graded scheme and has a terminating commutator filtration. Explicit nonlinear shears show why that kernel cannot be omitted. These statements are consequences of the unmarked inverse, not additional claims of novelty for the classical Pluecker map or for spectral cross-ratios.

**H.2. A single unmarked local algebra suffices.** The strongest form of the inverse needs no positive-dimensional support. Choose an  $n$ -dimensional source space  $U$ , write  $T = (t_{ij})$  for its universal map to  $V$ , and let  $\mathfrak{m} = (t_{ij})$ . Define

$$\mathfrak{A}_R^{[d]} = \mathbb{C}[t_{ij}] / ((\det T)I_p(\gamma_R \operatorname{Sym}^2 T) + \mathfrak{m}^{d+1}).$$

**Theorem H.2** (Local algebraic inverse). *For every pair of complex quadratic pencils, an ungraded algebra isomorphism  $\mathfrak{A}_R^{[d]} \simeq \mathfrak{A}_{R'}^{[d]}$  exists if and only if  $R' = gR$  for some  $g \in \operatorname{PGL}(V)$ . The order  $d = n^2 + 2n - 4$  is the smallest uniform order already for these single-point objects.*

The full statement and proof are Theorem 8.3. Proposition 2.1 separates the general truncation argument from the pencil-specific orbit rigidity: nonlinear coordinate changes form an explicitly described unipotent kernel, whereas the first relation space carries the reconstruction data. The extra local ingredient is that the residual coefficient module has factor-support ranks  $(1, \binom{N}{2})$ . Transposition reverses these unequal ranks and therefore cannot carry one pencil ideal to another. This orients the unordered matrix factors without a nontrivial projective bundle. Theorem 8.2 consequently removes that restriction from the moving-family theorem as well. This is not a claim that the reduced determinant cone alone remembers the pencil: the nonreduced first relation kernel and its residual coefficients are indispensable. Proposition 8.4 also identifies the entire tangent-identity kernel, including its affine-space parametrization and nonadditive group law.

**H.3. Conductors, collision singularities, and actual failure orbits.** Normalization alone does not describe the gluing of its branches. Theorems II.Y.1 and II.Y.3 determine two successive local models. On the coprime two-divisor stratum the completed image is

$$\mathbb{C}[[z, x_1, \dots, x_c, y_1, \dots, y_c]] / (x_i y_j).$$

Its conductor is  $(x, y)$ , its intrinsic singular ideal is  $(x, y)^c$ , and its depth is  $\dim(z) + 1$ . When two common binary roots collide, the model is instead

$$\mathbb{C}[[u, \Delta]] \oplus (u_1, \dots, u_m)t \subset \mathbb{C}[[u, t]], \quad \Delta = t^2,$$

up to a specified smooth factor. Its conductor has a ramified double cover in the normalization. These formulas determine the Cohen–Macaulay cases and distinguish the singular scheme from its reduced support. Proposition II.Z.1 organizes the reduced intersections of different divisor-degree images by the common part of two chosen divisors.

The boundary also has a direct role in the pencil inverse. Theorem II.AA.1 constructs a common point in every intrinsic first-relation orbit closure of a pencil failure algebra. A scalar-matrix contraction of the flag-pencil relation has  $\gcd a^{D-q_n}$ , where  $q_3 = 3$  and  $q_n = 4$  for  $n \geq 4$ . The selected degree- $n(n-1)$  divisor is unique but ramified; its normalization fibre has tangent dimension  $\binom{n^2+3n-5-q_n}{3n-4-q_n} - 1$ . Proposition II.AA.3 gives its entire Artin algebra by

reciprocal-polynomial equations. This specializes *actual* failure algebras through flat families; it does not merely place an unrelated common-factor example beside the inverse theorem. The exact order of the flag coefficient filtration, proved in Lemma II.AA.2, is the link between the pencil representation and the divisor collision.

**H.4. Factorization loci and the geometric comparison.** The normalization mechanism has classical precedents. Chipalkatti’s coincident-root locus parametrizes one binary form with a prescribed root partition. His Proposition 5.1, Theorem 5.4, and Corollary 5.8 distinguish multiple normalization points from failure of tangent injectivity and describe the singular support [19]. Kurmann gives local incidence equations, proves their scheme-theoretic equality in Theorem 1.5, and expresses smoothness through splittings of partitions in Proposition 2.1 [20]. In particular, uniqueness plus tangent injectivity is not a new general principle of the present paper.

Here the datum is an  $r$ -plane of forms in  $e$  variables, not one binary form; the selected divisor has prescribed degree, not a root partition. The residual datum is a Grassmannian subspace, and for  $e > 2$  divisor degrees need not fill an interval. These changes produce high-codimension conductor strata and non-Cohen–Macaulay branch gluings. Theorems II.Y.1 and II.Y.3 compute the image rings, not merely the incidence graph or singular support. Ferrand’s conductor-square formalism [21] is the standard algebraic framework; the particular conductor ideals, singular Fitting ideals, and depths are calculated here for these Grassmannian strata.

There is a second, closer binary comparison. Hu–Lin–Shao [23, §2.2, Corollary 3.6, and Theorem 1.1] study tuples of binary forms, define common-factor strata using resultant homomorphisms, and resolve the map-space boundary by successive blow-ups. On the open set of linearly independent  $r$ -tuples, passing to their span is the frame quotient onto  $\mathrm{Gr}(r, S_D)$ ; the common-divisor condition is invariant under this quotient. Thus our binary loci genuinely meet that theory rather than forming an unrelated example. We do not identify its resultant scheme structure with our image structure without a scheme comparison, nor claim a new general resolution. Our multivariate local rings and the pencil-orbit boundary are the specific additional conclusions.

Rigidification is used as infrastructure, not as an independent headline innovation. The appropriate primary general result is Abramovich–Olsson–Vistoli, Appendix A, Theorem A.1 [22], for compatible flat finitely presented normal inertia subgroups, without a centrality assumption. Our argument identifies the particular substitution subgroup and twisted right-factor subgroup and retains their actual group laws. The boundary statement is on the maximal-lower-Hilbert-function first-relation stratum throughout.

The comparison with regular-pencil Segre strata is made in Remark II.AA.5. The common closed flag orbit is not presented as a new classification of congruence orbits. The new pencil-side conclusion is its explicit ramified boundary under intrinsic cotangent changes, together with the complete normalization-fibre algebra. The separate documentary status of Ballico’s 1993 paper, discussed below, remains unchanged: absent its full theorem text, no nonanticipation conclusion is drawn.

**H.5. The intrinsic mechanism.** In the graph neighbourhood of  $B_R$ , with tautological bundle  $\mathcal{U}$  and graph map  $T : \mathcal{U} \rightarrow V$ , the ideal is

$$(\det T)I_p(\gamma_R \mathrm{Sym}^2 T).$$

Its generators have degree  $d$ . This degree calculation alone only explains why lower orders are constant. The inverse requires three further intrinsic operations. Multiplication in the finite scheme recovers the first relation kernel and hence the whole homogeneous cone. The rank-one Fano scheme of its reduced determinant cone then recovers the two tensor rulings. On the socle Grassmannian one may distinguish them by projective triviality, already on a Schubert line. In

general the unequal coefficient-support ranks orient them without that restriction. Properness forces the induced map to  $\mathrm{PGL}(V)$  to be constant. After choosing its linear lift, the actual normal-bundle map lies in the commutant of  $\mathrm{End}(V)$ , which recovers the right bundle without a line twist (Lemma 4.2). Finally, cancellation of the intrinsic determinant ideal recovers the coefficient module. For pencils its exterior representation is irreducible, and exterior duality identifies its single coefficient line with the Pluecker line of  $R$ .

Theorem 5.1 separates this geometric mechanism from the particular multiplication algebra. The curve theorem shows why the mechanism is not tied to the dimension of the original parameter Grassmannian. The geometric orientation used here has a local replacement: coefficient-support asymmetry. Theorem 8.3 uses it to reconstruct the pencil from one unmarked Artin local algebra.

Theorem 4.3 gives the broader principle. It reconstructs simultaneously varying subspaces in several Schur modules from the unmarked first-relation thickening, provided one coefficient support is nonzero and proper. The coefficient spaces may have arbitrary prescribed ranks; they need not be lines coming from quadratic pencils. Its proof isolates the actual factor-descent step from classical linear-preserver theory. Example 4.4 shows why the global descent is not equivalent to fibrewise reconstruction: two Zariski locally trivial families have the same Artin fibre and isomorphic nilradical graded vector bundles in every degree, but their abstract total schemes differ. The inverse detects the inequivalent degree-three coefficient maps. This is the mechanism behind the moving-pencil theorem, rather than an assertion of novelty for canonical grading or for the numerical value of the first relation degree.

**H.6. An exact image criterion and a pencil-native global consequence.** The coefficient inverse admits an intrinsic converse. Theorem I.1 starts with an unmarked first relation whose homogeneous radical is a determinant cone. After recovering its unordered rulings, invariance under one full factor group is necessary and sufficient for the residual relation to come from coefficient subspaces in the Cauchy modules. Both orderings satisfy this test precisely when every coefficient support is zero or full. Thus the strict-support condition is exactly the criterion for a unique admissible orientation in the recognized class. Theorem I.2 computes its entire affine stabilizer, including the transpose component and the universal nonlinear automorphism kernel. Corollary I.4 expresses the inverse as an equivalence of geometric orbit sets, with those ambiguities specified rather than suppressed.

A global consequence lies inside the quadratic-pencil problem itself. Fix a plane  $H = \langle x, y \rangle \subset V$ . For a degree- $m$  map  $f = [a : b] : \mathbb{P}^1 \rightarrow \mathbb{P}(H)$ , take the pencil subbundle

$$\mathcal{R}_f = (ax + by)H.$$

Every fibre is equivalent to  $\langle x^2, xy \rangle$ . Nevertheless, Theorem P.1 proves

$$Y_f \simeq Y_{f'} \iff f' = \alpha \circ f \circ \sigma, \quad \alpha, \sigma \in \mathrm{PGL}_2(\mathbb{C}),$$

for their unmarked order- $d$  failure thickenings with trivial source bundle. Their fibre algebras and every nilradical graded vector bundle depend only on  $n, m$ , not on  $f$ . Their abstract pencil, quotient and first-relation bundles are fixed as well. Thus the entire covering equivalence problem, not only one chosen pair, is retained in the global multiplication and is invisible to those additive data. The maps  $z^3$  and  $z^3 + z$  give an explicit separating pair of quadratic pencils (Corollary P.2).

Neither complete reducibility of Cauchy modules, linear preservers of a Segre cone, nor division by the common factor of a pencil is new in isolation. Their synthesis here has a precise output: recognition of the image and its exact orientation ambiguity after all matrix markings are forgotten, followed by recovery of a whole covering map from a locally isotrivial pencil thickening. A projective factor map alone would not recover the actual source bundle; the intrinsic normal map and its commutant do so. A pointwise pencil inverse would not recover a covering map; one constant left transformation over the entire proper base does so. These distinctions specify the

geometric content beyond the classical ingredients. The recognized coefficient class is not the class of all determinant-radical relations. Its geometric orbit statement alone does not describe nonreduced deformation groupoids; the relative comparison below uses a further scheme-theoretic inverse.

**H.7. Intrinsic strata, infinitesimal parameters, and pencil moduli.** The relative inverse begins with a geometric condition on the first relation, not invariance under a factor group. For a space  $K$  of homogeneous forms, the divisor-incidence functor defines a canonical scheme-theoretic image in the relation Grassmannian. Its exact-gcd open is the locally closed stratum of Lemma II.M.1, now identified by its incidence universal property in Corollary II.N.5. Multiplication by the universal common-divisor line is an isomorphism onto this open. Consequently the common factor is recovered even over a nonreduced parameter base. The condition is scheme-theoretic: merely asking all geometric fibres to have the specified common factor is insufficient. When that divisor is a power of a determinant, the remaining parameter is an arbitrary gcd-free residual subspace, modulo the full determinant-preserver group (Theorem II.M.3). This is a smooth quotient stack, and neither of the two factor symmetries is required.

For pencils the exact common divisor is  $\delta^{n-1}$ , and the primitive residual relation has degree  $3n - 4$  (Lemma L.1). The Cauchy projection and the Pluecker equations recognize its image as a locally closed scheme, not just a set of complex points. The resulting relative theorem is

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \simeq [\text{Gr}(2, \text{Sym}^2 V) / \text{PGL}(V)]. \quad (\text{H.2})$$

Here the superscript records two explicit operations on morphisms: first remove tangent-identity nonlinear substitutions, then remove the ineffective right matrix factor, and fppf stackify. Theorem L.2 proves this equivalence over arbitrary complex parameter schemes, retaining all singular pencils. It does not identify the unrigidified algebra stack with the pencil stack. Its tangent complex is  $[\mathfrak{pgl}(V) \rightarrow \text{Hom}(R, \text{Sym}^2 V/R)]$ , so the effective stack is smooth of dimension  $n - 3$ , with precisely the pencil stabilizer jumps. Rank-incidence and spectral Fitting strata retain their scheme structures and deformation equations by Corollary L.3. This is an exact-image and rigidification comparison; the displayed rank and spectral consequences are its functorial restrictions, not an independent classification of those loci. The two quotient operations have the universal properties of Theorem M.1 and Corollary M.2.

The ambient determinant-divisor stratum is substantially larger. Theorem O.1 computes its normal space to the pencil stratum in three parts: non-Pluecker lines, non-scalar right-factor directions, and other Cauchy summands. Theorem O.2 constructs through every pencil a projective Grassmannian of relations with the same determinant radical, containing a projective line whose other points admit neither factor symmetry. Thus the family inverse describes how the coefficient and pencil loci sit in, and can be left within, a larger intrinsic stratum. It does not falsely infer deformation stability from fibrewise recognition.

**H.8. The normalization of a natural common-divisor boundary.** There is a different geometric question beyond the pencil image: what happens when a divisor of a first relation is no longer uniquely recoverable in a family? Let  $E$  have dimension  $e \geq 2$ , set  $S_j = \text{Sym}^j E$  and  $s_j = \dim S_j$ , and fix  $g, h \geq 1$ ,  $D = g + h$ , and  $2 \leq r \leq s_h$ . Let  $Y_g \subset \text{Gr}(r, S_D)$  be the scheme-theoretic image of the incidence of a line of degree- $g$  divisors of a rank- $r$  relation bundle. This definition uses the vanishing of the universal quotient map; it does not first recover a pencil, impose a factor symmetry, or choose a reduced structure on a set.

**Theorem H.3** (The divisorial boundary). *The multiplication morphism*

$$\mathbb{P}(S_g) \times \text{Gr}(r, S_h) \longrightarrow Y_g, \quad ([f], J) \longmapsto fJ$$

is finite and is the normalization. The locus  $Y_g$  is integral, of dimension  $s_g - 1 + r(s_h - r)$ . At a geometric relation  $K$  it is normal exactly when  $F_K = \gcd(K)$  has one degree- $g$  divisor  $[f]$  and  $\gcd(f, F_K/f) = 1$ . At the preimage  $([f], K/f)$ , if this last gcd has degree  $k$ , the fibre tangent dimension is  $s_k - 1$ .

The theorem is proved in Section II.N. Lemma II.N.3 supplies equations for the entire normalization fibre, not only its tangent space. For example,  $K = \langle x^2y, x^2z \rangle$  has a unique linear divisor, but the normalization fibre is  $\text{Spec } \mathbb{C}[a, b]/(a, b)^2$ , of length three. A different relation has two reduced divisor lifts; yet another has a gcd-degree jump at a normal point. The exact criterion distinguishes these behaviours. Corollary II.N.8 carries this intrinsic boundary to the first-relation algebra stack. It is not a classification of all finite-flat smoothings or of all linear orbits in that boundary.

A direct consequence on the pencil side is proved separately in Section N. Every quadratic pencil admits an explicit congruence degeneration to  $\langle x_1^2, x_1x_2 \rangle$ ; its orbit is the closed flag variety  $\text{Fl}(1, 2; V)$ , and it is the unique closed orbit in the full pencil Grassmannian. The construction is uniform for regular and singular pencils and lifts to a finite-flat family of their failure algebras with exact common factor  $\delta^{n-1}$ . This proves a particular universal orbit-closure adjacency by equations, rather than by copying a prescribed invariant locus through the effective-stack equivalence. Its elementary highest-weight mechanism is distinct from the normalization theorem, which deals with a larger boundary of arbitrary first relations.

**H.9. Framed families and spectral consequences.** Once tensor coordinates are retained, the parameter map is simpler. The first relation spaces give a closed immersion of  $\text{Gr}(2, W)$  into a Grassmannian, by Theorem 7.3. Its proof factors through the classical Pluecker embedding, tensoring a line with a fixed space, and a fixed linear inclusion. This is the framed encoding consequence of the coefficient calculation, not a substitute for the unmarked inverse. Theorem R.1 glues the actual universal order- $d$  neighbourhood. It is finite locally free over the relative socle of rank  $\binom{n^2+d}{d} - \binom{N}{2}$ , and projective and flat over the pencil parameter scheme. It commutes with arbitrary complex base change with that socle retained. Over a nonreduced base, the specified relative socle ideal need not be the absolute nilradical. These marked statements alone do not identify unmarked moduli stacks or deformation groupoids. The common-divisor descent and ineffective-inertia analysis in Theorem L.2 supply that separate step.

For regular pencils, the spectral sheaf and its Fitting incidence schemes are read from the reconstructed pencil. We retain these consequences because they exhibit information genuinely lost by coarser invariants: fixed discriminants and fixed reduced rank loci can conceal different elementary divisors. In Theorem S.2, every classical dominance specialization is transported to a flat family of the finite neighbourhoods, with every strict transition detected at order  $d$ . The spectral orbit classification itself is classical, not a second independent Torelli theorem.

**H.10. Prior work and the publication object.** The Weierstrass–Segre classification is used in its root-labelled form; compare [4, Theorem 1.1 and Corollary 2.1]. The orthogonal nilpotent input is isolated in Lemma S.1, with precise references to [5, Proposition 1 and Theorem 1], an independent dimension calculation, and an explicit distinction between full orthogonal orbits and their special-orthogonal components.

Ballico’s relative multiplication construction [2, Theorem 0.2; §1, Remark 1.2 and equation (6)] is a substantive antecedent for studying quadratic-normality failure on finite linear sections of an embedded surface. Here the defining algebra varies, and the inverse starts from an unmarked infinitesimal failure scheme. The opening page of the distinct earlier paper [1, p. 5] concerns higher-order embedding properties and restriction maps to finite subschemes. Its remaining theorem and proof text has not been inspected; that first page does not establish the absence of

an inverse statement. No anticipation or nonanticipation conclusion relative to that article is asserted.

The Artin-local aspect is compared at theorem level with [7, 8] in Section 2, including the distinction between canonical grading of a nonhomogeneous local algebra and orbit rigidity of our already homogeneous first relation. The linear-preserver input is classical [9]; its precise hypotheses and a proof in the required case are included in Lemma 4.1. The algebraic-group quotient step is supplied by the homomorphism theorem [11, Theorem 5.39 and Remark 5.42]; its use in the exact stabilizer calculation distinguishes a scheme-theoretic quotient from a bare bijection of complex points.

The present article consists of intrinsic reconstruction, the canonical common-divisor boundary, relative rigidification, explicit pencil specialization, and spectral readout. The reciprocal-likelihood theorem and projective critical schemes are preserved in a separate applications manuscript. Its finite-flat critical divisor requires supplied split semisimple data and simple disjoint projective poles; its totally real cover requires a supplied real definite realization and separated data. None of those hypotheses enters the all-pencil theorem here. Independent web, exterior-operator, K3, polarized, and boundary results are preserved in an explicitly non-submitted research archive. They are not inputs to any proof in this article and are not part of its significance claim. Every proof required here is included here.

#### APPENDIX I. RECOGNITION AND THE EXACT AMBIGUITY OF COEFFICIENT SYSTEMS

The support inequality in the preceding section is not merely a sufficient test in the class under consideration. We characterize that class intrinsically and determine exactly when its orientation is unique. We then compute the full linear stabilizer, including the cases in which an orientation is lost. These are statements about unmarked first relations, not about a supplied matrix presentation.

Throughout this section  $n \geq 2$ ,  $q \geq 1$ , and  $D = n + q$ . Let  $E$  have dimension  $n^2$ , put  $S = \text{Sym } E$ , and let  $0 \neq K \subset S_D$ . The affine cone is in  $E^*$ . Suppose that the radical of the homogeneous ideal  $(K)$  is a reduced hypersurface linearly equivalent to the determinant hypersurface in square  $n$ -by- $n$  matrices. We call this the *determinant-radical condition*. The ideal  $(K)$  is taken in the full symmetric algebra, not in its Artin truncation. Its reduced equation gives a line  $\mathbb{C}\delta \subset S_n$ , and unique factorization gives the intrinsic residual subspace

$$J = K/\delta \subset S_q. \quad (\text{I.1})$$

Changing the generator  $\delta$  does not change this subspace. The reduced singular loci of the recovered cone determine its unordered tensor rulings. An ordering identifies  $E = V^* \otimes U$ , with  $\dim U = \dim V = n$ , up to separate factor changes. The subgroup induced by  $\text{GL}(U)$  on the second factor is independent of those changes. An ordering is called *admissible* when  $J$  is invariant under this full subgroup. Admissibility is therefore an intrinsic condition on an ordered ruling, and does not ask for a chosen basis of either factor.

**Theorem I.1** (Intrinsic recognition and orientation dichotomy). *Under the determinant-radical condition, an ordering is admissible if and only if its Cauchy decomposition has*

$$J = \bigoplus_{\lambda \vdash q, \ell(\lambda) \leq n} L_\lambda \otimes S_\lambda U, \quad L_\lambda \subset S_\lambda V^*. \quad (\text{I.2})$$

*The coefficient spaces are unique and are recovered by projection and contraction. Suppose at least one ordering is admissible and put  $r_\lambda = \dim L_\lambda$  and  $m_\lambda = \dim S_\lambda \mathbb{C}^n$ . Then exactly one of the following alternatives occurs.*

(i) *Some  $0 < r_\lambda < m_\lambda$ . Precisely one of the two orderings is admissible, and every linear isomorphism between such relation spaces preserves that ordering.*

(ii) Every  $r_\lambda$  is either zero or  $m_\lambda$ . Both orderings are admissible. The residual space is a sum of full Cauchy summands, and transposition preserves  $K$  after identifications of the two matrix factors.

Conversely, every nonzero system in (I.2) satisfies the determinant-radical condition after multiplication by  $\delta$ . Thus the theorem characterizes the image of the coefficient construction, without assuming a coefficient presentation as part of the input.

*Proof.* The determinant polynomial is irreducible for  $n \geq 2$ . Every element of  $K$  belongs to the radical  $(\delta)$ , so  $K \subset \delta S_q$  and the cancellation in (I.1) is legitimate. The intrinsic recovery of the unordered factors follows from Lemma 4.1 and the rank-one rulings. Changing either factor frame conjugates its full general linear group into itself; scalar changes act as scalars on  $S_q$ . Consequently the definition of admissibility is independent of all these choices.

For an ordering, the Cauchy formula is

$$S_q = \bigoplus_{\lambda \vdash q, \ell(\lambda) \leq n} \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U.$$

As representations of  $\mathrm{GL}(U)$ , the nonzero  $\mathbb{S}_\lambda U$  are irreducible and pairwise nonisomorphic. Finite-dimensional rational representations of this group over  $\mathbb{C}$  are completely reducible. Hence an invariant subspace splits across its isotypic components. In the  $\lambda$ -component, evaluation identifies that subspace with

$$\mathrm{Hom}_{\mathrm{GL}(U)}(\mathbb{S}_\lambda U, J) \otimes \mathbb{S}_\lambda U.$$

Schur's lemma embeds the displayed multiplicity space uniquely in  $\mathbb{S}_\lambda V^*$ ; this is  $L_\lambda$ . This proves necessity in (I.2), including the absence of unrecorded cross-summand graphs. Sufficiency follows since the second factor is full. Projection onto a Cauchy summand followed by contraction with the dual of its second factor recovers  $L_\lambda$ .

If the reverse ordering is also admissible,  $J$  is invariant under both full factor groups. Their product acts irreducibly on each Cauchy summand, and these product representations are pairwise nonisomorphic. Complete reducibility now makes  $J$  a sum of whole summands. Equivalently, all  $r_\lambda$  are zero or full. Conversely a sum of whole summands is invariant under both groups and under the factor exchange. This proves the dichotomy. One may also see uniqueness directly from the contraction ranks  $(r_\lambda, m_\lambda)$ : a factor exchange reverses them, which is impossible at a nonzero proper support. Every linear isomorphism of relation spaces preserves the determinant cone; Lemma 4.1 leaves just the two factor alternatives. It must therefore preserve the unique admissible ordering in case (i).

Finally choose a nonzero  $L_\lambda$ . At an invertible matrix  $T : U \rightarrow V$ , the map  $\mathbb{S}_\lambda T$  is invertible. A nonzero covector in  $L_\lambda$  consequently has a nonzero matrix coefficient at  $T$ . Thus the residual coefficients do not vanish simultaneously at any invertible matrix. All elements of  $\delta J$  vanish on the determinant hypersurface. Their common zero set is exactly that hypersurface; the Nullstellensatz and its reduced irreducible equation give  $\mathrm{rad}(\delta J) = (\delta)$ , as asserted.  $\square$

The complete reducibility and Cauchy statements here are classical representation theory; see [3]. Their role is to supply an intrinsic image criterion *after* the matrix factors have been recovered. For a relation space that fails the admissibility criterion, the theorem does not assert a one-sided coefficient description. In particular it is not a classification of arbitrary homogeneous ideals or arbitrary degeneracy schemes.

**I.1. Stabilizers and the full transpose ambiguity.** Use the normal action  $T \mapsto gTh^{-1}$ . Its action on polynomial functions is inverse pullback, so it gives an honest representation  $\rho$  on  $E = V^* \otimes U$ . The kernel of  $\rho : \mathrm{GL}(V) \times \mathrm{GL}(U) \rightarrow \mathrm{GL}(E)$  is the diagonal scalar subgroup. For a

recognized system define

$$G_L = \{g \in \mathrm{GL}(V) : (\mathbb{S}_\lambda g^{-1})^* L_\lambda = L_\lambda \text{ for every } \lambda\}, \quad H_K = \mathrm{Stab}_{\mathrm{GL}(E)}(K).$$

These are closed subgroup schemes, with the stabilizer scheme structures from their actions on Grassmannians.

**Theorem I.2** (Exact coefficient stabilizer). *In case (i) of Theorem I.1,*

$$H_K \simeq (G_L \times \mathrm{GL}(U))/\mathbb{G}_m. \quad (\text{I.3})$$

*In case (ii), after choosing bases for the two factors,*

$$H_K \simeq ((\mathrm{GL}_n \times \mathrm{GL}_n)/\mathbb{G}_m) \rtimes C_2, \quad (\text{I.4})$$

where  $C_2$  is generated by transposition. These are isomorphisms of affine algebraic groups over  $\mathbb{C}$ , and hence of their represented functors on commutative complex algebras.

For the ungraded algebra  $A(K) = S/((K) + \mathfrak{m}^{D+1})$ , they give the split exact sequence

$$1 \longrightarrow U_K \longrightarrow \mathrm{Aut}(A(K)) \longrightarrow H_K \longrightarrow 1. \quad (\text{I.5})$$

Here  $U_K$  is the universal higher-order substitution group from Proposition 2.2, of dimension  $n^2(\mathrm{length} A(K) - 1 - n^2)$ .

*Proof.* Every element of  $H_K(\mathbb{C})$  preserves the radical of  $(K)$ , and therefore the determinant cone. In case (i), the unique admissible orientation excludes its transposed component. The remaining map is  $\rho(g, h)$ . Cancellation of the determinant line and projection onto each Cauchy summand show that it preserves  $J$  if and only if  $g \in G_L$ ; the arbitrary right map acts on the full factor and introduces no additional condition. The kernel is the diagonal  $\mathbb{G}_m$ . In case (ii), every separate factor transformation and transposition preserves each active full summand. These exhaust the determinant preservers, so they exhaust  $H_K(\mathbb{C})$ . The transposition has order two and lies outside the preserving component for  $n \geq 2$ , giving the asserted semidirect product. Its action on the left-right quotient is conjugation by the matrix-transpose map; no particular formula for its action on chosen factor lifts is needed.

We justify the scheme statement as well. The homomorphism theorem for algebraic groups identifies the quotient by the kernel with the closed image [11, Theorem 5.39 and Remark 5.42]. Every finite-type group scheme over a characteristic-zero field is smooth [10, Lemma 39.8.2]. Thus the stabilizer and the image are reduced closed subgroup schemes. The equalities on complex points just proved imply equality of their defining radical ideals. This gives (I.3) and (I.4) as algebraic-group isomorphisms, not only as bijections on complex points. Finally Proposition 2.2 applies in degree  $D$  and gives (I.5) with the stated kernel, splitting and dimension.  $\square$

For completeness, the omitted zero system has  $K = 0$ . Then  $A(K)$  is the free truncated symmetric algebra,  $H_K = \mathrm{GL}(E)$ , and its linear automorphisms need not preserve any matrix factors. The determinant-radical condition deliberately excludes this third possibility. A full nonzero coefficient system and a zero system therefore lose information in different ways: the former retains unordered matrix geometry and has exactly a transpose ambiguity, while the latter retains no intrinsic matrix geometry at all.

**I.2. The zero/full case over a projective reduction.** The local transpose ambiguity also has an exact global form. Fix a nonempty set of active partitions and take the corresponding  $\mathcal{L}_\lambda$  to be full, with all other coefficient bundles zero. Use the same construction as in (4.4), without its strict-support hypothesis.

**Proposition I.3** (Global ambiguity for full coefficient systems). *Let  $Y_i$  be two such schemes, for the same  $n, q$  and active partitions, over smooth connected projective bases  $B_i$  with source bundles  $\mathcal{U}_i$ . They are isomorphic as abstract complex schemes if and only if there are an isomorphism*

$\sigma : B_1 \rightarrow B_2$  and an actual bundle isomorphism  $\mathcal{U}_1 \simeq \sigma^* \mathcal{U}_2$ . An isomorphism exchanging the two ruling families over a fixed  $\sigma$  exists precisely when, in addition, the common source bundle is projectively trivial. Equivalently,

$$\mathcal{U}_1 \simeq \mathcal{M}^{\oplus n}, \quad \sigma^* \mathcal{U}_2 \simeq \mathcal{M}^{\oplus n}$$

for a line bundle  $\mathcal{M}$  on  $B_1$ . Thus even the full systems recover the actual source-bundle class; the extra ambiguity is an exchange, not an undetected line twist.

*Proof.* An abstract isomorphism recovers the reduction, the actual normal bundle map, the first relation, and the unordered determinant rulings by the proof of Theorem 4.3, which uses strict support only to orient the rulings. Over the connected base the choice to preserve or exchange the two components is constant. In the preserving case, projectivity makes the left projective map constant. A constant lift and Lemma 4.2 give an actual isomorphism of the source bundles.

In the exchanging case each source projective bundle is identified with the constant left projective bundle on the other side. It is therefore trivial. A vector bundle with a trivialized projectivization is a line twist of a constant vector space: locally lift the given projective trivialization to linear frames; on intersections their differences are scalar matrices, whose scalar cocycle is the required line bundle. After pullback by  $\sigma$ , write  $\mathcal{U}_i^* = \mathcal{M}_i^* \otimes W_i$ , with  $W_i$  constant of dimension  $n$ . The normal bundles become  $\mathcal{M}_i^* \otimes (V_i \otimes W_i)$ . Their actual isomorphism induces a morphism from  $B_1$  to the projective-linear isomorphisms between two constant  $n^2$ -spaces. This target is affine; the morphism is constant. Choose its constant linear lift. On removing it, the actual bundle map is scalar on each projective fibre. In local line frames its off-diagonal entries vanish and all diagonal entries agree. Reducedness makes these equalities hold as bundle maps. Their common entry is a nowhere vanishing section of  $\mathcal{H}om(\mathcal{M}_1^*, \mathcal{M}_2^*)$ , and hence  $\mathcal{M}_1 \simeq \mathcal{M}_2$ . This proves necessity of the common actual line twist, and thus of the source-bundle isomorphism.

Conversely any actual source-bundle isomorphism, together with a constant left identification, preserves all active full Cauchy summands, so it gives an isomorphism of the schemes. If the source bundles have the common line-twist presentation above, choose identifications of the constant factor spaces and transpose their tensor factors, leaving the line factor intact. This is a global actual normal map. It preserves the determinant line, every active full summand, and the truncation, and supplies an exchanging isomorphism. These constructions prove all the claims.  $\square$

**I.3. The intrinsic orbit space.** Fix a rank vector  $\mathbf{r}$  with at least one proper nonzero entry and form the product of Grassmannians

$$X_{\mathbf{r}} = \prod_{\lambda \vdash \mathbf{r}, \ell(\lambda) \leq n} \text{Gr}(r_{\lambda}, \mathbb{S}_{\lambda} V^*).$$

Zero and full factors in this product are points. Let  $\mathcal{Z}_{\mathbf{r}}$  denote the set of subspaces of  $\text{Sym}^D \mathbb{C}^{n^2}$  satisfying the determinant-radical condition and the intrinsic admissibility test, with ranks  $\mathbf{r}$  in the unique admissible orientation.

**Corollary I.4** (Geometric orbit classification). *The coefficient construction induces bijections*

$$X_{\mathbf{r}}(\mathbb{C}) / \text{PGL}(V) \simeq \mathcal{Z}_{\mathbf{r}} / \text{GL}_{n^2}(\mathbb{C}) \simeq \{A(K) : K \in \mathcal{Z}_{\mathbf{r}}\} / \simeq. \quad (\text{I.6})$$

*Their stabilizers, including the ineffective right factor and the nonlinear kernel of the last passage, are precisely those in Theorem I.2. For a nonempty zero/full rank pattern there is just one relation orbit for that pattern, with the stabilizer in (I.4).*

*Proof.* The recognition theorem gives surjectivity of the first map. A linear isomorphism of two such relations is orientation-preserving; its left factor carries every  $L_{\lambda}$  to the corresponding coefficient space, and its right factor is unrestricted. Scalars act trivially on the Grassmannians.

This proves injectivity and the first bijection. The first-relation orbit proposition gives the second. For a zero/full pattern each coefficient Grassmannian is a point, and the full-stabilizer theorem supplies the last claim.  $\square$

Equation (I.6) is a statement about geometric orbit sets with the explicitly computed affine stabilizers. It does not identify nonreduced moduli stacks or assert that every infinitesimal deformation remains in this class. For smooth projective reduced bases, its uniquely oriented case combines with actual factor descent in Theorem 4.3. The following pencil-native application shows that this global step separates an entire covering problem that fibrewise classification cannot see.

## APPENDIX J. RECONSTRUCTION ON A PROJECTIVE CURVE

The global part of the inverse argument has a more economical geometric realization. It uses a proper base to make the projective left frame constant, and a nontrivial right projective bundle to distinguish the rulings. Neither step requires the whole socle Grassmannian. We now replace that Grassmannian by a curve. In particular, a neighbourhood supported on one Schubert line already retains every pencil.

Let  $C$  be a smooth connected projective complex curve, and let  $\mathcal{U}$  be a rank- $n$  bundle on  $C$ , where  $n \geq 3$ . Assume that  $\mathbb{P}_C(\mathcal{U}^*)$  is not projectively trivial. A sufficient numerical condition is

$$\deg \det \mathcal{U} \not\equiv 0 \pmod{n}. \quad (\text{J.1})$$

Indeed a projectively trivial rank- $n$  bundle is a line-bundle twist of a trivial bundle, so its determinant degree is divisible by  $n$ . Fix  $V$  of dimension  $n$ , put  $E = \text{Hom}(\mathcal{U}, V)$ , and let  $\mathfrak{a}$  be the zero-section ideal in  $\text{Sym } E^*$ . For a pencil  $R \subset \text{Sym}^2 V$ , the quotient  $\gamma_R : \text{Sym}^2 V \rightarrow S_R$  defines the homogeneous ideal

$$\mathcal{I}_{R,C} = (\det T) I_p(\gamma_R \text{Sym}^2 T), \quad p = \binom{n+1}{2} - 2, \quad (\text{J.2})$$

where  $T : \mathcal{U} \rightarrow V$  is the universal map on  $\text{Tot } E$ . The expression is independent of local frames: both determinant multiplication and the maximal-minor construction are bundle constructions. Define

$$Y_{R,C}^{[k]} = \text{Spec}_C(\text{Sym } E^* / (\mathcal{I}_{R,C} + \mathfrak{a}^{k+1})).$$

Only its abstract complex scheme is retained, not its projection to  $C$ , its embedding in  $\text{Tot } E$ , or either tensor factor.

**Theorem J.1** (Curve-supported reconstruction). *For fixed  $(C, \mathcal{U}, V)$  as above and  $d = n^2 + 2n - 4$ , every pair of quadratic pencils satisfies*

$$Y_{R,C}^{[d]} \simeq Y_{R',C}^{[d]} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (\text{J.3})$$

*All lower-order neighbourhoods are independent of  $R$ . The order- $d$  schemes are finite locally free over  $C$  of rank  $\binom{n^2+d}{d} - \binom{N}{2}$ , where  $N = \binom{n+1}{2}$ . No regularity condition is imposed on the pencil.*

*Proof.* On any frame chart, the ideal in (J.2) is generated in degree  $d$ . Its degree- $d$  piece  $K_R$  is a subbundle of rank  $\binom{N}{2}$ , by the fixed coefficient inclusion (7.8); its quotient is locally free. The quotient algebra has its full symmetric powers in degrees less than  $d$ , has  $\text{Sym}^d E^* / K_R$  in degree  $d$ , and is zero above that degree. This proves the rank and the assertion about lower orders.

Its reduction is  $C$ , and its nilradical  $\mathfrak{n}$  recovers  $L = \mathfrak{n}/\mathfrak{n}^2$  and

$$K_R = \ker[\text{Sym}^d L \longrightarrow \mathfrak{n}^d].$$

Thus the abstract finite-order scheme reconstructs the entire graded algebra  $\text{Sym } L/(K_R)$ , by the same argument as Lemma 7.2. The ideal here means the homogeneous ideal generated by the recovered kernel, not its radical. No splitting of the reduction or supplied projection is used.

The reduction of this reconstructed cone is the determinant cone. Indeed its ideal has the determinant factor, and when  $T$  is invertible the quadratic quotient has full rank. Its rank-one Fano scheme has two ruling components,  $\mathbb{P}(V) \times C$  and  $\mathbb{P}_C(\mathcal{U}^*)$ . The hypothesis distinguishes the first one intrinsically. An abstract isomorphism of neighbourhoods may induce a nonidentity automorphism of  $C$ ; pull back the second bundle along that automorphism. The oriented ruling map is nevertheless a morphism from  $C$  to  $\text{PGL}_n$ . It is constant, since  $C$  is connected projective and  $H^0(C, \mathcal{O}_C) = \mathbb{C}$ . Lemma 3.2 therefore supplies one constant projective left frame.

Cancel the intrinsic determinant ideal and apply the coefficient construction of Theorem 5.1. For pencils, Lemma 6.2 leaves just the single line  $\mathbb{C}\beta_R$ . Exterior duality (6.8) recovers  $R$  with respect to that same projective frame. This proves the forward implication in (J.3). Conversely a linear lift of  $g$  induces the bundle map on  $E$  and the quotient isomorphism on  $S_R$ , and carries all the displayed ideals and their truncations to the corresponding ideals for  $R'$ . This proves the converse.  $\square$

**Corollary J.2** (One Schubert line suffices). *Let  $\ell \simeq \mathbb{P}^1$  be any Schubert line in  $B_R = \text{Gr}(n, S_R)$ , and use the canonical graph projection  $Z_R^{[d]} \rightarrow B_R$  to form its restriction  $Z_{R,\ell}^{[d]}$ . Its abstract isomorphism class is independent of the choice of  $\ell$ , and*

$$Z_{R,\ell}^{[d]} \simeq Z_{R',\ell'}^{[d]} \iff R' = gR \text{ for some } g \in \text{PGL}(V).$$

*The smallest uniform reconstruction order for these curve-supported neighbourhoods is again  $d$ . Their reductions all have dimension one, instead of  $n(p - n)$ .*

*Proof.* The tautological bundle on a Schubert line is  $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}^{n-1}$ . Pulling back the homogeneous ideal of the graph neighbourhood gives exactly (J.2) with that bundle. Its isomorphism type depends on this bundle and  $R$ , not on the chosen inclusion of the line in the socle. The determinant degree is  $-1$ , so Theorem J.1 applies. Its lower-order schemes are all independent of the pencil, while  $\langle x_1^2, x_1x_2 \rangle$  and  $\langle x_1^2, x_2^2 \rangle$  are inequivalent. This proves the uniform minimality assertion.  $\square$

This restriction is made in the construction; it is not a claim that an abstract neighbourhood canonically chooses a Schubert line. No line is supplied to an isomorphism in the corollary: the reduced curve is recovered as the reduction of its source. Nor is the corollary a statement about one unmarked Artinian normal slice. The projective curve and the nontrivial ruling remain essential to this argument. The same theorem applies, for example, to  $\mathcal{U} = \mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$  on every smooth projective connected curve with a point  $P$ . It isolates the properness and ruling obstruction used in reconstruction, independently of the original Grassmannian realization.

## APPENDIX K. RECONSTRUCTION OF MOVING PENCILS AND THEIR ISOMORPHISMS

The inverse need not be applied separately to constant pencils. A single unmarked finite scheme can retain a pencil varying over its entire projective reduction. We prove this assertion, including the recovery of the normal bundle and the ambiguity in lifting the recovered data to an isomorphism. The coefficient lines in this section are subbundles; constancy of the pencil is not assumed.

**K.1. The family and its first relation bundle.** Fix  $n \geq 3$ , and write

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4, \quad M = \binom{N}{2}.$$

An admissible datum consists of a smooth connected projective complex variety  $B$ , a rank- $n$  vector bundle  $\mathcal{U}$  on  $B$ , a constant  $n$ -dimensional vector space  $V$ , and a rank-two locally direct-summand subbundle

$$\mathcal{R} \subset \mathrm{Sym}^2 V \otimes \mathcal{O}_B.$$

We require that  $\mathbb{P}_B(\mathcal{U}^*)$  is not a trivial projective bundle. Put  $\mathcal{S} = (\mathrm{Sym}^2 V \otimes \mathcal{O}_B)/\mathcal{R}$  and  $\mathcal{E} = \mathrm{Hom}(\mathcal{U}, V \otimes \mathcal{O}_B)$ . On its total space, let  $T$  be the universal map and let  $\gamma$  denote the quotient onto  $\mathcal{S}$ . Define the homogeneous ideal and its truncations by

$$\begin{aligned} \mathcal{I}_{\mathcal{R}} &= (\det T)I_p(\gamma \mathrm{Sym}^2 T), \\ \mathcal{A}_{\mathcal{R}}^{[k]} &= \mathrm{Sym}(\mathcal{E}^*)/(\mathcal{I}_{\mathcal{R}} + \mathfrak{a}^{k+1}), \quad Y_{\mathcal{R}}^{[k]} = \mathrm{Spec}_B \mathcal{A}_{\mathcal{R}}^{[k]}, \end{aligned} \tag{K.1}$$

where  $\mathfrak{a}$  is the positive-degree ideal. Determinant-line changes of frame multiply local generators by units, so this is an ideal sheaf, independent of all frames. This construction is the pullback of the coefficient construction in Theorem R.1, with an arbitrary rank- $n$  source bundle. It is not necessary to choose an embedding of  $B$  in a socle Grassmannian.

**Lemma K.1** (A varying coefficient line). *The ideal  $\mathcal{I}_{\mathcal{R}}$  has no component of degree less than  $d$  and is generated by its degree- $d$  component  $\mathcal{K}_{\mathcal{R}}$ . This component is a subbundle of  $\mathrm{Sym}^d \mathcal{E}^*$  of rank  $M$ . After cancellation of the determinant line, its degree- $2p$  coefficient module determines  $\mathcal{R}$  as a subbundle, once the two tensor factors of  $\mathcal{E}$  are supplied. In particular,  $Y_{\mathcal{R}}^{[d]} \rightarrow B$  is finite locally free of rank*

$$\binom{n^2 + d}{d} - M.$$

*Proof.* Put  $F(H) = \bigwedge^p \mathrm{Sym}^2 H$  for rank- $n$  vector spaces or bundles  $H$ . This is an irreducible polynomial representation by Lemma 6.2; its dimension is  $M$ . Write  $\mathcal{L}_{\mathcal{R}} = \bigwedge^p \mathcal{S}^*$ , viewed, by  $\bigwedge^p \gamma^*$ , as a line subbundle of  $F(V)^* \otimes \mathcal{O}_B$ . The residual minors are precisely the image of

$$\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U}) \longrightarrow \mathrm{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \tag{K.2}$$

The full map  $F(V)^* \otimes F(U) \rightarrow \mathrm{Sym}^{2p}(V^* \otimes U)$  is the nonzero inclusion of its unique Cauchy summand: evaluation at  $T = 1$  is nonzero, as in (6.6). Restricting to a nonzero line on the left therefore has rank  $M$  in every fibre. The coefficients vary algebraically. The constant-rank criterion makes the image a subbundle; equivalently one may choose a nonvanishing maximal minor of this coefficient map locally on the Grassmannian of pencils and pull it back to  $B$ . Multiplication by the determinant, a nonzero polynomial in every fibre, again has constant rank  $M$ .

All defining generators have degree  $d$ , proving the assertion about the ideal. For the recovery assertion, the residual image, inside the indicated Cauchy summand, is  $\mathcal{L}_{\mathcal{R}} \otimes F(\mathcal{U})$ . Contraction with  $F(\mathcal{U})^*$  has image exactly  $\mathcal{L}_{\mathcal{R}}$  as a line subbundle of  $F(V)^* \otimes \mathcal{O}_B$ . This statement is an equality of images of bundle maps, not just equality at closed points. Exterior duality identifies it with the Pluecker line  $\bigwedge^2 \mathcal{R}$ , up to the fixed determinant line of  $\mathrm{Sym}^2 V$ . The Grassmannian Pluecker embedding consequently recovers the morphism  $B \rightarrow \mathrm{Gr}(2, \mathrm{Sym}^2 V)$ , and its pullback of the tautological bundle recovers  $\mathcal{R}$ . Finally, the truncated algebra is the sum of its free graded pieces below  $d$  and the locally free degree- $d$  quotient by  $\mathcal{K}_{\mathcal{R}}$ . Its rank is the asserted binomial sum minus  $M$ .  $\square$

## K.2. The unmarked inverse.

**Theorem K.2** (Reconstruction of a moving pencil). *Let  $(B_i, \mathcal{U}_i, V_i, \mathcal{R}_i)$ ,  $i = 1, 2$ , be admissible data of the same rank  $n$ . Then*

$$Y_{\mathcal{R}_1}^{[d]} \simeq Y_{\mathcal{R}_2}^{[d]} \tag{K.3}$$

as abstract complex schemes if and only if there exist an isomorphism  $\sigma : B_1 \rightarrow B_2$ , a constant linear isomorphism  $g : V_1 \rightarrow V_2$ , and a bundle isomorphism  $h : \mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$  such that

$$\sigma^* \mathcal{R}_2 = (\text{Sym}^2 g) \mathcal{R}_1. \quad (\text{K.4})$$

The reduced base, normal tensor orientation, isomorphism class of the actual source bundle, and the varying pencil are recovered together. Neither the projection onto the base nor the ambient bundle is supplied in (K.3). The constant left map cannot be replaced by a separately chosen change of coordinates in each fibre. A pair  $(g, h)$  defining the same normal-bundle map is unique up to simultaneous multiplication by a nonzero constant.

*Proof.* Let  $\mathfrak{n}_i$  be the nilradical of  $\mathcal{O}_{Y_{\mathcal{R}_i}^{[d]}}$ . Since the positive-degree ideal is nilpotent and  $B_i$  is reduced, the reduction is exactly  $B_i$ , and this ideal is exactly  $\mathfrak{n}_i$ . There are no relations below degree  $d$ , so multiplication intrinsically recovers

$$\mathcal{E}_i^* = \mathfrak{n}_i / \mathfrak{n}_i^2, \quad \mathcal{K}_i = \ker \left[ \text{Sym}^d(\mathfrak{n}_i / \mathfrak{n}_i^2) \longrightarrow \mathfrak{n}_i^d \right]. \quad (\text{K.5})$$

The map is well defined since  $\mathfrak{n}_i^{d+1} = 0$ . An abstract isomorphism gives  $\sigma$ , an isomorphism of these normal bundles, and the matching relation subbundles. In particular it gives an isomorphism of the untruncated homogeneous cones

$$\text{Spec}_{B_i}(\text{Sym } \mathcal{E}_i^* / (\mathcal{K}_i)).$$

This cone is reconstructed from the first relation module; it is not the associated graded algebra of the finite scheme, whose degrees above  $d$  vanish.

The reduction of this cone is the determinant cone. Indeed the residual quadratic map is surjective wherever  $T$  is invertible, whereas the determinant factor annihilates the ideal wherever  $T$  is singular. On a frame chart the determinant hypersurface is reduced over the smooth base, so the radical ideal is precisely the determinant ideal. Successive relative reduced singular loci give the lower matrix-rank cones. The rank-one cone has the two ruling parameter schemes

$$\mathbb{P}(V_i) \times B_i, \quad \mathbb{P}_{B_i}(\mathcal{U}_i^*),$$

by Lemma 3.1. Only the first is projectively trivial. Thus an isomorphism cannot interchange them. The induced map on the trivial ruling is a morphism  $B_1 \rightarrow \text{Isom}(\mathbb{P}(V_1), \mathbb{P}(V_2))$ . After choosing one identification this target is the affine variety  $\text{PGL}_n$ . Since  $B_1$  is connected and projective, every such morphism is constant. Choose one constant linear lift  $g$ .

We check that no line-bundle ambiguity remains in the other factor. Locally trivialize both source bundles. The linear rank-one preserver with its left projective map fixed to  $[g]$  is of the form

$$T \longmapsto gTh^{-1} \quad (\text{K.6})$$

by Lemma 3.2. Once this particular  $g$  is fixed, the local linear map  $h$  is unique: if two right maps give the same value for every  $T$ , they are equal. The local maps therefore agree on overlaps and glue to an actual bundle isomorphism  $\mathcal{U}_1 \rightarrow \sigma^* \mathcal{U}_2$ . Alternatively, after removing  $g$  the normal-bundle map commutes with  $\text{End}(V_1)$ ; the commutant is the right-factor homomorphism bundle. Rescaling the constant lift  $g$  rescales  $h$  by the same constant. This proves both existence and the asserted uniqueness for a given normal-bundle map.

Cancel the intrinsic determinant ideal in the recovered homogeneous ideal. On a frame chart the determinant polynomial is a non-zero-divisor and  $((\det T)\mathcal{J} : (\det T)) = \mathcal{J}$ ; the identities glue without selecting a global determinant generator. Apply Lemma K.1 to the degree- $2p$  part. The right map  $h$  acts on the full factor  $F(\mathcal{U})$ , so contraction against its dual removes that change of frame. The remaining line subbundle transforms by the one constant left map  $g$ . Exterior duality yields exactly (K.4). This uses the subbundle version proved above, not the constant-coefficient hypothesis in Theorem 5.1.

Conversely  $\sigma, g, h$  satisfying (K.4) identify the quotient bundles  $\mathcal{S}_i$ , the quadratic maps, and their maximal-minor ideals. The determinant changes by a unit in local frames. The map (K.6) consequently identifies both the homogeneous ideals and every truncation, proving (K.3).  $\square$

**Corollary K.3** (Uniform sharpness over an admissible base). *For fixed  $B, \mathcal{U}, V$ , all schemes  $Y_{\mathcal{R}}^{[k]}$  with  $k < d$  are independent of  $\mathcal{R}$ . Order  $d$  is the smallest uniform order at which the moving pencil is recoverable in the sense of Theorem K.2.*

*Proof.* Below  $d$  the ideal  $\mathcal{I}_{\mathcal{R}}$  contributes no relations, and the algebra is  $\bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^*$ . Constant pencils are among the admissible subbundles. The pencils  $\langle x_1^2, x_1 x_2 \rangle$  and  $\langle x_1^2, x_2^2 \rangle$  are inequivalent: all members of the first have a common linear factor and those of the second do not. A base automorphism does not change either constant pencil. The theorem separates their order- $d$  schemes, whereas every lower truncation agrees. This proves uniform, not pairwise, minimality.  $\square$

**K.3. The part of an isomorphism that the inverse does not record.** The recovered data classify isomorphism classes, but an isomorphism of finite schemes need not itself be linear in the nilpotent directions. The following statement identifies this distinction without replacing it by a claim about moduli stacks.

Fix an admissible datum, put  $Y = Y_{\mathcal{R}}^{[d]}$ , and let  $\mathcal{L}_{\mathcal{R}}$  be the group of triples  $(\sigma, g, h)$  satisfying (K.4), modulo  $(g, h) \sim (cg, ch)$  for  $c \in \mathbb{C}^*$ . Its group law is composition of their induced normal-bundle maps. Write  $\text{Aut}_{\mathbb{C}}(Y)$  for the group of complex scheme automorphisms. For  $j \geq 1$ , define  $G_j$  to consist of the automorphisms inducing the identity on the reduction and satisfying

$$(\alpha^* - 1)(\mathfrak{n}^k) \subseteq \mathfrak{n}^{k+j} \quad \text{for every } k \geq 0, \quad \mathfrak{n}^0 = \mathcal{O}_Y. \quad (\text{K.7})$$

They act on sheaves on the same underlying topological space.

**Theorem K.4** (Linear data and the nilpotent kernel). *There is an exact sequence of groups*

$$1 \longrightarrow G_1 \longrightarrow \text{Aut}_{\mathbb{C}}(Y) \longrightarrow \mathcal{L}_{\mathcal{R}} \longrightarrow 1. \quad (\text{K.8})$$

*The displayed homogeneous presentation of  $Y$  supplies a splitting. Moreover  $G_{d+1} = 1$ ,  $[G_i, G_j] \subseteq G_{i+j}$ , and there are natural injections*

$$G_j/G_{j+1} \hookrightarrow H^0(B, \text{Der}_{\mathbb{C}}(\text{gr}_{\mathfrak{n}} \mathcal{O}_Y)_j) \quad (\text{K.9})$$

*into additive groups of degree- $j$  derivations. Thus the kernel is nilpotent and records precisely the automorphisms invisible on the associated graded scheme. No surjectivity onto the derivation spaces is asserted.*

*Proof.* An automorphism induces a base map and a degree-one bundle map. The proof of Theorem K.2 identifies these with a unique class of triples in  $\mathcal{L}_{\mathcal{R}}$ . The graded algebra is generated in degree one over degree zero, with relation subbundle  $\mathcal{K}_{\mathcal{R}}$ . Hence its entire graded automorphism is already determined by that triple. The kernel is the identity on every graded piece, which is exactly condition (K.7) for  $j = 1$ . Conversely the homogeneous maps  $T \mapsto gTh^{-1}$  give actual automorphisms of  $Y$ , respect composition, and induce the prescribed triples. This proves exactness and the splitting. A choice of the homogeneous presentation supplies that section; the assertion does not endow an arbitrary unmarked presentation with a preferred section.

For  $\alpha \in G_j$ , put  $D_{\alpha} = \alpha^* - 1$ . It raises filtration by  $j$ , and is therefore nilpotent as an additive operator. The inverse of  $1 + D_{\alpha}$  is the finite geometric series. Composition of filtration-raising operators shows that a commutator of elements of  $G_i, G_j$  raises filtration by at least  $i + j$ . Also  $G_{d+1} = 1$  follows by taking  $k = 0$ , since  $\mathfrak{n}^{d+1} = 0$ .

The leading term of  $D_{\alpha}$  defines maps  $\mathfrak{n}^k/\mathfrak{n}^{k+1} \rightarrow \mathfrak{n}^{k+j}/\mathfrak{n}^{k+j+1}$ . They are independent of lifts. The identity

$$D_{\alpha}(ab) = aD_{\alpha}(b) + D_{\alpha}(a)b + D_{\alpha}(a)D_{\alpha}(b)$$

reduces to the Leibniz identity in the indicated graded degree, since  $2j \geq j + 1$ . Thus the leading term is a sheaf derivation over  $\mathbb{C}$ , including its action on degree zero; it need not be an  $\mathcal{O}_B$ -linear derivation. Leading terms add under composition, and their kernel is exactly  $G_{j+1}$ . They glue globally, proving (K.9) and all assertions.  $\square$

**Example K.5** (A nontrivial invisible automorphism). Let  $B = \mathbb{P}^1$  and  $\mathcal{U} = \mathcal{O}(-1) \oplus \mathcal{O}^{n-1}$ . The bundle  $\mathcal{E}^* = V^* \otimes \mathcal{U}$  has at least two trivial direct summands. Choose their degree-one generators  $x, y$ . The substitution

$$x \mapsto x + y^2$$

fixing  $y$ , the other summands and  $\mathcal{O}_B$ , defines a nonidentity automorphism of  $Y_{\mathcal{R}}^{[d]}$  in  $G_1$ , for every admissible  $\mathcal{R}$ . Indeed each relation has degree  $d$ ; its change under this substitution has degree at least  $d + 1$  and vanishes in the truncation. The inverse is  $x \mapsto x - y^2$ . The automorphism is nonidentity because there are no degree-two relations. Therefore identifying all finite-scheme automorphisms with the stabilizer of the pencil would be false, even in this simple case. The inverse theorem and the exact sequence retain, rather than suppress, this higher-order freedom.

#### K.4. A nonconstant family across spectral collisions.

**Example K.6.** Take  $n = 4$ ,  $B = \mathbb{P}_{[s:t]}^1$ ,  $\mathcal{U} = \mathcal{O}(-1) \oplus \mathcal{O}^3$ , and  $V = \mathbb{C}^4$ . In diagonal quadratic coordinates set

$$A = \text{diag}(1, 1, 1, 1), \quad B_{s,t} = \text{diag}(0, s, t, 2s + 3t).$$

The map  $\mathcal{O} \oplus \mathcal{O}(-1) \rightarrow \text{Sym}^2 V \otimes \mathcal{O}$  with columns  $A, B_{s,t}$  has rank two everywhere. The first diagonal entry shows that any scalar relation involving  $A$  has zero coefficient on  $A$ ; the next two entries then force  $s = t = 0$ , which is impossible on  $\mathbb{P}^1$ . Its image  $\mathcal{R}$  is therefore a locally direct-summand pencil bundle. Since it contains the nondegenerate form  $A$ , it is regular at every point, including all spectral collisions.

On the open set of four distinct eigenvalues, an ordered cross-ratio of  $0, s, t, 2s + 3t$  is

$$\chi = \frac{t(s + 3t)}{(2s + 3t)(t - s)}.$$

The rational function  $J = (\chi^2 - \chi + 1)^3 / (\chi^2(1 - \chi)^2)$  is invariant under the six cross-ratio transformations arising from permutations of four points. It is nonconstant: at  $[s : t] = [1 : 2]$  and  $[1 : 3]$ , the values of  $\chi$  are  $7/4$  and  $15/11$ , and the corresponding values of  $J$  differ. Thus the pencil family is not projectively isotrivial, even allowing a change of parameter within each pencil. The discriminant of  $\det(zA - B_{s,t})$  is

$$4s^2t^2(2s + 3t)^2(s - t)^2(s + 3t)^2(s + t)^2. \quad (\text{K.10})$$

No factor is inverted in forming  $Y_{\mathcal{R}}^{[20]}$ . This finite scheme is locally free over its entire projective reduction of rank  $\binom{36}{20} - 45$ . By Theorem K.2, its unmarked isomorphism class recovers the whole morphism into the pencil Grassmannian, including the six collision points and the pencil there, up to one constant transformation of  $V$  and an automorphism of the base. This is a consequence about a genuinely moving family, not just the pointwise Pluecker encoding. The cross-ratio and discriminant calculations are classical elementary readouts; their role here is to exhibit a family to which the unmarked global inverse applies.

*Remark K.7* (Scope of the family statement). The reduction in this section is smooth, projective and reduced. The source projective bundle is assumed nontrivial, which orients the two rulings. The theorem allows all fibre pencils, singular or regular; regularity is used only in the spectral description of the last example. Theorem R.1 separately allows arbitrary complex base change with its specified relative socle ideal. The present intrinsic inverse does not replace that ideal by an absolute nilradical over a nonreduced base. Nor does (K.8) assert an equivalence of unmarked

moduli stacks or a description of every infinitesimal deformation. It is an exact classification of geometric isomorphisms, their graded data, and the residual automorphism kernel for the stated admissible objects.

#### APPENDIX L. THE EFFECTIVE MODULI STACK OF PENCIL FAILURE ALGEBRAS

We apply the common-divisor inverse to the actual first relations of quadratic pencils. Their common divisor has a higher multiplicity than the single determinant factor used to recover their support. Removing that multiplicity places every pencil, without a regularity hypothesis, in the smooth stratum of the preceding section.

Keep  $W = \text{Sym}^2 V$ ,  $N = \binom{n+1}{2}$ ,  $p = N - 2$ , and put

$$\begin{aligned} M &= \binom{N}{2}, & s &= n - 1, & h &= 3n - 4, \\ \mu &= (\underbrace{3, \dots, 3}_{n-2}, 2, 0), & A &= \mathbb{S}_\mu V^*, & F &= \mathbb{S}_\mu U. \end{aligned} \tag{L.1}$$

Both  $A$  and  $F$  have dimension  $M$ . The Cauchy inclusion  $A \otimes F \subset S_h(E)$  will be fixed; its subspace and all its projections are equivariant, independent of a choice of bases.

**Lemma L.1** (Exact common multiplicity). *For every  $R \in \text{Gr}(2, W)$ , including every singular pencil, the first relation space of its order- $d$  failure algebra is*

$$K_R = \delta^{n-1} \tilde{J}_R, \quad \tilde{J}_R = L_R^0 \otimes F \subset S_h(E), \quad \gcd(\tilde{J}_R) = 1. \tag{L.2}$$

Here  $L_R^0 \subset A$  is the Pluecker line of  $R$ , under the canonical projective identification  $\mathbb{P}(A) = \mathbb{P}(\wedge^2 W)$ . The common divisor of  $K_R$  is therefore exactly  $\delta^{n-1}$ , up to scalar. Its degree remains  $d = sn + h = n^2 + 2n - 4$ .

*Proof.* Exterior duality and  $\wedge^2 \text{Sym}^2 H = \mathbb{S}_{(3,1)} H$  give

$$\bigwedge^{N-2} \text{Sym}^2 H = \mathbb{S}_\lambda H, \quad \lambda = (\underbrace{n+1, \dots, n+1}_{n-2}, n, n-2) = \mu + (n-2)^n.$$

Thus this representation is  $(\det H)^{n-2} \otimes \mathbb{S}_\mu H$ . Applied to the universal matrix, its coefficient polynomials factor as  $\delta^{n-2}$  times the matrix coefficients of  $\mathbb{S}_\mu T$ . The additional determinant in the failure ideal gives (L.2). On the left, exterior duality also yields

$$A \simeq \bigwedge^2 W \otimes (\det V)^{-3},$$

which identifies the residual line with the Pluecker line.

We prove a slightly stronger assertion: the matrix coefficient space  $\beta \otimes \mathbb{S}_\mu U$  has greatest common divisor 1 for every nonzero  $\beta \in \mathbb{S}_\mu V^*$ . Its common divisor is taken to a scalar multiple of itself by the full right group. Its zero divisor is therefore invariant under right multiplication. A nonconstant invariant divisor cannot meet the open orbit of invertible matrices. Since the complement of that orbit is the irreducible determinant hypersurface, any such common divisor is a power of  $\delta$ .

It remains to exclude  $\delta$ . Choose a rank- $n - 1$  map  $T_0$ . The representation  $\mathbb{S}_\mu$  has length  $n - 1$ , so  $\mathbb{S}_\mu T_0 \neq 0$ . Choose a nonzero vector in its image. The span of its left  $\text{GL}(V)$ -orbit is the entire irreducible representation. Consequently some left translate  $gT_0$  has a coefficient not annihilated by  $\beta$ . Not all residual coefficients vanish on the rank- $n - 1$  locus. They have no determinant factor, proving the assertion. The same argument at an invertible matrix shows that the coefficient vector is nonzero everywhere on the invertible locus.  $\square$

Let  $\mathcal{P} = \text{Gr}(2, W)$ . The morphism

$$\iota : \mathcal{P} \longrightarrow X_{h,M}(E), \quad R \longmapsto L_R^0 \otimes F \quad (\text{L.3})$$

is a closed immersion. Indeed, the Pluecker embedding is followed by the closed immersion  $\mathbb{P}(A) \rightarrow \text{Gr}(M, A \otimes F)$ ,  $L \mapsto L \otimes F$ , and then by a linear Grassmannian inclusion. The inverse to the middle map is contraction with  $F^*$ . This factorization proves the assertion on schemes, including nonreduced test schemes, rather than only on points. Lemma L.1 shows that its image lies in the open set  $X_{h,M}(E)$ .

Define the *pencil relation stratum*  $Z_{\text{pen}}$  inside  $Z_{n-1, 3n-4, M}^{\det}$  by the following intrinsic scheme conditions. Recover the common-divisor line, its determinant root, and its unordered tensor rulings by Theorem II.M.3. In an ordering, the primitive residual relation is required to lie in the indicated Cauchy summand, to have the form  $L \otimes F$ , and to satisfy the Pluecker equations on  $L$  under  $A \simeq \bigwedge^2 W \otimes (\det V)^{-3}$ . These are equations on subbundles; a test on the fibres alone is not their definition. Use both orderings and descend. The two ordered images are disjoint: their support ranks are respectively  $(1, M)$  and  $(M, 1)$ , with  $M > 1$ . Each is closed in the residual Grassmannian. Thus this prescription defines a locally closed scheme, independent of the tensor frames. It is an intrinsic image criterion for a natural family of first relations, not a definition by an algebra isomorphism to a preselected pencil.

**Theorem L.2** (Effective all-pencil moduli and deformations). *The stack of first relations in  $Z_{\text{pen}}$  is*

$$[Z_{\text{pen}} / \text{GL}(E)] \simeq [\mathcal{P} / G^\circ]. \quad (\text{L.4})$$

*There is an exact sequence of groups*

$$1 \longrightarrow \text{GL}(U) \longrightarrow G^\circ \longrightarrow \text{PGL}(V) \longrightarrow 1, \quad (\text{L.5})$$

*and its kernel acts trivially on  $\mathcal{P}$ . Quotient the morphisms in the algebra stack first by the nonlinear tangent-identity kernel of Proposition II.M.4, and then by this ineffective right group, and fppf stackify. The resulting effective pencil failure stack satisfies*

$$\mathcal{F}_{\text{pen}}^{\text{eff}} \simeq [\text{Gr}(2, \text{Sym}^2 V) / \text{PGL}(V)]. \quad (\text{L.6})$$

*This equivalence is valid over all complex parameter schemes. It identifies automorphisms after the specified quotients, infinitesimal deformation groupoids, and families of degenerating pencils, including singular pencils.*

*The stack is smooth of dimension  $n - 3$ . At  $R$ , its tangent complex is*

$$[\mathfrak{pgl}(V) \xrightarrow{d_R} \text{Hom}(R, W/R)], \quad d_R(X)(r) = (X \otimes 1 + 1 \otimes X)r \bmod R. \quad (\text{L.7})$$

*In particular, the dimension of its space of first-order isomorphism classes is  $n - 3 + \dim \text{Stab}_{\text{PGL}(V)}(R)$ . Within this stratum, every infinitesimal pencil deformation and its effective failure-algebra deformation lift together across a small extension of local Artinian complex algebras.*

*Proof.* In the determinant frame of Theorem II.M.3, the residual pencil locus is  $\iota(\mathcal{P}) \amalg \tau\iota(\mathcal{P})$ , where  $\tau$  exchanges the two factors. The disjointness just proved and the equivariance of the Cauchy and Pluecker maps give, as schemes with group action,

$$\iota(\mathcal{P}) \amalg \tau\iota(\mathcal{P}) \simeq G \times^{G^\circ} \mathcal{P}.$$

Quotienting by  $G$  gives (L.4). All maps used here are morphisms and all inverse operations are contractions, common-divisor descent, or the inverse on the Pluecker image. Consequently this calculation remains valid on a nonreduced base. It is not an inference from equality of complex-point stabilizers.

The map in (L.5) sends the class of  $(g, h)$  to  $[g]$ . If  $g = aI$ , multiplication by the diagonal scalar  $a^{-1}$  writes its class uniquely as  $(I, a^{-1}h)$ . This proves the asserted kernel. The action on the Pluecker line is the projective action of  $g$ ; the full right factor has no effect on  $R$ . Every  $\mathrm{PGL}(V)$ -valued isomorphism lifts fppf locally to  $G^\circ$ , and two lifts differ by that kernel. The quotient of morphisms and stackification therefore gives  $[\mathcal{P}/\mathrm{PGL}(V)]$ . Combining this with Proposition II.M.4 proves (L.6). No splitting of (L.5) over the base is required.

A Grassmannian chart writes a nearby pencil as the graph of a map  $R \rightarrow W/R$ . Infinitesimal coordinate changes act by  $d_R$ , giving (L.7). Grassmannian charts and  $\mathrm{PGL}(V)$  are smooth, so the quotient is smooth and the graph coefficients lift over every small extension. A coordinate change can also be lifted in the smooth group, and its action determines a compatible lift of the corresponding object. Torsors over a local Artinian complex algebra can be trivialized, since smooth torsors lift a point of their algebraically closed residue field. This proves the lifting assertion for deformation groupoids. Finally,

$$\dim \mathcal{P} - \dim \mathrm{PGL}(V) = 2(N - 2) - (n^2 - 1) = n - 3.$$

Taking kernel and cokernel of  $d_R$  gives the stated first-order dimension, including stabilizer jumps at singular pencils.  $\square$

**Corollary L.3** (Native specialization loci). *Every  $\mathrm{PGL}(V)$ -invariant locally closed subscheme  $Q \subset \mathrm{Gr}(2, \mathrm{Sym}^2 V)$ , with its specified scheme structure, has a corresponding intrinsic effective failure stratum isomorphic to  $[Q/\mathrm{PGL}(V)]$ . Its infinitesimal equations and its specializations are retained, not just its closed points. In particular, this applies to rank-incidence and spectral Fitting conditions on the universal pencil, including their intersections. A family in such a locus specializes to a pencil in its closure exactly when its effective failure family has the corresponding specialization in the associated closure stratum.*

*Proof.* Restrict the equivariant closed immersion (L.3) and its inverse to  $Q$ . The construction of  $Z_{\mathrm{pen}}$  and its two quotient steps then restrict to give the stated isomorphism. Equations for a rank or a Fitting condition are equivariant equations in the universal pencil, so they are included in this argument with their scheme structures. Apply the same construction to the closure of  $Q$  to obtain the specialization assertion.  $\square$

*Remark L.4.* Theorem L.2 does not say that the unrigidified algebra stack is the pencil stack. At a complex pencil its linear stabilizer remains  $(G_R \times \mathrm{GL}(U))/\mathbb{G}_m$ , and its full algebra automorphism group also has the nonlinear kernel in Proposition 2.2. Both are real automorphisms, not redundant notation. They are explicitly removed only in the effective comparison. Nor is the quotient asserted to be proper, although its framed Grassmannian is projective.

The parameter base in this theorem is distinct from the intrinsic projective reduction in Theorem 8.2. Here projective coordinate changes may vary over the parameter scheme. In that moving theorem, properness of the intrinsic reduction forces one constant left map and recovers an actual source bundle. Neither assertion is substituted for the other. The present equivalence upgrades the geometric orbit statement to a family statement without changing the hypotheses or conclusions of the moving inverse.

## APPENDIX M. UNIVERSAL PROPERTIES OF THE TWO RIGIDIFICATIONS

The effective stacks used above have a universal characterization. This specifies what is forgotten from isomorphisms, as opposed to what is retained from objects. We use rigidification in its standard sense of removing a flat normal subgroup of inertia; see [18, Tag 04V2]. The argument below gives the universal property directly in the present situation, including the nonconstant subgroup obtained by descent.

Let  $H_0 \subset \mathrm{Gr}(r, S_D(E))$  be a  $\mathrm{GL}(E)$ -invariant locally closed subscheme. Let  $\mathcal{A}_{H_0}$  be the stack of algebras satisfying the trace, filtration and Hilbert-function conditions of Proposition II.M.4, whose intrinsic kernel is in  $H_0$ . Write  $\mathcal{R}_{H_0} = [H_0 / \mathrm{GL}(E)]$ .

**Theorem M.1** (First-relation rigidification). *Taking the first relation defines an fppf gerbe*

$$q : \mathcal{A}_{H_0} \longrightarrow \mathcal{R}_{H_0}.$$

*Its relative inertia  $\mathcal{U}$  consists exactly of the substitutions which induce the identity on  $\mathcal{N}/\mathcal{N}^2$ . It is a smooth affine normal subgroup of the inertia. In a homogeneous presentation its underlying scheme is  $\mathrm{Hom}(\mathcal{E}, \mathcal{N}^2)$ , with the substitution group law. For every stack  $\mathcal{Z}$  in groupoids on complex schemes, precomposition with  $q$  identifies*

$$\mathrm{Hom}(\mathcal{R}_{H_0}, \mathcal{Z}) \simeq \{F \in \mathrm{Hom}(\mathcal{A}_{H_0}, \mathcal{Z}) : F(\mathcal{U}) = 1\}. \quad (\text{M.1})$$

*The right side is the full subgroupoid, including all natural isomorphisms, of functors killing the indicated relative inertia. The construction is compatible with base change on the relation stack.*

*Proof.* A relation bundle has its homogeneous truncated algebra as a lift. Every algebra over that relation bundle is locally isomorphic to this lift. The first-relation proposition shows that an isomorphism of relation bundles lifts fppf locally, and any two lifts differ by a unique element of the indicated kernel. These assertions are exactly the gerbe and relative-inertia assertions. The kernel is intrinsic, so it is stable under conjugation. Its local affine-space description shows smoothness and flatness; no additive group law is being asserted.

For clarity, we prove the descent implicit in (M.1). Given  $F$  on the right and an object  $y$  of  $\mathcal{R}_{H_0}$ , choose local lifts  $a_i$  over an fppf cover. Locally choose comparison isomorphisms on overlaps above the identity of  $y$ . Two choices differ by  $\mathcal{U}$ , hence have the same image under  $F$ . On triple overlaps the failure of the chosen lifts to satisfy the cocycle condition is again in  $\mathcal{U}$ . Their images therefore give a genuine descent datum for  $F(a_i)$ , which is effective because  $\mathcal{Z}$  is a stack. Refinements and different lift choices give canonically isomorphic descended objects. An arrow between two objects of  $\mathcal{R}_{H_0}$  is treated in the same way, using local lifts of that arrow. This constructs the descended functor. Natural isomorphisms descend by the same argument, proving full faithfulness as well as essential surjectivity. Conversely every functor pulled back by  $q$  kills its relative inertia. All the constructions use covers and isomorphism sheaves, so commute with base change.  $\square$

**Corollary M.2** (The effective pencil stack is a successive rigidification). *On the pencil relation stratum the second morphism*

$$[\mathcal{P}/G^\circ] \longrightarrow [\mathcal{P}/\mathrm{PGL}(V)], \quad \mathcal{P} = \mathrm{Gr}(2, \mathrm{Sym}^2 V),$$

*is the rigidification by the normal relative inertia induced from  $\ker(G^\circ \rightarrow \mathrm{PGL}(V)) = \mathrm{GL}(U)$ . Consequently Theorem L.2 has the universal property of first killing  $\mathcal{U}$  and then this right-factor inertia. It does not kill the stabilizer of the pencil in  $\mathrm{PGL}(V)$ .*

*Proof.* The right factor acts trivially on  $\mathcal{P}$ , and the quotient map of groups is fppf surjective. Objects and isomorphisms of the second quotient therefore lift locally to the first, with differences in that kernel. Apply the descent proof of Theorem M.1. Over an object represented by a  $G^\circ$ -torsor the subgroup is its conjugation-associated  $\mathrm{GL}(U)$ -bundle; it need not be a constant group scheme on the base. This describes the actual subgroup stack and shows that no global splitting is assumed. Composing with the first rigidification gives the asserted successive universal property. All remaining pencil automorphisms survive in the target.  $\square$

## APPENDIX N. A COMMON CLOSED SPECIALIZATION OF ALL QUADRATIC PENCILS

We give an orbit-closure consequence directly on the pencil side. Unlike restricting an equivalence to a previously specified locus, the following argument constructs the specialization. It works for every pencil, including pencils with singular generic member. Its mechanism is elementary projective representation theory; the additional failure algebra statement uses the all-pencil, exact-multiplicity construction.

**Theorem N.1** (Explicit common closed orbit). *Let  $\dim V = n \geq 3$ . The action of  $\mathrm{PGL}(V)$  on  $\mathcal{P} = \mathrm{Gr}(2, \mathrm{Sym}^2 V)$  has a unique closed orbit, namely*

$$\mathcal{C} = \{\ell H : \ell \subset H \subset V, \dim \ell = 1, \dim H = 2\}. \quad (\text{N.1})$$

*It is the flag variety  $\mathrm{Fl}(1, 2; V)$ , of dimension  $2n - 3$ . More precisely, after a fixed change of coordinates every pencil has a degeneration along the one-parameter subgroup*

$$\lambda(t) = \mathrm{diag}(1, t, t^2, \dots, t^2)$$

*to  $\langle x_1^2, x_1 x_2 \rangle$ . The pencil family is a subbundle on the whole affine line.*

*Proof.* Take independent forms  $q_0, q_1$ . The rational function  $q_1/q_0$  on  $\mathbb{P}(V^*)$  is nonconstant. In characteristic zero its differential is not identically zero: on an affine coordinate chart the common kernel of the coordinate derivations of the rational function field is  $\mathbb{C}$ . Choose a point  $[v]$  where  $q_0(v) \neq 0$  and this differential is nonzero. Replace  $q_1$  by  $q_1 - (q_1(v)/q_0(v))q_0$ . Then  $q_1(v) = 0$ , whereas its differential has a nonzero value on some vector  $w$  independent of  $v$ . Euler's identity shows that its value on the radial vector  $v$  is zero. Extending  $v, w$  to a basis of  $V^*$  gives dual coordinates in which

$$[x_1^2]q_0 = a \neq 0, \quad [x_1^2]q_1 = 0, \quad [x_1 x_2]q_1 = b \neq 0.$$

Here brackets denote a coefficient.

Apply  $\lambda(t)$  to the variables. The forms

$$Q_0(t) = q_0(x_1, tx_2, t^2 x_3, \dots, t^2 x_n), \quad Q_1(t) = t^{-1} q_1(x_1, tx_2, t^2 x_3, \dots, t^2 x_n)$$

have polynomial coefficients: the only possible weight-zero term in  $q_1$  was removed. Their limits are  $ax_1^2$  and  $bx_1 x_2$ . For  $t \neq 0$  they span a coordinate transform of  $R$ ; at zero they are independent. Thus their coefficient matrix has rank two everywhere, defines a subbundle, and supplies the asserted morphism from  $\mathbb{A}^1$  to  $\mathcal{P}$ .

The orbit of  $\langle x_1^2, x_1 x_2 \rangle$  is exactly  $\mathcal{C}$ . A pencil of the form  $\ell H$  recovers  $\ell$  as its gcd and  $H$  by division. These are scheme-theoretic inverse operations on the exact-gcd-one-degree stratum by Corollary II.N.5, applied to  $g = h = 1, r = 2$ . Under that inverse the condition  $\ell \subset H$  is the usual closed flag incidence. The flag-to-pencil map is therefore a locally closed immersion; since the flag variety is projective it is a closed immersion. It identifies  $\mathcal{C}$  with the flag variety, and the dimension is  $(n - 1) + (n - 2)$ . Every pencil orbit closure contains  $\mathcal{C}$ , by the constructed family and equivariance. Any closed orbit must consequently be  $\mathcal{C}$ , proving uniqueness.  $\square$

**Corollary N.2** (Flat closed specialization of failure algebras). *For every quadratic pencil  $R$  there is a finite locally free family over  $\mathbb{A}^1$ , of rank  $\binom{n^2+d}{d} - \binom{N}{2}$ , whose nonzero fibres are isomorphic to  $\mathfrak{A}_R^{[d]}$  and whose zero fibre is  $\mathfrak{A}_{\langle x_1^2, x_1 x_2 \rangle}^{[d]}$ . The intrinsic first relations throughout the family have exact gcd  $\delta^{n-1}$ . In a fixed determinant frame, the two ordered copies of the flag orbit form the unique closed  $G$ -orbit in the pencil relation locus. Equivalently the effective pencil failure stack has the common closed specialization described in Theorem N.1.*

*Proof.* Let  $R_t$  be the polynomial subbundle just constructed. Its Pluecker line gives, under the fixed exterior-duality identification, a line bundle  $L_t^0 \subset \mathbb{S}_\mu V^* \otimes \mathcal{O}_{\mathbb{A}^1}$ . Set

$$\mathcal{K}_t = \delta^{n-1}(L_t^0 \otimes \mathbb{S}_\mu U), \quad \mathcal{A}_t = \text{Sym}(E \otimes \mathcal{O}_{\mathbb{A}^1}) / ((\mathcal{K}_t) + \mathfrak{m}^{d+1}).$$

The first is a rank- $M$  subbundle. The second is finite locally free by its graded direct-sum presentation, with the claimed rank. Lemma L.1 gives the exact gcd at every fibre, including zero; in fact this family lies scheme-theoretically in the fixed-divisor exact stratum. Equivariance and the local inverse identify its nonzero fibres as asserted. In a determinant frame the pencil locus is  $G \times^{G^\circ} \mathcal{P}$ . The previous closed-orbit calculation therefore gives the assertion for  $G$ . This is a statement inside that locus; it does not assert closedness under arbitrary degenerations of the determinant form in the full relation Grassmannian.  $\square$

The closed specialization is very coarse compared with the unmarked local inverse: many pairwise nonisomorphic finite algebras have this same limiting fibre. The full first relation, not its common limit, retains the original pencil. The explicit family also differs from transport of spectral rank loci: it crosses from arbitrary pencils to a singular common-factor pencil without a regularity hypothesis. Theorem II.N.4, on the other hand, describes divisor collisions in the larger natural first-relation boundary, outside the exact pencil image. These are two distinct geometric statements, and neither is a classification of every congruence-orbit adjacency of quadratic pencils.

#### APPENDIX O. TRANSVERSE DEFORMATIONS WITH UNCHANGED DETERMINANT SUPPORT

The larger determinant-divisor stratum has deformations which are not pencil deformations and need not satisfy either factor-invariance condition. We now identify their tangent directions and construct complete families in which the homogeneous radical remains exactly the determinant ideal. Thus the intrinsic stratum theorem has content outside the coefficient class, not merely a different description of the same symmetry hypothesis.

Let  $J = L \otimes F$  be the primitive relation of a pencil  $R \in \text{Gr}(2, W)$ . In degree  $h = 3n - 4$ , write the equivariant Cauchy decomposition as

$$S_h(E) = (A \otimes F) \oplus H_{\text{other}},$$

where  $A, F, L$  are as in (L.1). The summand  $H_{\text{other}}$  contains every partition different from  $\mu$ ; this notation denotes its full vector space, not a quotient chosen separately at each pencil.

**Theorem O.1** (Normal directions to the pencil stratum). *The normal space at  $J$  of the pencil locus in  $X_{h,M}(E)$  has the following equivariant direct-sum description:*

$$\begin{aligned} N_J &\simeq \text{Hom}\left(\bigwedge^2 R, \bigwedge^2 (W/R)\right) \\ &\quad \oplus (\text{Hom}(L, A/L) \otimes \mathfrak{sl}(F)) \\ &\quad \oplus \text{Hom}(L \otimes F, H_{\text{other}}). \end{aligned} \tag{O.1}$$

*The summands respectively record departure from the Pluecker locus, departure from a full right multiplicity factor inside the same Cauchy summand, and departure into other Cauchy summands. They are all transverse directions in the effective first-relation stack: the infinitesimal  $G^\circ$ -orbit lies in the pencil tangent space, so no vector in  $N_J$  is removed by that orbit. Every such direction is the first-order term of a local algebraic family in the determinant-divisor stratum. This last assertion refers to the common-divisor stratum; preservation of the entire homogeneous radical for complete families is supplied below.*

*Proof.* The tangent space of a Grassmannian and the Cauchy splitting give

$$\text{Hom}(J, S_h(E)/J) = (\text{Hom}(L, A/L) \otimes \text{End}(F)) \oplus \text{Hom}(L \otimes F, H_{\text{other}}). \tag{O.2}$$

Under  $L \mapsto L \otimes F$ , the tangent to  $\mathbb{P}(A)$  is the first factor tensored with the identity of  $F$ . In characteristic zero the trace splits  $\text{End}(F) = \mathbb{C}I \oplus \mathfrak{sl}(F)$ . It remains to identify the normal space of the Pluecker embedding. The natural map  $\wedge^2 W \rightarrow \wedge^2(W/R)$  has kernel  $R \wedge W$ . At the line  $\wedge^2 R$ , the tangent image is  $\text{Hom}(\wedge^2 R, (R \wedge W)/\wedge^2 R)$ , which is the image of  $\text{Hom}(R, W/R)$ . Its quotient in the projective tangent space is  $\text{Hom}(\wedge^2 R, \wedge^2(W/R))$ . The common determinant twist in  $A$  cancels in this Hom space. Substitution in (O.2) proves (O.1) as a normal-space isomorphism. No splitting of the tangent sequence of the Pluecker embedding is being asserted.

The left group preserves the Pluecker locus, and the right group fixes  $L \otimes F$ ; hence the differential of its orbit is in the pencil tangent. Passing to the quotient tangent complexes in (II.M.4) does not change this normal quotient. Finally  $X_{h,M}(E)$  is open in a Grassmannian. A tangent vector is integrated by a graph with coefficients linear in one parameter, after restricting to a neighbourhood of zero which stays in this open set. Multiplication by  $\delta^{n-1}$  and Proposition II.M.4 give the claimed family of finite flat algebras.  $\square$

**Theorem O.2** (Complete fixed-support families through every pencil). *Put  $e = n^2$ . For every pencil  $R$  there is an  $e$ -dimensional subspace  $B \subset J = \tilde{J}_R$  such that  $\gcd(B) = 1$  and the forms in  $B$  have no common zero on the invertible matrix locus. The projective Grassmannian*

$$\Sigma_B = \{J' \in \text{Gr}(M, S_h(E)) : B \subset J'\} \simeq \text{Gr}(M - e, S_h(E)/B) \quad (\text{O.3})$$

*contains  $J$ , is contained in  $X_{h,M}(E)$ , and satisfies*

$$\sqrt{(\delta^{n-1} J')} = (\delta) \quad \text{for every geometric } J' \in \Sigma_B. \quad (\text{O.4})$$

*Its universal first relation gives a finite flat family of the prescribed Hilbert function. It contains a projective line through  $J$  whose every other point is invariant under neither full factor group. Consequently determinant-radical relations through each pencil admit positive-dimensional complete families outside the recognized coefficient class, and their intrinsic deformation problem is described by Theorem II.M.3.*

*Proof.* By Lemma L.1, the residual coefficient vector is nonzero on every invertible matrix. Evaluation gives a morphism

$$\text{GL}_n / \mathbb{G}_m \longrightarrow \mathbb{P}(J^*).$$

Its projective image closure  $Y$  has dimension at most  $e - 1$ . Since  $M > e$  for  $n \geq 3$ , a general projective linear subspace of codimension  $e$  in  $\mathbb{P}(J^*)$  misses  $Y$ . For such a subspace  $\mathbb{P}(B^\perp)$ , the associated  $e$ -plane  $B \subset J$  has no common zero on the invertible locus. To justify the generic avoidance directly, the incidence of pairs consisting of a point of  $Y$  and an  $e$ -plane annihilated by it has dimension strictly smaller than  $\text{Gr}(e, J)$ . Its projection is closed because  $Y$  is projective.

The condition  $\gcd(B) = 1$  is a second nonempty open condition on  $\text{Gr}(e, J)$ . To see nonemptiness, take any nonzero  $b_1 \in J$ . It has only finitely many irreducible factors. Since  $\gcd(J) = 1$ , a general  $b_2 \in J$  avoids divisibility by each such factor; then  $\gcd(b_1, b_2) = 1$ . Extend their span to an  $e$ -plane. Openness follows from Lemma II.M.1. The two nonempty opens intersect, as  $\text{Gr}(e, J)$  is irreducible.

Every  $J'$  containing this  $B$  has greatest common divisor 1, and its forms cannot vanish simultaneously on an invertible matrix. The common zero set of  $\delta^{n-1} J'$  is thus exactly  $V(\delta)$ . The Nullstellensatz over an algebraically closed field proves (O.4). This is a statement about geometric homogeneous radicals. The scheme-theoretic family used here is the universal subbundle  $\delta^{n-1} \mathcal{J}'$ , not an operation of taking radicals on a nonreduced parameter ring. Multiplication by this fixed nonzero form gives a subbundle; its quotient in degree  $d$  and all the lower symmetric powers are locally free. The truncated algebras are therefore finite flat.

For the final assertion, choose  $j_0 \in J \setminus B$ , a complement  $C$  to  $\mathbb{C}j_0$  in  $J$  containing  $B$ , and a nonzero decomposable vector

$$\eta \in \operatorname{Sym}^h V^* \otimes \operatorname{Sym}^h U \subset H_{\text{other}}.$$

The inequality  $M > e$  allows the choice of  $C$ . The family

$$J_{[a:b]} = C \oplus \mathbb{C}(aj_0 + b\eta), \quad [a:b] \in \mathbb{P}^1,$$

is a projective line in  $\Sigma_B$  through  $J = J_{[1:0]}$ . For  $b \neq 0$  its projection to the  $(h)$ -Cauchy summand is exactly the line  $\mathbb{C}\eta$ . A decomposable line in  $\operatorname{Sym}^h V^* \otimes \operatorname{Sym}^h U$  is invariant under neither full factor group: each of the two irreducible factors has dimension greater than one. Equivariance of the Cauchy projection therefore excludes both possible admissible orientations of  $J_{[a:b]}$ . The determinant-preserver theorem says that these are the only orientations after any change of the recovered matrix coordinates. This proves the assertion intrinsically, not only in the chosen matrix frame.  $\square$

*Remark O.3.* The three normal modes and the complete fixed-support families separate two questions. The effective pencil stratum has exactly the deformation theory of quadratic pencils, including their singular boundary. The larger determinant-divisor stratum contains additional directions, explicitly described here, which cease to be pencil or coefficient relations. Retaining the determinant radical does not suppress these directions. Conversely, no classification of all ideals with a prescribed radical is needed to describe this larger locally closed stratum and its family inverse. These assertions enlarge the reconstruction problem without replacing the all-pencil theorem by a generic statement.

#### APPENDIX P. COVERING MAPS INSIDE ONE FIBRE ORBIT OF QUADRATIC PENCILS

We give a consequence native to the pencil construction. It embeds the equivalence problem for maps  $\mathbb{P}^1 \rightarrow \mathbb{P}^1$  into unmarked finite failure schemes, while keeping the fibre algebra and every nilradical graded vector bundle fixed. Unlike a varying spectral invariant, the phenomenon occurs within a single orbit of singular pencils.

Fix  $n \geq 3$ , a two-dimensional subspace  $H = \langle x, y \rangle \subset V$ , and the trivial source bundle  $\mathcal{U} = \mathcal{O}_{\mathbb{P}^1}^n$ . Let  $f = [a:b] : \mathbb{P}^1 \rightarrow \mathbb{P}(H)$  be a morphism of degree  $m \geq 1$ , where  $a, b$  are homogeneous of degree  $m$  with no common zero. Its tautological line subbundle is

$$\mathcal{L}_f = \mathcal{O}(-m) \xrightarrow{(a,b)} H \otimes \mathcal{O}_{\mathbb{P}^1}.$$

Multiplication in  $\operatorname{Sym} V$  defines a rank-two subbundle

$$\mathcal{R}_f = \mathcal{L}_f H \subset \operatorname{Sym}^2 V \otimes \mathcal{O}_{\mathbb{P}^1}, \quad \mathcal{R}_f|_{[s:t]} = \langle (ax + by)x, (ax + by)y \rangle. \quad (\text{P.1})$$

Let  $Y_f = Y_{\mathcal{R}_f}^{[d]}$  be its order- $d$  failure thickening, with

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n^2 + 2n - 4, \quad M = \binom{N}{2}, \quad e = n^2.$$

All schemes below are compared without projection, grading, embedding, matrix factors, or coefficient markings.

**Theorem P.1** (Unmarked realization of the covering equivalence problem). *For fixed  $n \geq 3$  and  $m \geq 1$ , the schemes  $Y_f$  have the following properties.*

(i) *They are finite locally free over their reduction  $\mathbb{P}^1$ , of rank  $\binom{e+d}{d} - M$ , and are Zariski locally trivial with the same Artin algebra as fibre. Every fibre pencil is equivalent to  $\langle x^2, xy \rangle$ .*

(ii) Writing  $\mathfrak{n}_f$  for their nilradicals, their graded vector bundles are independent of  $f$ :

$$\begin{aligned} \mathfrak{n}_f^j / \mathfrak{n}_f^{j+1} &\simeq \mathcal{O}^{\binom{e+j-1}{j}} \quad (1 \leq j < d), \\ \mathfrak{n}_f^d &\simeq \mathcal{O}^{\binom{e+d-1}{d}-3M} \oplus \mathcal{O}(m)^{\oplus 2M}. \end{aligned} \quad (\text{P.2})$$

Their abstract first-relation bundles are all  $\mathcal{O}(-2m)^{\oplus M}$ , and their pencil and quotient bundles are respectively  $\mathcal{O}(-m)^2$  and  $\mathcal{O}(2m) \oplus \mathcal{O}^{N-3}$ .

(iii) For two degree- $m$  maps,

$$Y_f \simeq Y_{f'} \iff f' = \alpha \circ f \circ \sigma \quad \text{for some } \alpha, \sigma \in \text{PGL}_2(\mathbb{C}). \quad (\text{P.3})$$

Thus the double-orbit set of degree- $m$  morphisms  $\mathbb{P}^1 \rightarrow \mathbb{P}^1$  is realized faithfully by unmarked quadratic-pencil failure schemes with all the additive data in (ii) fixed. This is a statement about isomorphism classes, not an equivalence of their automorphism groupoids.

*Proof. The pencil and local triviality.* Multiplication by a nonzero linear form is injective on  $H$ . Since  $(a, b)$  never vanishes, the map  $\mathcal{O}(-m) \otimes H \rightarrow \text{Sym}^2 H \otimes \mathcal{O}$  has rank two everywhere. Its image is a subbundle with locally free quotient. In the basis  $x^2, xy, y^2$  its two columns are

$$\begin{pmatrix} a & 0 \\ b & a \\ 0 & b \end{pmatrix}, \quad \text{with Pluecker coordinates } (a^2, ab, b^2). \quad (\text{P.4})$$

It follows that  $\mathcal{R}_f \simeq \mathcal{O}(-m)^2$ . The quotient of  $\text{Sym}^2 H \otimes \mathcal{O}$  is a line bundle of degree  $2m$ . A constant complement to  $\text{Sym}^2 H$  in  $\text{Sym}^2 V$  gives  $\mathcal{S}_f \simeq \mathcal{O}(2m) \oplus \mathcal{O}^{N-3}$ . On an open set where  $\mathcal{L}_f$  is trivialized, complete its nonvanishing generator to a frame of  $H \otimes \mathcal{O}$ . The inverse frame map takes  $\mathcal{R}_f$  to  $\langle x^2, xy \rangle$ . Extend it by the identity on a fixed complement of  $H$  in  $V$ . With the source bundle trivial, it identifies the defining minors and the truncated algebra with a product on that open set. This proves local triviality, not merely equality of fibre isomorphism classes. Lemma K.1 gives finite local freeness and the asserted rank.

*The entire graded bundle calculation.* The first relations occur in degree  $d$ , so all lower positive graded pieces are the full constant symmetric powers of  $E = V^* \otimes \mathbb{C}^n$ . Put  $F(V) = \bigwedge^p \text{Sym}^2 V$ , of dimension  $M$ . The coefficient line  $\mathcal{L}_{\mathcal{R}_f} = \bigwedge^p \mathcal{S}_f^*$  is  $\mathcal{O}(-2m)$ . Under exterior duality

$$F(V)^* \simeq \bigwedge^2 (\text{Sym}^2 V) \otimes \det(\text{Sym}^2 V)^*,$$

its inclusion is the Pluecker line of  $\mathcal{R}_f$ . The constant determinant twist does not affect the bundle calculation. Thus it lies in a fixed three-dimensional subspace of  $F(V)^*$ , with column  $(a^2, ab, b^2)$ . There is an exact sequence

$$0 \longrightarrow \mathcal{O}(-2m) \xrightarrow{(a^2, ab, b^2)^t} \mathcal{O}^3 \xrightarrow{\begin{pmatrix} -b & a & 0 \\ 0 & -b & a \end{pmatrix}} \mathcal{O}(m)^2 \longrightarrow 0. \quad (\text{P.5})$$

Indeed the composite is zero; the maximal minors of the second map are  $b^2, -ab, a^2$ , which have no common zero. It is therefore everywhere surjective. Its rank-one kernel is the image of the displayed nowhere vanishing column. Taking a constant complement proves

$$(F(V)^* \otimes \mathcal{O}) / \mathcal{L}_{\mathcal{R}_f} \simeq \mathcal{O}^{M-3} \oplus \mathcal{O}(m)^2. \quad (\text{P.6})$$

The fixed Cauchy inclusion, followed by multiplication by the fixed determinant, embeds  $F(V)^* \otimes F(\mathbb{C}^n)$  as an  $M^2$ -dimensional constant subspace of  $\text{Sym}^d E$ . The first relation bundle is precisely  $\mathcal{L}_{\mathcal{R}_f} \otimes F(\mathbb{C}^n)$  in that subspace. Choose a constant complement to the whole subspace, not to its varying relation subbundle. The top graded quotient is therefore

$$\mathcal{O}^{\binom{e+d-1}{d}-M^2} \oplus \left( (F(V)^* \otimes \mathcal{O}) / \mathcal{L}_{\mathcal{R}_f} \right)^{\oplus M}.$$

Equation (P.6) gives exactly (P.2). It also shows that the first relation bundle is  $\mathcal{O}(-2m)^{\oplus M}$ . All assertions in (ii) concern vector bundles; their multiplication maps have not been identified.

*The global inverse is the covering equivalence.* Suppose  $Y_f \simeq Y_{f'}$ . Their reductions give an automorphism of  $\mathbb{P}^1$ . By Theorem 8.2, or the actual descent theorem with the pencil coefficient line, the varying pencil subbundles are identified by that base map and a single constant  $g \in \mathrm{GL}(V)$ . For a nonzero  $\ell \in H$ , the common polynomial divisor of the two-dimensional space  $\ell H \subset \mathrm{Sym}^2 V$  is precisely  $\ell$ , up to scalar:  $x$  and  $y$  have no common nonconstant divisor in  $\mathrm{Sym} V$ . Dividing the pencil space by this common factor recovers  $H$ . These operations are equivariant under  $\mathrm{GL}(V)$ . Consequently an equality between  $g(\ell H)$  and a pencil of the form  $\ell' H$  forces  $gH = H$  and  $[g\ell] = [\ell']$ . The constant restriction  $g|_H$  therefore supplies a single  $\alpha \in \mathrm{PGL}(H)$  for the whole map. Replacing the recovered base map by its inverse if necessary gives the right-hand side of (P.3).

Conversely choose linear lifts of  $\alpha$  on  $H$  and extend by the identity on a complement in  $V$ . Together with the base map and the identity identification of the trivial source bundles, this gives equivalent pencil data. Their homogeneous normal maps induce an isomorphism of the finite schemes. This proves (iii) and the theorem.  $\square$

**Corollary P.2** (A pencil-native separating pair). *For every  $n \geq 3$ , the degree-three maps*

$$f_0 = [s^3 : t^3], \quad f_1 = [s^3 + st^2 : t^3]$$

*give nonisomorphic unmarked order- $d$  failure schemes of quadratic pencils, although they have the same Artin fibres, the same conormal and first-relation bundles, the same pencil and quotient bundles, and isomorphic nilradical graded vector bundles in every degree. For  $n = 3$  their common finite rank is 167945, and their top graded bundle is*

$$\mathcal{O}^{75537} \oplus \mathcal{O}(3)^{\oplus 30}.$$

*Proof.* The pairs of cubics have no common zero. In the affine coordinate  $z = s/t$  the maps are  $z^3$  and  $z^3 + z$ . The first has two ramification points, both of index three. The second has the two distinct zeros of  $3z^2 + 1$ , each of index two, and infinity of index three. Independent projective changes in source and target preserve this multiset of indices, so the two maps are inequivalent. The theorem proves all the geometric assertions. For  $n = 3$ ,  $e = 9$ ,  $d = 11$ ,  $M = 15$ , whence  $\binom{20}{11} - 15 = 167945$  and  $\binom{19}{11} - 45 = 75537$ .  $\square$

This consequence uses the global inverse essentially. Pointwise local reconstruction returns the same pencil orbit at every point for every  $f$ ; it cannot recover the map  $f$ . Even supplementing it with all the abstract bundles displayed in the theorem leaves the covering equivalence problem unresolved. It is the multiplication in the total unmarked thickening, followed by descent to one constant left transformation, that recovers that problem. The double-orbit description is not an assertion that the scheme has the automorphism group of the cover: the unused complement of  $H$ , the right factor, and the higher-order kernel remain present, as recorded by the previous automorphism theorems.

## APPENDIX Q. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 7.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

**Q.1. The spectral torsion sheaf.** Let  $R \subset \text{Sym}^2 V$  be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line  $L_R = \mathbb{P}(R)$ , the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (\text{Q.1})$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve  $L_R$ . Thus  $\mathcal{T}_R$  is a torsion sheaf of length  $n$ , whose zeroth Fitting divisor is the full discriminant. The pair  $(L_R, \mathcal{T}_R)$  is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of  $R$ . The relation with elementary divisors is classical; see [4, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

**Lemma Q.1** (Sheaves on the affine spectral line). *Let  $A, B$  be symmetric complex matrices with  $A$  invertible. On the affine line with coordinate  $z$ , the cokernel of  $zA - B$  is the finite-dimensional  $\mathbb{C}[z]$ -module on  $\mathbb{C}^n$  for which multiplication by  $z$  is  $C = A^{-1}B$ . An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

*Proof.* Multiplication on the target by  $A^{-1}$  identifies the presentation with  $zI - C$ . Reduction of a polynomial vector modulo these relations sends  $\sum_j z^j v_j$  to  $\sum_j C^j v_j$ ; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism  $G$  satisfying  $GC = C'G$ , which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data.  $\square$

**Lemma Q.2** (Polynomial square roots of invertible operators). *For every invertible complex matrix  $K$  there is  $h \in \mathbb{C}[x]$  with  $h(K)^2 = K$ . If  $K$  is self-adjoint for a nondegenerate symmetric form and commutes with  $C$ , then  $h(K)$  has both properties.*

*Proof.* Write the minimal polynomial as  $\prod_\lambda (x - \lambda)^{a_\lambda}$ , where  $\lambda \neq 0$ . In its local Artinian factor choose a square root  $\mu_\lambda^2 = \lambda$ , and put

$$h_\lambda(x) = \mu_\lambda \sum_{j=0}^{a_\lambda-1} \binom{1/2}{j} \left( \frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo  $(x - \lambda)^{a_\lambda}$ , gives  $h_\lambda(x)^2 = x$ . The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial. Evaluation at  $K$  proves the assertion even when  $K$  is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in  $K$  commutes with every operator commuting with  $K$ .  $\square$

**Theorem Q.3** (Spectral reconstruction at the first relation). *Let  $n \geq 3$ ,  $d = n^2 + 2n - 4$ , and let  $R, R'$  be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes  $Z_R^{[d]}$  and  $Z_{R'}^{[d]}$  are isomorphic;*
- (2) *there is an isomorphism  $f : L_R \rightarrow L_{R'}$  with  $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$ ;*
- (3) *the pencils  $R, R'$  are projectively equivalent.*

*In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.*

*Proof.* The equivalence of the first and third conditions is Theorem 7.1. A congruence of pencils induces an isomorphism of (Q.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using  $f$  to identify the lines. Write the corresponding symmetric pairs as  $(A, B)$  and  $(A', B')$ , with  $A, A'$  invertible, and set  $C = A^{-1}B$ ,  $C' = A'^{-1}B'$ . Choose the presentation  $zA - B$  for the affine coordinate on both lines. Lemma Q.1 identifies the two sheaves with the modules of  $C, C'$ , so their isomorphism gives a similarity. After applying that change of coordinates, assume  $C' = C$ . Both  $A$  and  $A'$  are nondegenerate symmetric forms for which  $C$  is self-adjoint. Then  $K = A^{-1}A'$  commutes with  $C$  and is self-adjoint for  $A$ . Lemma Q.2 provides a polynomial  $h$  with  $h(K)^2 = K$ . Put  $H = h(K)$ . It commutes with  $C$ , is self-adjoint for  $A$ , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over  $\mathbb{C}$ .  $\square$

For  $x \in \text{Supp } \mathcal{T}_R$ , put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x}/\mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \quad (\text{Q.2})$$

If the elementary-divisor exponents at  $x$  are  $e_1, \dots, e_b$ , the structure theorem over the discrete valuation ring  $\mathcal{O}_{L_R,x}$  gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (\text{Q.3})$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at  $x$  gives only  $\ell_1(x)$ , and the multiplicity of the discriminant gives only the eventual value  $\ell_a(x) = \sum_j e_j$ . The intermediate values are genuinely additional information.

**Q.2. A specialization with constant discriminant and coranks.** The following family keeps both ends of this length profile fixed. Fix  $i \in \mathbb{C}$  with  $i^2 = -1$ , and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2} N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (\text{Q.4})$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (\text{Q.5})$$

Both matrices are symmetric and  $A_0$  is invertible. They are independent for every  $t$ . For  $n > 4$ , append  $I_{n-4}$  to  $A_0$  and append  $\text{diag}(\mu_5, \dots, \mu_n)$  to  $B_t$ , where the  $\mu_j$  are fixed distinct nonzero numbers. Denote the resulting matrices by  $A^{(n)}, B_t^{(n)}$  and their pencil again by  $R_t$ .

**Theorem Q.4** (Separation inside fixed reduced rank data). *For every  $n \geq 4$ , the family  $R_t$ ,  $t \in \mathbb{A}^1$ , has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of  $t$ :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (\text{Q.6})$$

*For every minor size, the reduced determinantal locus on the marked parameter line is also independent of  $t$ .*

- (2) *At the root  $s = 0$ , the spectral module has exponents  $(3, 1)$  for  $t \neq 0$  and  $(2, 2)$  for  $t = 0$ . Consequently  $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$  for every  $t \neq 0$ , whereas all the punctured fibres are isomorphic.*

- (3) The schemes  $Z_{R_t}^{[d]}$  form a flat projective family over  $\mathbb{A}^1$ , with fixed reduction. After one identification of the socle quotient bundles, every truncation of order  $k < d$  is a constant family. Thus the first separating order remains exactly  $d = n^2 + 2n - 4$  even under the constraints in part (1).

*Proof.* The identities  $N_0^2 = 0$  and  $\text{rank } N_0 = 1$  give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} (P_t \quad I_2). \quad (\text{Q.7})$$

The right factor in the last expression is surjective and the left factor is injective. Hence  $\text{rank } C_t = 2$  for every  $t$ , whereas  $\text{rank } C_t^2 = 1$  precisely when  $t \neq 0$ . The nilpotent partitions are therefore  $(3, 1)$  and  $(2, 2)$ . Also  $\det A_0 = 1$  and  $\det(sI_4 + uC_t) = s^4$ , proving (Q.6). At  $s = 0$  the full pencil has corank two for all  $t$ ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on  $\mathbb{P}^1(\mathbb{C})$  implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (Q.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem Q.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on  $\mathbb{A}^1 \setminus 0$  is needed.

For flatness we use the relative gluing of Theorem R.1; the following coordinates identify its pullback in the present family. First note that  $t \mapsto R_t$  is a morphism to  $\text{Gr}(2, \text{Sym}^2 V)$ , not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy  $a(A^{(n)}) = b(B_t^{(n)}) = 1$  and  $a(B_t^{(n)}) = b(A^{(n)}) = 0$ . Their common kernel is a fixed complement to every  $R_t$ . It trivializes  $S_{R_t} = \text{Sym}^2 V / R_t$ , and hence identifies the relative socle Grassmannian with  $B \times \mathbb{A}^1$ .

On a frame chart of its tautological bundle, the graph description (7.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \text{Sym}^j E^* \oplus (\text{Sym}^d E^* / K_{R_t}). \quad (\text{Q.8})$$

By Theorem 7.3,  $K_{R_t}$  is a subbundle of constant rank  $\binom{N}{2}$ , with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (Q.8) is locally free on  $B \times \mathbb{A}^1$ , of rank  $\binom{n^2+d}{d} - \binom{N}{2}$ . The graph projection is a finite flat morphism to that base. Since  $B$  is projective, its composition with projection to  $\mathbb{A}^1$  is flat and projective. Below degree  $d$  there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes.  $\square$

The qualifier “reduced” in part (1) is necessary. At  $s = 0$ , in a local parameter  $z = s/u$ , the gcd of the three-row minors of the four-dimensional block is a unit times  $z$  for  $t \neq 0$  and a unit times  $z^2$  for  $t = 0$ . Both radicals are  $(z)$ . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

**Q.3. An independently visible change of reciprocal geometry.** For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [4, §3].

**Corollary Q.5** (A conic specializes to a line). *For the four-dimensional family (Q.5), the reciprocal curve is a smooth plane conic when  $t \neq 0$  and a projective line when  $t = 0$ . The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

*Proof.* For  $s \neq 0$ , multiplication on the right by the fixed invertible matrix  $A_0^{-1}$  identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If  $t \neq 0$ , the matrices  $I_4, C_t, C_t^2$  are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image  $[s : u] \mapsto [s^2 : -su : u^2]$ , a smooth conic in its spanning plane. At  $t = 0$  the last term vanishes, and  $I_4, C_0$  are independent, giving a line. The remaining assertions follow from Theorem Q.4, with  $d = 20$ . The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods.  $\square$

#### APPENDIX R. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put  $P = \text{Gr}(2, \text{Sym}^2 V)$ , and write  $\mathcal{R}$  and  $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$  for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where  $\mathcal{U}$  is tautological of rank  $n$ . As before,  $N = n(n+1)/2$ ,  $p = N - 2$ ,  $d = n + 2p$ , and  $M = \binom{N}{2}$ .

**Theorem R.1** (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- $d$  neighbourhood  $\mathcal{Z}^{[d]}$  of  $\mathcal{B}$ . There is a rank- $M$  vector subbundle  $\mathcal{K} \subset \text{Sym}^d \mathcal{E}^*$  such that*

$$\mathcal{Z}^{[d]} = \text{Spec}_{\mathcal{B}} \left( \text{Sym} \mathcal{E}^* / (\mathcal{K}, \text{Sym}^{d+1} \mathcal{E}^*) \right). \quad (\text{R.1})$$

*This is finite locally free over  $\mathcal{B}$ , of rank*

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (\text{R.2})$$

*and is projective and flat over  $P$ . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes  $T \rightarrow P$ , including nonreduced ones. For  $k < d$ , its order- $k$  truncation is  $\text{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^*$ , with truncated multiplication. After trivializing  $\mathcal{S}$  on the parameter base, these lower-order families are constant.*

*Proof.* The locus in  $\text{Gr}_P(n, V \oplus \mathcal{S})$  on which projection to  $\mathcal{S}$  is injective is canonically  $\text{Tot } \mathcal{E}$ . It is an open neighbourhood of  $\mathcal{B}$ , not a choice of formal coordinates. On a frame chart of  $\mathcal{U}$ , let  $T : \mathcal{U} \rightarrow V$  be the universal graph map and let  $\gamma : \text{Sym}^2 V \rightarrow \mathcal{S}$  be the quotient. The same block elimination that proves (7.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T)I_p(\gamma \text{Sym}^2 T).$$

Every generator is homogeneous of degree  $d$ .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \mathrm{Sym}^2 \mathcal{U} \longrightarrow \mathrm{Sym}^d \mathcal{E}^*. \quad (\text{R.3})$$

The inclusion used in this expression is  $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \mathrm{Sym}^2 V)^*$ . The determinant factors evaluate  $\det T$ , and the remaining factors evaluate  $\bigwedge^p \mathrm{Sym}^2 T$  followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map  $\nu$  of Theorem 7.3, restricted to a line subbundle tensored with the fixed space  $\bigwedge^p \mathrm{Sym}^2 \mathbb{C}^n$ . Its image is a subbundle of rank  $M$ : locally split the line in the source, and split the fixed complex linear inclusion  $\nu$ . Both splittings remain splittings over any complex algebra. Denote its image by  $\mathcal{K}$ . A right change of frame of  $\mathcal{U}$  changes the graph coordinates and this coefficient map contragrediently. The determinant character in (R.3) is exactly the one arising from  $\det T$ . Hence these local ideals and their degree- $d$  subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let  $\mathfrak{b}$  be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by  $\mathfrak{b}^{d+1}$  therefore gives precisely (R.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in  $n^2$  variables gives (R.2). The morphism  $\mathcal{B} \rightarrow P$  is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree  $d$  the ideal contributes no relation. This proves the final assertion as well.  $\square$

*Remark R.2.* Over a nonreduced base, the theorem concerns the specified relative socle subscheme  $\mathcal{B}_T$  and its ideal. It does not identify that ideal with the absolute nilradical of  $\mathcal{Z}_T^{[d]}$ , which can also contain nilpotents from the base. On geometric fibres  $\mathcal{B}_t$  is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

**R.1. Spectral incidence schemes from the first relation space.** On the regular-pencil locus, pass locally to ordered generators  $(A, B)$  with  $A$  invertible, and put  $C = A^{-1}B$ . For a marked point with affine coordinate  $z$ , set  $X = C - zI$ . The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

**Lemma R.3** (Truncated spectral presentation). *Let  $D$  be a commutative ring,  $X \in \mathrm{Mat}_n(D)$ , and  $a \geq 1$ . There is a natural isomorphism*

$$\mathrm{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \mathrm{coker}(X^a : D^n \rightarrow D^n). \quad (\text{R.4})$$

*Equivalently, the  $a$ -square block presentation with diagonal blocks  $X$  and subdiagonal blocks  $-I$  is equivalent, by invertible row and column operations over  $D$ , to  $I_{n(a-1)} \oplus X^a$ .*

*Proof.* Map the class of  $\sum_{j=0}^{a-1} w^j v_j$  to the class of  $\sum_{j=0}^{a-1} X^j v_j$  modulo  $X^a D^n$ . The relation  $wv = Xv$  and the relation  $w^a v = 0$  show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient  $-I$  gives the

stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms.  $\square$

**Theorem R.4** (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 7.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (\text{R.5})$$

*This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to  $\text{length}(\mathcal{T}_z/\mathfrak{m}_z^a \mathcal{T}_z) \geq h$ . Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- $d$  relation subspace.*

*Proof.* The closed immersion  $\Phi$  of Theorem 7.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to  $P$ , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation  $(z+w)I - C = wI - X$ . Lemma R.3 gives the presentation  $X^a$  for restriction to the order- $a$  fat point centred at  $z$ . Its  $(h-1)$ -st Fitting ideal is exactly the ideal in (R.5); Fitting ideals of finitely presented modules commute with base change. On a field the cokernel has dimension at least  $h$  exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in  $V$  conjugates  $X$ ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (Q.3) then identifies its geometric fibres with the prescribed elementary-divisor partition.  $\square$

**R.2. Realizing the complete Segre stratification.** Let  $P_\sigma$  denote the reduced closure in  $P$  of a regular Segre stratum, and let  $J_\sigma = \Phi(P_\sigma)$ . Because  $\Phi$  is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (\text{R.6})$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem R.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [4, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem Q.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (R.6) is the image of  $\Phi$ , not a moduli space of every possible unmarked deformation of a finite scheme.

## APPENDIX S. FIXED-DISCRIMINANT STRATA AND ALL ADMISSIBLE SPECIALIZATIONS

The universal family of Section R and the example of Section Q lead to a classification of the entire fixed-discriminant, fixed-reduced-rank locus. The spectral orbit order is classical. Every allowed specialization is realized here by a flat family of the original finite multiplication-failure neighbourhoods, and the same first relation detects every strict transition.

Fix distinct  $\alpha_1, \dots, \alpha_s \in \mathbb{C}$ , positive  $m_i$  with  $\sum m_i = n \geq 3$ , and  $1 \leq c_i \leq m_i$ . Fix a nonsingular symmetric  $A$ , and let  $\mathfrak{p}_A$  be its self-adjoint endomorphisms. Consider the reduced locally closed

variety

$$X_{m,c}(A) = \left\{ C \in \mathfrak{p}_A : \det(zI - C) = \prod_i (z - \alpha_i)^{m_i}, \quad \dim \ker(C - \alpha_i I) = c_i \right\}. \quad (\text{S.1})$$

Exclude  $s = 1, c_1 = n$ , for which  $C$  is scalar and  $\langle A, AC \rangle$  is not a pencil. Every other datum is realized. Put  $R_C = \langle A, AC \rangle$ ,  $d = n^2 + 2n - 4$ , and  $Z_C = Z_{R_C}^{[d]}$ . The full binary determinant is  $\det(A) \prod_i (u + \alpha_i v)^{m_i}$ . The coranks and labelled roots specify all reduced determinantal loci on the line.

Write  $\mathcal{P}(m, c)$  for partitions into exactly  $c$  positive parts. For equal-size partitions,  $\mu \preceq \lambda$  means  $\sum_{j \leq a} \mu_j \leq \sum_{j \leq a} \lambda_j$  for every  $a$ ; write  $\lambda'$  for the conjugate partition.

**S.1. The classical orthogonal input.** We isolate the orbit theorem used below, with its precise group and source. Let  $(H, A)$  be a nondegenerate complex symmetric space of dimension  $m$ . The acting group is the full orthogonal group  $O(A)$ , and the operators are  $A$ -self-adjoint, not elements of the skew-adjoint Lie algebra  $\mathfrak{o}(A)$ .

**Lemma S.1** (Self-adjoint nilpotent orbits). *The full  $O(A)$ -orbits of self-adjoint nilpotent operators are indexed by all partitions  $\lambda$  of  $m$ . If  $\mathcal{O}_\lambda$  denotes the orbit, then*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left( m^2 - \sum_a (\lambda'_a)^2 \right), \quad \mathcal{O}_\mu \subset \overline{\mathcal{O}_\lambda} \iff \mu \preceq \lambda.$$

*An  $O(A)$ -orbit is one  $SO(A)$ -orbit when  $\lambda$  has an odd part. When every part is even, it is the disjoint union of two  $SO(A)$ -orbits, interchanged by an isometry of determinant  $-1$ . The closure statement above is for the full orthogonal orbit and does not identify the closures of its two components.*

*Proof.* The classification and closure assertions are [5, Proposition 1, p. 444; Theorem 1, p. 447], respectively, with  $\varepsilon = 1$  and  $G(V)$  as defined in [5, §1.1, p. 443]. Remark 1 on p. 444 specifies that all partitions occur; the order in §1.4 on p. 446 is precisely the row-dominance order used here. The proof of Theorem 1 occupies pp. 447–451. These references concern the full group, not its identity component.

For the dimension formula there is a short independent calculation. The trace pairing on  $\text{End } H$  is nondegenerate on both its self-adjoint and skew-adjoint summands. For a self-adjoint nilpotent  $N$ , the commutator with  $N$  exchanges the summands, and the two resulting maps are adjoint up to sign for that pairing. They have equal rank. Consequently  $\dim \text{GL}(H)N = 2 \dim O(A)N$ , also the identity in [5, §2.4, Remark 8(i), p. 456]. The general linear centralizer has dimension  $\sum_{i,j} \min(\lambda_i, \lambda_j) = \sum_a (\lambda'_a)^2$ , as follows by writing homomorphisms between the cyclic  $\mathbb{C}[t]$ -modules of the Jordan decomposition. This proves the stated dimension.

We give the component argument explicitly. The good bases of [5, Lemma 1, p. 444] decompose the symmetric space orthogonally into cyclic Jordan blocks, each with a reversed-identity Gram matrix. If one block has odd size, its negative identity, extended by the identity on the other blocks, centralizes  $N$  and has determinant  $-1$ . The centralizer then meets both components of  $O(A)$ , so the orbit is a single  $SO(A)$ -orbit.

For the converse, write

$$E_k = \ker N^k / (\ker N^{k-1} + N \ker N^{k+1}).$$

The form induced by  $A(u, N^{k-1}v)$  on  $E_k$  is nondegenerate and symmetric; this is immediate on the same cyclic blocks. A commuting isometry induces  $g_k \in O(E_k)$ . A filtration of the Jordan module by its cyclic lengths and powers of  $t$  gives

$$\det g = \prod_k (\det g_k)^k.$$

Indeed the multiplicity-space action of a length- $k$  block occurs  $k$  times on the associated graded; the strictly triangular terms of a module automorphism do not change its determinant. If all lengths are even, this product is one. The centralizer lies in  $SO(A)$ , and the index-two quotient of  $O(A)$  therefore gives exactly two  $SO(A)$ -orbits. Since  $SO(A)$  is connected, these are the connected components of the full orbit. This does not preclude their closures from meeting.  $\square$

**Theorem S.2** (Classification and specialization at the first relation). *The following statements hold.*

- (1) *The  $O(A)$ -orbits in  $X_{\mathbf{m},c}(A)$  are indexed by  $\lambda \in \prod_i \mathcal{P}(m_i, c_i)$ , with dimensions*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left( n^2 - \sum_i \sum_a (\lambda'_{i,a})^2 \right). \quad (\text{S.2})$$

*Unmarked isomorphism classes of  $Z_C$  are these tuples modulo the permutation group induced on the roots by projective transformations preserving their multiplicities and coranks.*

- (2) *A specialization from  $\lambda$  to  $\mu$  inside (S.1) exists if and only if*

$$\mu_i \preceq \lambda_i \quad \text{for every } i. \quad (\text{S.3})$$

*Every such pair is realized over a smooth pointed algebraic curve, after shrinking at the special point, with  $A$  fixed and with constant full determinant and reduced rank loci.*

- (3) *Every such pencil family induces a flat projective family of  $Z_C$ . After shrinking the curve, every truncation of order  $k < d$  is constant. If at least one dominance inequality is strict, the generic and special  $Z_C$  are nonisomorphic; their first distinguishing order is  $d$ .*
- (4) *The recovered spectral module at  $\alpha_i$  has length profile*

$$L_{i,a} = \sum_j \min(a, \lambda_{i,j}). \quad (\text{S.4})$$

*Every  $L_{i,a}$  can only increase in the stated specialization. All these inequalities together are equivalent to (S.3), giving an intrinsic numerical criterion for every admissible transition.*

*Proof.* We use Lemma S.1. Its full-orthogonal classification and dominance order have exactly the group and self-adjoint convention needed here. The comparison for Grassmannian Segre strata is [4, Theorem 5.1].

The primary spaces  $V_i = \ker(C - \alpha_i I)^{m_i}$  are an orthogonal direct sum: the Chinese remainder idempotents are polynomials in the self-adjoint  $C$ , hence self-adjoint projections. Each restricted form is nondegenerate, and  $C|_{V_i} = \alpha_i I + N_i$ , with  $N_i$  self-adjoint nilpotent. The prescribed corank is its number of Jordan blocks. The classical theorem gives the indexing. Conversely, a Jordan block is self-adjoint for the reversed-identity Gram matrix; an isometry with the chosen nondegenerate form realizes each partition.

The space of orthogonal decompositions has dimension  $(n^2 - \sum_i m_i^2)/2$ . Adding the nilpotent orbit dimensions proves (S.2). These are dimensions in the fixed- $A$  matrix space, not in an unmarked moduli stack. Full orthogonal orbits need not be connected; no irreducibility assertion is needed. Theorem Q.3 identifies the unmarked neighbourhood with the pencil, and its spectral module determines the partition at each root. A projective change of the line can therefore do exactly the stated root permutations. For one or two roots the projective stabilizer may be infinite; only its finite permutation image is used.

For necessity, the same polynomial idempotents decompose a family with fixed characteristic polynomial into subbundles of ranks  $m_i$ . The rank of  $N_i^a$  cannot increase at specialization. Since  $m_i - \text{rank } N_i^a = \sum_j \min(a, \lambda_{i,j})$ , this gives the length inequalities. Summing the first  $a$  columns of conjugate Young diagrams shows that they are equivalent to dominance: conjugation reverses the dominance order.

For sufficiency, fix an orthogonal decomposition of  $(V, A)$ . The closure criterion puts a chosen representative of  $\mu_i$  in the closure of the orbit for  $\lambda_i$ . Thus the special tuple lies in the closure of the product of the generic orbits. Choose an irreducible component of that closure containing the point and meeting the generic orbit product. Successive general hyperplane sections through the point, avoiding the closed complement, give an integral algebraic curve with generic point in that open orbit product. Normalize it and choose a point over the prescribed point. A normal complex curve is smooth. Remove the other points outside the generic orbit product. A constant curve suffices in the equality case. The resulting block operator  $\bigoplus_i(\alpha_i I + N_i(t))$  is self-adjoint for the same  $A$ . Its determinant is unchanged, and its coranks agree generically and specially, hence throughout the shrunk curve. This gives the required specialization. It does not assert constancy of the full higher Fitting ideals.

For part (3), pull back Theorem R.1 along the curve's morphism to  $P = \text{Gr}(2, \text{Sym}^2 V)$ . That theorem identifies the family with the actual ideal-adic neighbourhoods and proves flatness and projectivity under arbitrary base change. No inference from fibrewise isomorphisms is needed. An affine Grassmannian chart around the special pencil supplies a fixed complement to nearby  $R_C$ , identifying  $S_{R_C}$ ,  $B$ , and  $E$ . There is no relation below degree  $d$ , so all lower neighbourhoods become constant. For a strict dominance transition,  $\sum_{i,a}(\lambda'_{i,a})^2$  strictly increases; this sum is invariant under permutations of the decorated roots. The generic and special tuples therefore cannot be related by such a permutation. Part (1) separates their unmarked degree- $d$  neighbourhoods. Flatness is asserted only for these finite neighbourhoods, not unrestricted failure schemes.  $\square$

**Corollary S.3** (Exactly when the coarse data lose a labelled class). *The number of root-labelled congruence classes with the prescribed data is  $\prod_i \#\mathcal{P}(m_i, c_i)$ . It is one precisely when, for every  $i$ , either  $c_i = 1$  or  $m_i - c_i \leq 1$ . The first possible ambiguity with fixed full determinant and all reduced rank loci occurs in dimension four, at  $(m, c) = (4, 2)$ , with partitions  $(3, 1)$  and  $(2, 2)$ .*

*Proof.* Subtracting one from every part identifies  $\mathcal{P}(m, c)$  with partitions of  $m - c$  into at most  $c$  parts. There is one if  $c = 1$  or  $m - c \leq 1$ . Otherwise  $(m - c)$  and  $(m - c - 1, 1)$  are distinct admissible partitions. The theorem gives the count and the minimal example.  $\square$

**S.2. The first relation as a standard graded Betti space.** The finite invariant has a usual Hilbert-scheme interpretation. The first Betti space means the generator space in the first free module of a minimal graded resolution of the quotient, not the space of relations among generators of its defining ideal. The following statement retains the marked quotient presentation.

**Proposition S.4** (A graded Hilbert locus and its first Betti space). *In the fixed coordinates of Theorem 7.3, let  $S = \text{Sym } H^*$ ,  $N = \binom{n+1}{2}$ ,  $M = \binom{N}{2}$ , and  $Q_R = S/((K_R) + S_{\geq d+1})$ . Then*

$$\text{Tor}_1^S(Q_R, \mathbb{C})_d = K_R. \quad (\text{S.5})$$

*This identity commutes with arbitrary complex base change. The functor of graded quotients with Hilbert function*

$$h_j = \begin{cases} \dim S_j, & 0 \leq j < d, \\ \dim S_d - M, & j = d, \\ 0, & j > d \end{cases}$$

*is represented by  $\text{Gr}(M, S_d)$ . The pencil locus is a closed copy of  $\text{Gr}(2, \text{Sym}^2 V)$ . Its degree- $d$  first Betti space recovers every scheme-theoretic pencil parameter, including parameters over nonreduced bases.*

*Proof.* Put  $J = (K_R) + S_{\geq d+1}$  and  $S_+ = \bigoplus_{j>0} S_j$ . The augmented sequence  $0 \rightarrow J \rightarrow S \rightarrow Q_R \rightarrow 0$  gives  $\text{Tor}_1^S(Q_R, \mathbb{C}) = J/S_+J$ , since its map to  $S/S_+$  is zero. Its degree- $d$  part is  $K_R$ . The

same argument over a base algebra applies to the degree- $d$  subbundle with locally free quotient and commutes with pullback. A graded quotient with the indicated locally free components has zero ideal below degree  $d$ , a rank- $M$  subbundle in degree  $d$ , and the full ideal in higher degrees. Conversely, every such subbundle defines an ideal, as its positive-degree products lie above degree  $d$ . This is the Grassmannian functor. The closed immersion and its inverse on the image are Theorem 7.3. This is not an inference from closed-point Torelli to unmarked deformation-groupoid equivalence.  $\square$

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AI assistance was used in developing and drafting the arguments and finite verification scripts. All structural proofs are supplied for independent mathematical scrutiny; computational receipts are not formal proof certificates. No AI system is listed as an author.

For the present revision, GPT-6 Astra Pro assisted specifically with the incidence normalization and overlap criterion in Section II.N, the universal-property proof in Section M, the explicit specialization in Section N, and the associated exact tests and exposition. These arguments require independent human verification. No statement here represents a completed journal review or a formal proof-assistant certification.

Additional disclosure for the present revision. GPT-6 Astra Pro assisted specifically with the conductor and collision models, the scalar-matrix degeneration, the flag-jet calculation, the reciprocal-polynomial fibre presentation, literature comparisons, and exact computational checks. The proofs above state the complete mathematical arguments; finite computations are consistency checks, not universal proof certificates. No journal decision or historical priority determination is inferred from the build and source audits.

#### ASSISTANCE IN THE PRESENT REVISION

An AI assistant helped formulate and write the binary conductor stratification, the simultaneous incidence atlas, the evaluated singular Fitting ideal, and the minimal-relation-module argument for pure-power fibres. It also prepared the source-preserving reorganization and finite exact consistency checks. The written proofs, rather than those finite checks, are the mathematical basis of the statements and are submitted to independent scrutiny.

Revision 154. AI assistance was used for the polarized moduli compatibility, spectral divisor-fibre construction, homological calculations and proof expansions. The fixed five-variable Betti table uses exact computer-assisted rational elimination with its complete finite inputs recorded. The general theorems rely on the written arguments and the explicitly attributed classical results, not on regression counts or source-preservation receipts.

Revision 155. AI assistance was used to formulate and draft the exact quadratic coefficient-section theorem and the reciprocal power-ideal description of spectral Fitting schemes, to check their finite examples, and to prepare the companion-paper organization. Classical apolar and orthogonal-orbit results are explicitly attributed. The general claims rely on the written proofs; finite exact tests and source preservation are not formal proof certificates.

Revision 157. AI assistance was used in deriving and drafting the universal power-ideal formula, its complete-quadric graph interpretation, the contact-divisor and resolved-pencil statements, and their exact finite checks. Classical determinantal, apolar and complete-quadric results are attributed at their points of use. The proofs, rather than the computations, support the general statements. No formal proof certification or journal endorsement is claimed.

Scope of the new intrinsic construction. The source-normalized envelope is an algebra bundle with an intrinsic projective power diagram. It is not a claimed preferred subquotient of the sharp failure algebra. Singular-pencil classification is read through the classical complex symmetric

Kronecker form. The explicit Hilbert tail is a boundary calculation for the stated transverse family, not a classification of every fibre of the incidence compactification. Exact computations accompanying the source check finite instances; they do not certify the general proofs.

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